

Tối ưu hóa Danh mục Cổ phiếu Ngân hàng Việt Nam

Ứng dụng Mô hình Black-Litterman kết hợp Phân cụm K-means và Monte Carlo

Universe: 25 cổ phiếu ngân hàng niêm yết tại HOSE / HNX

Dữ liệu: Giá đóng cửa tháng & Vol hóa trị trung bình | 2014-06 → 2026-05

Cửa sổ huấn luyện: 36 tháng (rolling) | OOS: 106 tháng (2017-08 → 2026-05)

Lãi suất phi rủi ro: VN10Y / 12 (time-varying)

Danh mục: MKT · TAN · MV · BL · EW (1/N) · RP (Risk Parity)

Danh mục	Ann. Return	Ann. Vol	Sharpe	Sortino	MDD
MKT	+23.40%	27.05%	0.729	0.763	-32.51%
TAN	+19.20%	27.81%	0.558	0.549	-51.54%
MV	+16.24%	25.40%	0.494	0.539	-38.44%
BL	+25.29%	29.77%	0.726	0.715	-51.13%
EW	+26.62%	27.38%	0.837	0.904	-36.55%
RP	+26.62%	27.38%	0.837	0.904	-36.55%

1. Hiệu suất Out-of-Sample (Tóm tắt)

Lãi nhuận netm hoá, biến động, Sharpe, Sortino, MDD, Calmar và tail risk của 6 danh mục trên 106 tháng OOS (08/2017 – 05/2026). Lãi suất phi rủi ro sử dụng VN10Y / 12 theo tháng netm.

Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar	Win_Rate
MKT	0.234033	0.270493	0.728858	0.762929	-0.325098	0.719883	0.613208
TAN	0.191990	0.278101	0.557740	0.549311	-0.515448	0.372472	0.622642
MV	0.162408	0.253993	0.494209	0.539221	-0.384445	0.422448	0.575472
BL	0.252876	0.297692	0.725561	0.715253	-0.511309	0.494566	0.622642
EW	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
RP	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642

2. Kiểm định Thống kê

2a. Paired t-test ($H_0: E[r_{BL}] = E[r_{other}]$)

Kiểm định t ghép cặp hai phía so sánh lãi nhuận tháng OOS của BL với từng danh mục còn lại.

Pair	t_statistic	p_value	is_significant_5pct
BL vs MKT	0.361026	0.718805	False
BL vs TAN	1.1679	0.245477	False
BL vs MV	1.5319	0.128552	False
BL vs EW	-0.284069	0.776917	False
BL vs RP	-0.284069	0.776917	False

2b. Jobson-Korkie Test ($H_0: SR_{BL} = SR_{other}$)

Kiểm định Jobson & Korkie (1981) so sánh trực tiếp hai hệ số Sharpe (dùng thống kê Sharpe theo tháng, không annualise). Thống kê z kiểm định phân phối chuẩn $N(0,1)$.

Pair	z_statistic	p_value	is_significant_5pct
BL vs MKT	-0.049603	0.960439	False
BL vs TAN	0.911789	0.361880	False
BL vs MV	1.0882	0.276495	False
BL vs EW	-0.726267	0.467675	False
BL vs RP	-0.726267	0.467675	False

3. Kiểm định Phân phối Lợi nhuận

3a. Jarque-Bera (H0: phân phối chuẩn)

Kiểm tra liệu lợi nhuận tháng OOS có phân phối chuẩn không. Bác bỏ H0 ($p < 0.05$) nghĩa là lợi nhuận có thể lệch hoặc đuôi dày.

Portfolio	JB_stat	JB_pval	Skewness	Excess_Kurtosis	is_normal_5pct
MKT	10.1416	0.006277	-0.175466	1.6061	False
TAN	16.2824	0.000291	-0.212219	2.0242	False
MV	83.7728	0.000000	0.143990	4.6169	False
BL	7.2160	0.027106	-0.228615	1.3133	False
EW	7.2799	0.026253	0.198406	1.3413	False
RP	7.2799	0.026253	0.198406	1.3413	False

3b. Ljung-Box (H0: không có tự tương quan)

Kiểm tra tự tương quan chuỗi lợi nhuận tháng tại các trễ 5, 10, 20. $p > 0.05$ nghĩa là không có bằng chứng tự tương quan.

Portfolio	LB5_stat	LB5_pval	LB5_no_autocorr	LB10_stat	LB10_pval	LB10_no_autocorr	LB20_stat	LB20_pval	LB20_no_autocorr
MKT	8.8677	0.114457	True	19.7177	0.032039	False	34.2652	0.024385	False
TAN	4.5857	0.468501	True	11.5802	0.314135	True	25.5470	0.181299	True
MV	2.0715	0.839169	True	16.2845	0.091772	True	25.7963	0.172645	True
BL	5.3942	0.369681	True	10.5265	0.395572	True	24.1239	0.237024	True
EW	3.0475	0.692660	True	12.8746	0.230769	True	23.9440	0.244848	True
RP	3.0475	0.692660	True	12.8746	0.230769	True	23.9440	0.244848	True

4. Phân tích Drawdown

4a. Tóm tắt Drawdown

Số lần sụt giảm, độ sâu trung bình, thời gian phục hồi trung bình, số tháng danh mục vượt ngưỡng -20%.

Portfolio	N_Drawdowns	Max_DD_pct	Avg_DD_Depth_pct	Avg_Recovery_Months	Months_Below_Neg20
MKT	9	-32.5100	-15.8300	6.9000	19
TAN	5	-51.5400	-24.9000	9.6000	51
MV	5	-38.4400	-21.3200	11.8000	26
BL	6	-51.1300	-23.6000	12.4000	40
EW	7	-36.5500	-15.3500	9.0000	34
RP	7	-36.5500	-15.3500	9.0000	34

4b. Chi tiết từng Drawdown

Ngày bắt đầu, ngày đáy, ngày phục hồi, độ sâu (%), số tháng liên tiếp và số tháng phục hồi cho từng đợt sụt giảm.

Portfolio	DD_Start	DD_Trough	DD_Recovery	Depth_pct	Months_to_Trough	Recovery_Months	Total_Months
MKT	2018-04-30	2018-06-30	2019-12-31 00:00:00	-26.9400	2	18.0000	20
MKT	2020-02-29	2020-03-31	2020-11-30 00:00:00	-31.0700	1	8.0000	9
MKT	2021-01-31	2021-01-31	2021-02-28 00:00:00	-7.8800	0	1.0000	1
MKT	2021-07-31	2021-08-31	2022-01-31 00:00:00	-14.3800	1	5.0000	6
MKT	2022-02-28	2022-10-31	2024-02-29 00:00:00	-32.5100	8	16.0000	24
MKT	2024-04-30	2024-06-30	2024-08-31 00:00:00	-6.9900	2	2.0000	4
MKT	2024-10-31	2024-11-30	2025-01-31 00:00:00	-2.8100	1	2.0000	3
MKT	2025-04-30	2025-04-30	2025-07-31 00:00:00	-7.6900	0	3.0000	3
MKT	2025-09-30	2026-03-31	Not recovered	-12.2300	6	nan	8

Portfolio	DD_Start	DD_Trough	DD_Recovery	Depth_pct	Months_to_Trough	Recovery_Months	Total_Months
TAN	2018-04-30	2020-03-31	2020-12-31 00:00:00	-38.0500	23	9.0000	32
TAN	2021-01-31	2021-01-31	2021-02-28 00:00:00	-8.0800	0	1.0000	1
TAN	2021-07-31	2021-08-31	2021-12-31 00:00:00	-15.6200	1	4.0000	5
TAN	2022-02-28	2022-12-31	2025-08-31 00:00:00	-51.5400	10	32.0000	42
TAN	2025-09-30	2026-03-31	2026-05-31 00:00:00	-11.1900	6	2.0000	8
MV	2018-04-30	2020-03-31	2020-12-31 00:00:00	-38.4400	23	9.0000	32
MV	2021-01-31	2021-01-31	2021-03-31 00:00:00	-8.0800	0	2.0000	2
MV	2021-06-30	2021-08-31	2022-01-31 00:00:00	-13.0300	2	5.0000	7
MV	2022-02-28	2022-12-31	2025-07-31 00:00:00	-31.6700	10	31.0000	41
MV	2025-09-30	2026-03-31	Not recovered	-15.3700	6	nan	8
BL	2018-04-30	2018-07-31	2020-01-31 00:00:00	-28.6100	3	18.0000	21
BL	2020-02-29	2020-03-31	2020-11-30 00:00:00	-31.6200	1	8.0000	9
BL	2021-01-31	2021-01-31	2021-02-28 00:00:00	-10.4600	0	1.0000	1
BL	2021-06-30	2022-10-31	2025-06-30 00:00:00	-51.1300	16	32.0000	48
BL	2025-09-30	2025-11-30	2026-02-28 00:00:00	-11.6400	2	3.0000	5
BL	2026-03-31	2026-03-31	Not recovered	-8.1200	0	nan	2
EW	2018-04-30	2018-12-31	2020-09-30 00:00:00	-30.3600	8	21.0000	29
EW	2021-01-31	2021-01-31	2021-02-28 00:00:00	-6.2900	0	1.0000	1
EW	2021-06-30	2021-08-31	2022-01-31 00:00:00	-13.5800	2	5.0000	7
EW	2022-02-28	2022-10-31	2024-09-30 00:00:00	-36.5500	8	23.0000	31
EW	2024-10-31	2024-11-30	2025-02-28 00:00:00	-4.6700	1	3.0000	4
EW	2025-04-30	2025-04-30	2025-05-31 00:00:00	-4.8700	0	1.0000	1
EW	2025-09-30	2026-03-31	Not recovered	-11.1000	6	nan	8
RP	2018-04-30	2018-12-31	2020-09-30 00:00:00	-30.3600	8	21.0000	29
RP	2021-01-31	2021-01-31	2021-02-28 00:00:00	-6.2900	0	1.0000	1

Portfolio	DD_Start	DD_Trough	DD_Recovery	Depth_pct	Months_to_Trough	Recovery_Months	Total_Months
RP	2021-06-30	2021-08-31	2022-01-31 00:00:00	-13.5800	2	5.0000	7
RP	2022-02-28	2022-10-31	2024-09-30 00:00:00	-36.5500	8	23.0000	31
RP	2024-10-31	2024-11-30	2025-02-28 00:00:00	-4.6700	1	3.0000	4
RP	2025-04-30	2025-04-30	2025-05-31 00:00:00	-4.8700	0	1.0000	1
RP	2025-09-30	2026-03-31	Not recovered	-11.1000	6	nan	8

5. Phân tích One-at-a-time

Mỗi tham số thay đổi một lần, các tham số còn lại giữ nguyên giá trị cũ (lookback=36, max_weight=30%, k=3, rf=VN10Y). Chọn BL Sharpe để dùng để đánh giá mức hiệu quả.

Tham số: lookback

Value	Is_Baseline	Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar	Win_Rate
24	False	MKT	0.235121	0.265019	0.739276	0.799771	-0.326455	0.720227	0.593220
24	False	TAN	0.232108	0.274176	0.703592	0.834590	-0.517867	0.448199	0.593220
24	False	MV	0.208152	0.240141	0.703555	0.869632	-0.334275	0.622695	0.567797
24	False	BL	0.268213	0.297371	0.770127	0.913148	-0.343371	0.781117	0.567797
24	False	EW	0.270876	0.263778	0.878302	0.952496	-0.379934	0.712955	0.652542
24	False	RP	0.270876	0.263778	0.878302	0.952496	-0.379934	0.712955	0.652542
36	True	MKT	0.234033	0.270493	0.728858	0.762929	-0.325098	0.719883	0.613208
36	True	TAN	0.191990	0.278101	0.557740	0.549310	-0.515448	0.372472	0.622642
36	True	MV	0.162408	0.253993	0.494210	0.539222	-0.384445	0.422449	0.575472
36	True	BL	0.252876	0.297692	0.725560	0.715252	-0.511309	0.494566	0.622642
36	True	EW	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
36	True	RP	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
48	False	MKT	0.195444	0.251826	0.636748	0.642588	-0.317482	0.615607	0.595745
48	False	TAN	0.228688	0.271121	0.714048	0.692491	-0.481909	0.474545	0.617021
48	False	MV	0.141531	0.227842	0.467152	0.486528	-0.316631	0.446992	0.563830
48	False	BL	0.186781	0.274338	0.552917	0.518623	-0.480045	0.389090	0.617021
48	False	EW	0.241854	0.252678	0.818272	0.844153	-0.362445	0.667283	0.606383
48	False	RP	0.241854	0.252678	0.818272	0.844153	-0.362445	0.667283	0.606383

Tham số: max_weight

Value	Is_Baseline	Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar	Win_Rate
20%	False	MKT	0.234033	0.270493	0.728858	0.762929	-0.325098	0.719883	0.613208
20%	False	TAN	0.220025	0.281697	0.650139	0.666817	-0.440590	0.499386	0.632075
20%	False	MV	0.179781	0.260903	0.547709	0.578302	-0.405647	0.443195	0.594340
20%	False	BL	0.254806	0.286682	0.760160	0.769591	-0.478955	0.532004	0.641509
20%	False	EW	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
20%	False	RP	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
30%	True	MKT	0.234033	0.270493	0.728858	0.762929	-0.325098	0.719883	0.613208
30%	True	TAN	0.191990	0.278101	0.557740	0.549310	-0.515448	0.372472	0.622642
30%	True	MV	0.162408	0.253993	0.494210	0.539222	-0.384445	0.422449	0.575472
30%	True	BL	0.252876	0.297692	0.725560	0.715252	-0.511309	0.494566	0.622642
30%	True	EW	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
30%	True	RP	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
40%	False	MKT	0.234033	0.270493	0.728858	0.762929	-0.325098	0.719883	0.613208
40%	False	TAN	0.180892	0.281645	0.511316	0.515126	-0.516288	0.350370	0.603774
40%	False	MV	0.155090	0.257685	0.458730	0.510449	-0.385704	0.402096	0.566038
40%	False	BL	0.253213	0.305199	0.708819	0.708351	-0.512443	0.494129	0.613208
40%	False	EW	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
40%	False	RP	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642

Tham số: k_clusters

Value	Is_Baseline	Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar	Win_Rate
2	False	MKT	0.234033	0.270493	0.728858	0.762929	-0.325098	0.719883	0.613208
2	False	TAN	0.191990	0.278101	0.557740	0.549310	-0.515448	0.372472	0.622642
2	False	MV	0.162408	0.253993	0.494210	0.539222	-0.384445	0.422449	0.575472

Value	Is_Baseline	Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar	Win_Rate
2	False	BL	0.176466	0.279951	0.498602	0.522913	-0.478276	0.368963	0.584906
2	False	EW	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
2	False	RP	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
3	True	MKT	0.234033	0.270493	0.728858	0.762929	-0.325098	0.719883	0.613208
3	True	TAN	0.191990	0.278101	0.557740	0.549310	-0.515448	0.372472	0.622642
3	True	MV	0.162408	0.253993	0.494210	0.539222	-0.384445	0.422449	0.575472
3	True	BL	0.252876	0.297692	0.725560	0.715252	-0.511309	0.494566	0.622642
3	True	EW	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
3	True	RP	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
4	False	MKT	0.234033	0.270493	0.728858	0.762929	-0.325098	0.719883	0.613208
4	False	TAN	0.191990	0.278101	0.557740	0.549310	-0.515448	0.372472	0.622642
4	False	MV	0.162408	0.253993	0.494210	0.539222	-0.384445	0.422449	0.575472
4	False	BL	0.248939	0.295860	0.716746	0.766222	-0.475690	0.523322	0.575472
4	False	EW	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
4	False	RP	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
5	False	MKT	0.234033	0.270493	0.728858	0.762929	-0.325098	0.719883	0.613208
5	False	TAN	0.191990	0.278101	0.557740	0.549310	-0.515448	0.372472	0.622642
5	False	MV	0.162408	0.253993	0.494210	0.539222	-0.384445	0.422449	0.575472
5	False	BL	0.287408	0.307397	0.814993	0.889591	-0.467403	0.614905	0.594340
5	False	EW	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
5	False	RP	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642

Tham số: rf_rate

Value	Is_Baseline	Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar	Win_Rate
4% fixed	False	MKT	0.234033	0.270493	0.717331	0.748941	-0.325098	0.719883	0.613208

Value	Is_Baseline	Portfolio	Ann_Return	Ann_Vol	Sharpe	Sortino	MDD	Calmar	Win_Rate
4% fixed	False	TAN	0.192536	0.278191	0.548314	0.545425	-0.515448	0.373532	0.622642
4% fixed	False	MV	0.162408	0.253993	0.481935	0.524301	-0.384445	0.422449	0.575472
4% fixed	False	BL	0.261400	0.299016	0.740428	0.723239	-0.508595	0.513965	0.632075
4% fixed	False	EW	0.266157	0.273821	0.825931	0.889631	-0.365473	0.728253	0.622642
4% fixed	False	RP	0.266157	0.273821	0.825931	0.889631	-0.365473	0.728253	0.622642
5% fixed	False	MKT	0.234033	0.270493	0.680361	0.712600	-0.325098	0.719883	0.613208
5% fixed	False	TAN	0.192949	0.277950	0.514298	0.513135	-0.514923	0.374715	0.622642
5% fixed	False	MV	0.162408	0.253993	0.442564	0.477545	-0.384445	0.422449	0.575472
5% fixed	False	BL	0.255281	0.296405	0.692569	0.653675	-0.504946	0.505561	0.632075
5% fixed	False	EW	0.266157	0.273821	0.789410	0.843007	-0.365473	0.728253	0.622642
5% fixed	False	RP	0.266157	0.273821	0.789410	0.843007	-0.365473	0.728253	0.622642
6% fixed	False	MKT	0.234033	0.270493	0.643392	0.668470	-0.325098	0.719883	0.613208
6% fixed	False	TAN	0.194646	0.277630	0.484982	0.486037	-0.513095	0.379356	0.632075
6% fixed	False	MV	0.162408	0.253993	0.403192	0.435981	-0.384445	0.422449	0.575472
6% fixed	False	BL	0.275864	0.300802	0.717629	0.699606	-0.499881	0.551859	0.632075
6% fixed	False	EW	0.266157	0.273821	0.752890	0.825780	-0.365473	0.728253	0.622642
6% fixed	False	RP	0.266157	0.273821	0.752890	0.825780	-0.365473	0.728253	0.622642
VN10Y (dynamic)	True	MKT	0.234033	0.270493	0.728858	0.762929	-0.325098	0.719883	0.613208
VN10Y (dynamic)	True	TAN	0.191990	0.278101	0.557740	0.549310	-0.515448	0.372472	0.622642
VN10Y (dynamic)	True	MV	0.162408	0.253993	0.494210	0.539222	-0.384445	0.422449	0.575472
VN10Y (dynamic)	True	BL	0.252876	0.297692	0.725560	0.715252	-0.511309	0.494566	0.622642
VN10Y (dynamic)	True	EW	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642
VN10Y (dynamic)	True	RP	0.266157	0.273821	0.837317	0.903664	-0.365473	0.728253	0.622642

6. Monte Carlo – b n v ng Sharpe tr c sai s d báo

V i m i m c nhi u $\delta \in \{-10\%, -5\%, 0\%, +5\%, +10\%\}$, view vector q c nhi u hoá 2000 l n theo phân ph i chu n. μ_{BL} c tính closed-form (tuy n tính trong q), Sharpe c a w_{BL} c nh c ánh giá d i m i q nhi u. C t TAN là Sharpe c nh (không dùng views).

Delta_Noise	Port_Type	Mean_Sharpe	Std_Sharpe	Min_Sharpe	Max_Sharpe
-0.100000	BL	0.607500	0.313800	-0.026800	1.7312
-0.100000	TAN	1.0271	0.398100	0.319600	2.3520
-0.050000	BL	0.609200	0.315100	-0.027800	1.7374
-0.050000	TAN	1.0271	0.398100	0.319600	2.3520
0.000000	BL	0.609000	0.315000	-0.027700	1.7368
0.000000	TAN	1.0271	0.398100	0.319600	2.3520
0.050000	BL	0.608800	0.314800	-0.027600	1.7359
0.050000	TAN	1.0271	0.398100	0.319600	2.3520
0.100000	BL	0.608700	0.314700	-0.027500	1.7355
0.100000	TAN	1.0271	0.398100	0.319600	2.3520

Full Monte Carlo data (first 60 rows shown):

Date	Delta_Noise	Port_Type	Expected_Sharpe
2017-08-31	-0.100000	BL	0.484699
2017-08-31	-0.100000	TAN	0.852097
2017-08-31	-0.050000	BL	0.486166
2017-08-31	-0.050000	TAN	0.852097
2017-08-31	0.000000	BL	0.486011
2017-08-31	0.000000	TAN	0.852097
2017-08-31	0.050000	BL	0.485814
2017-08-31	0.050000	TAN	0.852097

Date	Delta_Noise	Port_Type	Expected_Sharpe
2017-08-31	0.100000	BL	0.485701
2017-08-31	0.100000	TAN	0.852097
2017-09-30	-0.100000	BL	0.529385
2017-09-30	-0.100000	TAN	0.917878
2017-09-30	-0.050000	BL	0.530792
2017-09-30	-0.050000	TAN	0.917878
2017-09-30	0.000000	BL	0.530644
2017-09-30	0.000000	TAN	0.917878
2017-09-30	0.050000	BL	0.530455
2017-09-30	0.050000	TAN	0.917878
2017-09-30	0.100000	BL	0.530346
2017-09-30	0.100000	TAN	0.917878
2017-10-31	-0.100000	BL	0.443248
2017-10-31	-0.100000	TAN	0.879333
2017-10-31	-0.050000	BL	0.444761
2017-10-31	-0.050000	TAN	0.879333
2017-10-31	0.000000	BL	0.444601
2017-10-31	0.000000	TAN	0.879333
2017-10-31	0.050000	BL	0.444398
2017-10-31	0.050000	TAN	0.879333
2017-10-31	0.100000	BL	0.444282
2017-10-31	0.100000	TAN	0.879333
2017-11-30	-0.100000	BL	0.529816
2017-11-30	-0.100000	TAN	0.975314
2017-11-30	-0.050000	BL	0.531260

Date	Delta_Noise	Port_Type	Expected_Sharpe
2017-11-30	-0.050000	TAN	0.975314
2017-11-30	0.000000	BL	0.531108
2017-11-30	0.000000	TAN	0.975314
2017-11-30	0.050000	BL	0.530914
2017-11-30	0.050000	TAN	0.975314
2017-11-30	0.100000	BL	0.530803
2017-11-30	0.100000	TAN	0.975314
2017-12-31	-0.100000	BL	0.585216
2017-12-31	-0.100000	TAN	1.0086
2017-12-31	-0.050000	BL	0.586784
2017-12-31	-0.050000	TAN	1.0086
2017-12-31	0.000000	BL	0.586618
2017-12-31	0.000000	TAN	1.0086
2017-12-31	0.050000	BL	0.586408
2017-12-31	0.050000	TAN	1.0086
2017-12-31	0.100000	BL	0.586287
2017-12-31	0.100000	TAN	1.0086
2018-01-31	-0.100000	BL	0.764168
2018-01-31	-0.100000	TAN	1.2179
2018-01-31	-0.050000	BL	0.765295
2018-01-31	-0.050000	TAN	1.2179
2018-01-31	0.000000	BL	0.765176
2018-01-31	0.000000	TAN	1.2179
2018-01-31	0.050000	BL	0.765025
2018-01-31	0.050000	TAN	1.2179

Date	Delta_Noise	Port_Type	Expected_Sharpe
2018-01-31	0.100000	BL	0.764938
2018-01-31	0.100000	TAN	1.2179

7. Chuỗi Lợi nhuận Out-of-Sample (106 tháng)

Lợi nhuận tháng (simple return) của 6 danh mục và lãi suất phi rủi ro cho từng tháng OOS.

Date	RF_monthly	MKT	TAN	MV	BL	EW	RP
2017-08-31	0.004833	-0.026592	0.021803	0.000532	0.004067	-0.024700	-0.024700
2017-09-30	0.004833	0.008600	0.017614	0.012287	0.012754	0.014080	0.014080
2017-10-31	0.004833	0.059921	0.050255	0.029226	0.065712	0.017326	0.017326
2017-11-30	0.004833	0.184409	0.152730	0.163976	0.175310	0.187071	0.187071
2017-12-31	0.004833	0.048090	0.050632	0.034358	0.051444	0.018884	0.018884
2018-01-31	0.003817	0.243752	0.230535	0.294043	0.221359	0.267098	0.267098
2018-02-28	0.003817	0.120337	0.099277	0.077586	0.136180	0.098473	0.098473
2018-03-31	0.003817	0.016710	0.017842	0.015931	0.045105	0.020543	0.020543
2018-04-30	0.003817	-0.146733	-0.126264	-0.123802	-0.147346	-0.131849	-0.131849
2018-05-31	0.003817	-0.114886	-0.073255	-0.076191	-0.113053	-0.111764	-0.111764
2018-06-30	0.003817	-0.032586	-0.053069	-0.064188	-0.054541	-0.069962	-0.069962
2018-07-31	0.003817	0.007452	0.022622	0.022947	-0.001573	0.006366	0.006366
2018-08-31	0.003817	0.123037	0.060966	0.055417	0.170872	0.095768	0.095768
2018-09-30	0.003817	0.018549	0.002697	0.004246	0.012967	0.040664	0.040664
2018-10-31	0.003817	-0.121796	-0.095982	-0.095374	-0.136805	-0.115741	-0.115741
2018-11-30	0.003817	0.007542	-0.013142	-0.013195	0.011568	-0.005717	-0.005717
2018-12-31	0.003817	-0.018578	-0.045817	-0.044849	-0.011678	-0.037550	-0.037550
2019-01-31	0.003817	0.012980	0.048089	0.056572	0.026190	0.021086	0.021086
2019-02-28	0.003817	0.041454	0.019705	0.016662	0.024266	0.024012	0.024012
2019-03-31	0.003817	0.087474	0.063028	0.052068	0.078411	0.052151	0.052151
2019-04-30	0.003817	-0.009915	-0.002335	-0.001570	-0.013223	-0.014232	-0.014232

Date	RF_monthly	MKT	TAN	MV	BL	EW	RP
2019-05-31	0.003817	-0.040294	-0.036642	-0.028837	-0.047458	-0.045280	-0.045280
2019-06-30	0.003817	0.018816	0.015486	0.011442	0.011194	-0.006345	-0.006345
2019-07-31	0.003817	0.103573	0.079358	0.061666	0.091141	0.045768	0.045768
2019-08-31	0.003817	0.003178	0.004126	-0.007868	0.007728	-0.003207	-0.003207
2019-09-30	0.003817	0.044382	0.054395	0.054182	0.043244	0.042613	0.042613
2019-10-31	0.003817	0.047669	0.035522	0.034822	0.041424	0.028980	0.028980
2019-11-30	0.003817	-0.019920	-0.028705	-0.038766	-0.014220	-0.042676	-0.042676
2019-12-31	0.003817	0.067450	0.011788	-0.001566	0.055241	0.032970	0.032970
2020-01-31	0.002500	0.040052	0.009322	-0.000004	0.038354	0.064014	0.064014
2020-02-29	0.002500	-0.047792	-0.041175	0.029620	-0.032779	0.089194	0.089194
2020-03-31	0.002500	-0.276107	-0.300868	-0.301494	-0.293025	-0.240255	-0.240255
2020-04-30	0.002500	0.137555	0.150589	0.155972	0.137314	0.178490	0.178490
2020-05-31	0.002500	0.168756	0.089504	0.089163	0.064632	0.104038	0.104038
2020-06-30	0.002500	-0.046926	-0.055452	-0.035664	-0.053627	-0.018955	-0.018955
2020-07-31	0.002500	-0.040333	-0.053315	-0.048023	-0.030081	-0.035201	-0.035201
2020-08-31	0.002500	0.121801	0.114106	0.125604	0.149732	0.146584	0.146584
2020-09-30	0.002500	0.034012	0.054371	0.061032	0.061133	0.097577	0.097577
2020-10-31	0.002500	-0.000661	0.013751	0.034318	0.041503	0.034211	0.034211
2020-11-30	0.002500	0.114471	0.091426	0.111259	0.071170	0.107254	0.107254
2020-12-31	0.002500	0.100638	0.157318	0.165052	0.108781	0.139943	0.139943
2021-01-31	0.002108	-0.078755	-0.080811	-0.080838	-0.104636	-0.062929	-0.062929
2021-02-28	0.002108	0.128424	0.104445	0.056616	0.152345	0.144913	0.144913
2021-03-31	0.002108	0.043929	0.195558	0.224080	0.216741	0.147558	0.147558
2021-04-30	0.002108	0.069686	0.087668	0.039030	0.099892	0.089302	0.089302
2021-05-31	0.002108	0.154716	0.220957	0.144040	0.236005	0.233709	0.233709

Date	RF_monthly	MKT	TAN	MV	BL	EW	RP
2021-06-30	0.002108	0.044947	0.004665	-0.018138	-0.046715	-0.004647	-0.004647
2021-07-31	0.002108	-0.113116	-0.123328	-0.111084	-0.083136	-0.091124	-0.091124
2021-08-31	0.002108	-0.034625	-0.037530	-0.003530	-0.075167	-0.044764	-0.044764
2021-09-30	0.002108	0.000996	0.011400	0.025518	-0.036968	0.004185	0.004185
2021-10-31	0.002108	0.030575	0.031964	0.028788	0.061750	0.040120	0.040120
2021-11-30	0.002108	0.032911	0.057504	0.030239	0.054810	0.052475	0.052475
2021-12-31	0.002108	0.020766	0.083291	0.030334	-0.004583	0.032872	0.032872
2022-01-31	0.003333	0.104257	0.031341	0.043366	0.033405	0.076226	0.076226
2022-02-28	0.003333	-0.043408	-0.019388	-0.027255	-0.010184	-0.038160	-0.038160
2022-03-31	0.003333	-0.020615	-0.022193	-0.016968	-0.014785	0.004651	0.004651
2022-04-30	0.003333	-0.082787	-0.147377	-0.099271	-0.127443	-0.109177	-0.109177
2022-05-31	0.003333	-0.072449	-0.103551	-0.044184	-0.144848	-0.081414	-0.081414
2022-06-30	0.003333	-0.068432	-0.136007	-0.070779	-0.096287	-0.087762	-0.087762
2022-07-31	0.003333	0.044292	0.090316	0.018105	0.069541	0.059262	0.059262
2022-08-31	0.003333	0.074039	0.013024	0.046948	0.043295	0.050773	0.050773
2022-09-30	0.003333	-0.143466	-0.127780	-0.109730	-0.146202	-0.136735	-0.136735
2022-10-31	0.003333	-0.053868	-0.057153	-0.033293	-0.130204	-0.084503	-0.084503
2022-11-30	0.003333	0.083200	-0.084619	0.002064	0.082071	0.028596	0.028596
2022-12-31	0.003333	-0.006743	-0.079643	-0.028363	0.029065	-0.010396	-0.010396
2023-01-31	0.002917	0.148118	0.157561	0.143657	0.156780	0.152493	0.152493
2023-02-28	0.002917	-0.047656	-0.080255	0.005233	-0.107926	-0.061735	-0.061735
2023-03-31	0.002917	0.047184	0.070172	-0.007017	0.114593	0.052515	0.052515
2023-04-30	0.002917	-0.020110	-0.037522	-0.006488	-0.044418	-0.006776	-0.006776
2023-05-31	0.002917	0.021645	0.031663	0.037142	0.038866	0.023716	0.023716
2023-06-30	0.002917	0.046573	0.048782	0.045352	0.054050	0.056366	0.056366

Date	RF_monthly	MKT	TAN	MV	BL	EW	RP
2023-07-31	0.002917	0.072933	0.058160	0.042184	0.052379	0.064792	0.064792
2023-08-31	0.002917	-0.002617	0.011854	-0.017438	0.044115	0.016439	0.016439
2023-09-30	0.002917	-0.031677	-0.030950	-0.019040	-0.041012	-0.045869	-0.045869
2023-10-31	0.002917	-0.059835	-0.003531	-0.033969	-0.055288	-0.075824	-0.075824
2023-11-30	0.002917	0.024465	0.009890	0.007350	0.018604	0.044460	0.044460
2023-12-31	0.002917	0.023527	0.008199	0.017215	0.007707	0.036214	0.036214
2024-01-31	0.002550	0.091943	0.075801	0.078301	0.092206	0.072711	0.072711
2024-02-29	0.002550	0.095842	0.043038	0.063421	0.040724	0.065798	0.065798
2024-03-31	0.002550	0.003849	0.003879	-0.000978	0.022444	0.004252	0.004252
2024-04-30	0.002550	-0.048663	-0.013038	-0.042120	0.014609	-0.029587	-0.029587
2024-05-31	0.002550	-0.003662	0.002642	0.010818	0.071023	0.022505	0.022505
2024-06-30	0.002550	-0.018775	-0.022561	-0.020244	-0.004611	-0.012736	-0.012736
2024-07-31	0.002550	0.048634	0.052790	0.019355	0.042727	0.037203	0.037203
2024-08-31	0.002550	0.025287	0.051975	0.001160	0.041370	0.018478	0.018478
2024-09-30	0.002550	0.026921	0.012138	-0.015353	0.025754	0.039055	0.039055
2024-10-31	0.002550	-0.008231	-0.015224	-0.011221	-0.019030	-0.024802	-0.024802
2024-11-30	0.002550	-0.020013	-0.021911	-0.013213	-0.010509	-0.022484	-0.022484
2024-12-31	0.002550	0.018309	0.093132	0.011599	0.007447	0.030247	0.030247
2025-01-31	0.002583	0.018841	0.049506	0.037886	0.030785	0.011546	0.011546
2025-02-28	0.002583	0.028168	0.014633	0.032052	0.044863	0.045352	0.045352
2025-03-31	0.002583	0.000756	-0.039288	0.006031	-0.018431	0.000248	0.000248
2025-04-30	0.002583	-0.076861	-0.054662	-0.063103	-0.055894	-0.048695	-0.048695
2025-05-31	0.002583	0.042136	0.026155	0.015781	0.034939	0.077463	0.077463
2025-06-30	0.002583	0.038606	0.018894	0.012406	0.060004	0.029487	0.029487
2025-07-31	0.002583	0.101024	0.086195	0.093231	0.090637	0.156467	0.156467

Date	RF_monthly	MKT	TAN	MV	BL	EW	RP
2025-08-31	0.002583	0.191455	0.174109	0.093140	0.184158	0.155768	0.155768
2025-09-30	0.002583	-0.061845	-0.026854	-0.084397	-0.031925	-0.059003	-0.059003
2025-10-31	0.002583	-0.042226	-0.006632	-0.027556	-0.050097	-0.019916	-0.019916
2025-11-30	0.002583	-0.019884	-0.018711	-0.037211	-0.039125	-0.016465	-0.016465
2025-12-31	0.002583	0.025236	-0.006659	-0.001797	0.086259	0.016349	0.016349
2026-01-31	0.003625	0.098177	0.000622	0.047368	0.034152	0.034101	0.034101
2026-02-28	0.003625	-0.023014	0.007064	-0.013919	0.026943	-0.004318	-0.004318
2026-03-31	0.003625	-0.093967	-0.064750	-0.042369	-0.081196	-0.063432	-0.063432
2026-04-30	0.003625	0.029591	0.091541	0.029153	0.064458	0.036212	0.036212
2026-05-31	0.003625	0.029712	0.079321	0.011041	-0.000056	0.011828	0.011828

8. T n hi u Ph n c m K-means

Nh n c m v  th  view (Long / Short / Neutral) c a t ng c  phi u t i m i k  t i c n b ng. Ph n c m d a tr n (l i nhu n n m ho , bi n   ng n m ho )    c chu n ho  tr  c khi    a v o K-means (k=3).

T ng s  quan s t: 1,561 d ng

Date	Ticker	Cluster_Label	Assigned_View_Position
2017-08-31	VCB	1	Long
2017-08-31	BID	2	Neutral
2017-08-31	CTG	2	Neutral
2017-08-31	MBB	1	Long
2017-08-31	ACB	1	Long
2017-08-31	STB	0	Short
2017-08-31	SHB	0	Short
2017-09-30	VCB	0	Long
2017-09-30	BID	2	Neutral
2017-09-30	CTG	2	Neutral
2017-09-30	MBB	0	Long
2017-09-30	ACB	0	Long
2017-09-30	STB	1	Short
2017-09-30	SHB	1	Short
2017-10-31	VCB	1	Long
2017-10-31	BID	2	Neutral
2017-10-31	CTG	2	Neutral
2017-10-31	MBB	1	Long
2017-10-31	ACB	1	Long
2017-10-31	STB	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2017-10-31	SHB	0	Short
2017-11-30	VCB	1	Long
2017-11-30	BID	2	Neutral
2017-11-30	CTG	2	Neutral
2017-11-30	MBB	1	Long
2017-11-30	ACB	1	Long
2017-11-30	STB	0	Short
2017-11-30	SHB	0	Short
2017-12-31	VCB	1	Long
2017-12-31	BID	2	Neutral
2017-12-31	CTG	2	Neutral
2017-12-31	MBB	1	Long
2017-12-31	ACB	1	Long
2017-12-31	STB	0	Short
2017-12-31	SHB	0	Short
2018-01-31	VCB	1	Long
2018-01-31	BID	2	Neutral
2018-01-31	CTG	2	Neutral
2018-01-31	MBB	1	Long
2018-01-31	ACB	1	Long
2018-01-31	STB	0	Short
2018-01-31	SHB	0	Short
2018-02-28	VCB	1	Long
2018-02-28	BID	2	Neutral
2018-02-28	CTG	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2018-02-28	MBB	1	Long
2018-02-28	ACB	1	Long
2018-02-28	STB	0	Short
2018-02-28	SHB	0	Short
2018-03-31	VCB	1	Long
2018-03-31	BID	0	Neutral
2018-03-31	CTG	0	Neutral
2018-03-31	MBB	1	Long
2018-03-31	ACB	1	Long
2018-03-31	STB	2	Short
2018-03-31	SHB	0	Neutral
2018-04-30	VCB	1	Long
2018-04-30	BID	0	Neutral
2018-04-30	CTG	0	Neutral
2018-04-30	MBB	1	Long
2018-04-30	ACB	1	Long
2018-04-30	STB	2	Short
2018-04-30	SHB	0	Neutral
2018-05-31	VCB	2	Long
2018-05-31	BID	0	Neutral
2018-05-31	CTG	0	Neutral
2018-05-31	MBB	2	Long
2018-05-31	ACB	2	Long
2018-05-31	STB	1	Short
2018-05-31	SHB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2018-06-30	VCB	1	Long
2018-06-30	BID	0	Neutral
2018-06-30	CTG	0	Neutral
2018-06-30	MBB	1	Long
2018-06-30	ACB	1	Long
2018-06-30	STB	2	Short
2018-06-30	SHB	0	Neutral
2018-07-31	VCB	1	Long
2018-07-31	BID	0	Neutral
2018-07-31	CTG	0	Neutral
2018-07-31	MBB	1	Long
2018-07-31	ACB	1	Long
2018-07-31	STB	2	Short
2018-07-31	SHB	0	Neutral
2018-08-31	VCB	1	Long
2018-08-31	BID	0	Neutral
2018-08-31	CTG	0	Neutral
2018-08-31	MBB	1	Long
2018-08-31	ACB	1	Long
2018-08-31	STB	2	Short
2018-08-31	SHB	0	Neutral
2018-09-30	VCB	1	Long
2018-09-30	BID	0	Neutral
2018-09-30	CTG	0	Neutral
2018-09-30	MBB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2018-09-30	ACB	1	Long
2018-09-30	STB	2	Short
2018-09-30	SHB	0	Neutral
2018-10-31	VCB	1	Long
2018-10-31	BID	0	Neutral
2018-10-31	CTG	0	Neutral
2018-10-31	MBB	1	Long
2018-10-31	ACB	1	Long
2018-10-31	STB	2	Short
2018-10-31	SHB	0	Neutral
2018-11-30	VCB	1	Long
2018-11-30	BID	0	Neutral
2018-11-30	CTG	0	Neutral
2018-11-30	MBB	1	Long
2018-11-30	ACB	1	Long
2018-11-30	STB	2	Short
2018-11-30	SHB	0	Neutral
2018-12-31	VCB	1	Long
2018-12-31	BID	2	Neutral
2018-12-31	CTG	0	Short
2018-12-31	MBB	1	Long
2018-12-31	ACB	1	Long
2018-12-31	STB	0	Short
2018-12-31	SHB	2	Neutral
2019-01-31	VCB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2019-01-31	BID	2	Neutral
2019-01-31	CTG	0	Short
2019-01-31	MBB	1	Long
2019-01-31	ACB	1	Long
2019-01-31	STB	0	Short
2019-01-31	SHB	2	Neutral
2019-02-28	VCB	1	Long
2019-02-28	BID	0	Neutral
2019-02-28	CTG	2	Short
2019-02-28	MBB	1	Long
2019-02-28	ACB	1	Long
2019-02-28	STB	2	Short
2019-02-28	SHB	0	Neutral
2019-03-31	VCB	1	Long
2019-03-31	BID	2	Neutral
2019-03-31	CTG	0	Short
2019-03-31	MBB	1	Long
2019-03-31	ACB	1	Long
2019-03-31	STB	0	Short
2019-03-31	SHB	0	Short
2019-04-30	VCB	1	Long
2019-04-30	BID	2	Neutral
2019-04-30	CTG	0	Short
2019-04-30	MBB	1	Long
2019-04-30	ACB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2019-04-30	STB	0	Short
2019-04-30	SHB	0	Short
2019-05-31	VCB	1	Long
2019-05-31	BID	2	Neutral
2019-05-31	CTG	0	Short
2019-05-31	MBB	1	Long
2019-05-31	ACB	1	Long
2019-05-31	STB	0	Short
2019-05-31	SHB	0	Short
2019-06-30	VCB	1	Long
2019-06-30	BID	2	Neutral
2019-06-30	CTG	0	Short
2019-06-30	MBB	1	Long
2019-06-30	ACB	1	Long
2019-06-30	STB	0	Short
2019-06-30	SHB	0	Short
2019-07-31	VCB	1	Long
2019-07-31	BID	2	Neutral
2019-07-31	CTG	0	Short
2019-07-31	MBB	1	Long
2019-07-31	ACB	1	Long
2019-07-31	STB	0	Short
2019-07-31	SHB	0	Short
2019-08-31	VCB	1	Long
2019-08-31	BID	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2019-08-31	CTG	0	Short
2019-08-31	MBB	1	Long
2019-08-31	ACB	1	Long
2019-08-31	STB	0	Short
2019-08-31	SHB	0	Short
2019-09-30	VCB	1	Long
2019-09-30	BID	2	Neutral
2019-09-30	CTG	0	Short
2019-09-30	MBB	1	Long
2019-09-30	ACB	1	Long
2019-09-30	STB	0	Short
2019-09-30	SHB	0	Short
2019-10-31	VCB	1	Long
2019-10-31	BID	2	Neutral
2019-10-31	CTG	0	Short
2019-10-31	MBB	1	Long
2019-10-31	ACB	1	Long
2019-10-31	STB	0	Short
2019-10-31	SHB	0	Short
2019-11-30	VCB	1	Long
2019-11-30	BID	0	Neutral
2019-11-30	CTG	2	Short
2019-11-30	MBB	1	Long
2019-11-30	ACB	1	Long
2019-11-30	STB	2	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2019-11-30	SHB	0	Neutral
2019-12-31	VCB	1	Long
2019-12-31	BID	2	Neutral
2019-12-31	CTG	0	Short
2019-12-31	MBB	1	Long
2019-12-31	ACB	1	Long
2019-12-31	STB	0	Short
2019-12-31	SHB	0	Short
2020-01-31	VCB	1	Long
2020-01-31	BID	2	Neutral
2020-01-31	CTG	0	Short
2020-01-31	MBB	1	Long
2020-01-31	ACB	1	Long
2020-01-31	STB	0	Short
2020-01-31	SHB	0	Short
2020-02-29	VCB	1	Neutral
2020-02-29	BID	2	Long
2020-02-29	CTG	0	Short
2020-02-29	MBB	1	Neutral
2020-02-29	ACB	1	Neutral
2020-02-29	STB	0	Short
2020-02-29	SHB	0	Short
2020-03-31	VCB	1	Long
2020-03-31	BID	2	Neutral
2020-03-31	CTG	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-03-31	MBB	1	Long
2020-03-31	ACB	1	Long
2020-03-31	VIB	1	Long
2020-03-31	STB	0	Short
2020-03-31	SHB	2	Neutral
2020-04-30	VCB	2	Long
2020-04-30	BID	1	Neutral
2020-04-30	CTG	0	Short
2020-04-30	MBB	2	Long
2020-04-30	ACB	2	Long
2020-04-30	VIB	2	Long
2020-04-30	STB	0	Short
2020-04-30	SHB	1	Neutral
2020-05-31	VCB	1	Neutral
2020-05-31	BID	2	Long
2020-05-31	CTG	0	Short
2020-05-31	MBB	1	Neutral
2020-05-31	ACB	1	Neutral
2020-05-31	VIB	1	Neutral
2020-05-31	STB	0	Short
2020-05-31	SHB	2	Long
2020-06-30	VCB	2	Neutral
2020-06-30	BID	1	Long
2020-06-30	CTG	0	Short
2020-06-30	MBB	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-06-30	ACB	2	Neutral
2020-06-30	VIB	0	Short
2020-06-30	STB	0	Short
2020-06-30	SHB	1	Long
2020-07-31	VCB	2	Long
2020-07-31	BID	1	Neutral
2020-07-31	CTG	0	Short
2020-07-31	MBB	0	Short
2020-07-31	ACB	2	Long
2020-07-31	VIB	0	Short
2020-07-31	STB	0	Short
2020-07-31	SHB	1	Neutral
2020-08-31	VCB	0	Long
2020-08-31	BID	2	Neutral
2020-08-31	CTG	1	Short
2020-08-31	MBB	1	Short
2020-08-31	ACB	0	Long
2020-08-31	VIB	1	Short
2020-08-31	STB	1	Short
2020-08-31	SHB	2	Neutral
2020-08-31	KLB	1	Short
2020-09-30	VCB	1	Neutral
2020-09-30	BID	0	Long
2020-09-30	CTG	2	Short
2020-09-30	MBB	2	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-09-30	ACB	1	Neutral
2020-09-30	VIB	1	Neutral
2020-09-30	STB	2	Short
2020-09-30	SHB	0	Long
2020-09-30	KLB	1	Neutral
2020-10-31	VCB	1	Long
2020-10-31	BID	2	Neutral
2020-10-31	CTG	0	Short
2020-10-31	MBB	0	Short
2020-10-31	ACB	1	Long
2020-10-31	VPB	0	Short
2020-10-31	VIB	1	Long
2020-10-31	STB	0	Short
2020-10-31	SHB	2	Neutral
2020-10-31	KLB	0	Short
2020-11-30	VCB	2	Neutral
2020-11-30	BID	0	Long
2020-11-30	CTG	1	Short
2020-11-30	MBB	1	Short
2020-11-30	ACB	2	Neutral
2020-11-30	VPB	1	Short
2020-11-30	VIB	0	Long
2020-11-30	STB	1	Short
2020-11-30	SHB	0	Long
2020-11-30	KLB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2020-12-31	VCB	0	Neutral
2020-12-31	BID	2	Long
2020-12-31	CTG	1	Short
2020-12-31	MBB	1	Short
2020-12-31	ACB	0	Neutral
2020-12-31	VPB	1	Short
2020-12-31	VIB	2	Long
2020-12-31	STB	1	Short
2020-12-31	SHB	2	Long
2020-12-31	LPB	1	Short
2020-12-31	KLB	0	Neutral
2021-01-31	VCB	0	Neutral
2021-01-31	BID	2	Long
2021-01-31	CTG	1	Short
2021-01-31	MBB	1	Short
2021-01-31	ACB	0	Neutral
2021-01-31	VPB	1	Short
2021-01-31	VIB	2	Long
2021-01-31	STB	1	Short
2021-01-31	SHB	2	Long
2021-01-31	LPB	1	Short
2021-01-31	KLB	0	Neutral
2021-02-28	VCB	1	Neutral
2021-02-28	BID	0	Long
2021-02-28	CTG	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-02-28	MBB	1	Neutral
2021-02-28	ACB	1	Neutral
2021-02-28	VPB	1	Neutral
2021-02-28	VIB	0	Long
2021-02-28	STB	1	Neutral
2021-02-28	SHB	0	Long
2021-02-28	LPB	1	Neutral
2021-02-28	BAB	2	Short
2021-02-28	KLB	0	Long
2021-03-31	VCB	2	Long
2021-03-31	BID	1	Short
2021-03-31	CTG	1	Short
2021-03-31	MBB	1	Short
2021-03-31	ACB	2	Long
2021-03-31	VPB	1	Short
2021-03-31	HDB	1	Short
2021-03-31	VIB	2	Long
2021-03-31	STB	1	Short
2021-03-31	SHB	1	Short
2021-03-31	LPB	1	Short
2021-03-31	BAB	0	Neutral
2021-03-31	KLB	2	Long
2021-04-30	VCB	0	Short
2021-04-30	BID	0	Short
2021-04-30	CTG	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-04-30	MBB	0	Short
2021-04-30	ACB	0	Short
2021-04-30	VPB	0	Short
2021-04-30	HDB	0	Short
2021-04-30	VIB	1	Long
2021-04-30	STB	0	Short
2021-04-30	SHB	0	Short
2021-04-30	LPB	0	Short
2021-04-30	BAB	2	Neutral
2021-04-30	KLB	1	Long
2021-05-31	VCB	0	Neutral
2021-05-31	BID	2	Short
2021-05-31	CTG	2	Short
2021-05-31	MBB	0	Neutral
2021-05-31	ACB	0	Neutral
2021-05-31	VPB	2	Short
2021-05-31	HDB	2	Short
2021-05-31	VIB	1	Long
2021-05-31	STB	2	Short
2021-05-31	SHB	1	Long
2021-05-31	LPB	0	Neutral
2021-05-31	BAB	0	Neutral
2021-05-31	KLB	1	Long
2021-06-30	VCB	2	Neutral
2021-06-30	BID	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-06-30	CTG	0	Short
2021-06-30	MBB	0	Short
2021-06-30	ACB	2	Neutral
2021-06-30	VPB	0	Short
2021-06-30	HDB	0	Short
2021-06-30	VIB	1	Long
2021-06-30	TPB	2	Neutral
2021-06-30	STB	0	Short
2021-06-30	SHB	1	Long
2021-06-30	LPB	0	Short
2021-06-30	BAB	2	Neutral
2021-06-30	KLB	0	Short
2021-07-31	VCB	1	Neutral
2021-07-31	BID	2	Short
2021-07-31	CTG	2	Short
2021-07-31	MBB	2	Short
2021-07-31	ACB	1	Neutral
2021-07-31	VPB	0	Long
2021-07-31	HDB	2	Short
2021-07-31	VIB	0	Long
2021-07-31	TPB	1	Neutral
2021-07-31	STB	0	Long
2021-07-31	SHB	0	Long
2021-07-31	LPB	0	Long
2021-07-31	BAB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-07-31	KLB	0	Long
2021-08-31	VCB	1	Neutral
2021-08-31	BID	2	Short
2021-08-31	CTG	2	Short
2021-08-31	MBB	2	Short
2021-08-31	TCB	2	Short
2021-08-31	ACB	1	Neutral
2021-08-31	VPB	2	Short
2021-08-31	HDB	2	Short
2021-08-31	VIB	0	Long
2021-08-31	TPB	1	Neutral
2021-08-31	STB	2	Short
2021-08-31	SHB	0	Long
2021-08-31	LPB	0	Long
2021-08-31	BAB	1	Neutral
2021-08-31	KLB	2	Short
2021-09-30	VCB	2	Neutral
2021-09-30	BID	0	Short
2021-09-30	CTG	0	Short
2021-09-30	MBB	0	Short
2021-09-30	TCB	0	Short
2021-09-30	ACB	2	Neutral
2021-09-30	VPB	0	Short
2021-09-30	HDB	0	Short
2021-09-30	VIB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-09-30	TPB	2	Neutral
2021-09-30	STB	0	Short
2021-09-30	SHB	1	Long
2021-09-30	LPB	1	Long
2021-09-30	BAB	2	Neutral
2021-09-30	KLB	0	Short
2021-10-31	VCB	0	Neutral
2021-10-31	BID	2	Short
2021-10-31	CTG	2	Short
2021-10-31	MBB	2	Short
2021-10-31	TCB	2	Short
2021-10-31	ACB	0	Neutral
2021-10-31	VPB	1	Long
2021-10-31	HDB	2	Short
2021-10-31	VIB	1	Long
2021-10-31	TPB	0	Neutral
2021-10-31	STB	1	Long
2021-10-31	SHB	1	Long
2021-10-31	LPB	1	Long
2021-10-31	BAB	0	Neutral
2021-10-31	KLB	1	Long
2021-11-30	VCB	1	Neutral
2021-11-30	BID	0	Short
2021-11-30	CTG	0	Short
2021-11-30	MBB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-11-30	TCB	0	Short
2021-11-30	ACB	1	Neutral
2021-11-30	VPB	2	Long
2021-11-30	HDB	0	Short
2021-11-30	VIB	2	Long
2021-11-30	TPB	1	Neutral
2021-11-30	STB	0	Short
2021-11-30	SHB	2	Long
2021-11-30	LPB	2	Long
2021-11-30	BAB	1	Neutral
2021-11-30	KLB	0	Short
2021-12-31	VCB	2	Neutral
2021-12-31	BID	0	Short
2021-12-31	CTG	0	Short
2021-12-31	MBB	0	Short
2021-12-31	TCB	0	Short
2021-12-31	ACB	2	Neutral
2021-12-31	VPB	1	Long
2021-12-31	HDB	0	Short
2021-12-31	VIB	1	Long
2021-12-31	TPB	2	Neutral
2021-12-31	STB	0	Short
2021-12-31	SHB	1	Long
2021-12-31	LPB	1	Long
2021-12-31	BAB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2021-12-31	KLB	0	Short
2022-01-31	VCB	0	Neutral
2022-01-31	BID	2	Short
2022-01-31	CTG	2	Short
2022-01-31	MBB	2	Short
2022-01-31	TCB	2	Short
2022-01-31	ACB	0	Neutral
2022-01-31	VPB	1	Long
2022-01-31	HDB	2	Short
2022-01-31	VIB	1	Long
2022-01-31	TPB	0	Neutral
2022-01-31	STB	2	Short
2022-01-31	SHB	1	Long
2022-01-31	LPB	1	Long
2022-01-31	BAB	0	Neutral
2022-01-31	KLB	2	Short
2022-02-28	VCB	0	Short
2022-02-28	BID	0	Short
2022-02-28	CTG	1	Neutral
2022-02-28	MBB	1	Neutral
2022-02-28	TCB	1	Neutral
2022-02-28	ACB	0	Short
2022-02-28	VPB	2	Long
2022-02-28	HDB	1	Neutral
2022-02-28	VIB	2	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-02-28	TPB	1	Neutral
2022-02-28	STB	1	Neutral
2022-02-28	SHB	2	Long
2022-02-28	LPB	1	Neutral
2022-02-28	BAB	0	Short
2022-02-28	KLB	1	Neutral
2022-03-31	VCB	2	Neutral
2022-03-31	BID	1	Short
2022-03-31	CTG	1	Short
2022-03-31	MBB	1	Short
2022-03-31	TCB	1	Short
2022-03-31	ACB	2	Neutral
2022-03-31	VPB	0	Long
2022-03-31	HDB	1	Short
2022-03-31	VIB	0	Long
2022-03-31	TPB	2	Neutral
2022-03-31	STB	1	Short
2022-03-31	SHB	0	Long
2022-03-31	LPB	0	Long
2022-03-31	BAB	2	Neutral
2022-03-31	KLB	1	Short
2022-04-30	VCB	0	Neutral
2022-04-30	BID	2	Short
2022-04-30	CTG	2	Short
2022-04-30	MBB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-04-30	TCB	2	Short
2022-04-30	ACB	0	Neutral
2022-04-30	VPB	1	Long
2022-04-30	HDB	2	Short
2022-04-30	VIB	1	Long
2022-04-30	TPB	0	Neutral
2022-04-30	STB	1	Long
2022-04-30	SHB	1	Long
2022-04-30	LPB	1	Long
2022-04-30	BAB	0	Neutral
2022-04-30	KLB	1	Long
2022-05-31	VCB	0	Short
2022-05-31	BID	2	Neutral
2022-05-31	CTG	2	Neutral
2022-05-31	MBB	2	Neutral
2022-05-31	TCB	2	Neutral
2022-05-31	ACB	2	Neutral
2022-05-31	VPB	1	Long
2022-05-31	HDB	2	Neutral
2022-05-31	VIB	1	Long
2022-05-31	TPB	2	Neutral
2022-05-31	STB	1	Long
2022-05-31	SHB	1	Long
2022-05-31	LPB	2	Neutral
2022-05-31	BAB	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-05-31	KLB	1	Long
2022-06-30	VCB	1	Short
2022-06-30	BID	2	Neutral
2022-06-30	CTG	2	Neutral
2022-06-30	MBB	2	Neutral
2022-06-30	TCB	2	Neutral
2022-06-30	ACB	1	Short
2022-06-30	VPB	0	Long
2022-06-30	HDB	2	Neutral
2022-06-30	VIB	0	Long
2022-06-30	TPB	2	Neutral
2022-06-30	STB	2	Neutral
2022-06-30	SHB	0	Long
2022-06-30	LPB	2	Neutral
2022-06-30	BAB	1	Short
2022-06-30	KLB	0	Long
2022-07-31	VCB	1	Short
2022-07-31	BID	0	Neutral
2022-07-31	CTG	0	Neutral
2022-07-31	MBB	0	Neutral
2022-07-31	TCB	0	Neutral
2022-07-31	ACB	0	Neutral
2022-07-31	VPB	2	Long
2022-07-31	HDB	0	Neutral
2022-07-31	VIB	2	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-07-31	TPB	0	Neutral
2022-07-31	STB	0	Neutral
2022-07-31	SHB	2	Long
2022-07-31	LPB	0	Neutral
2022-07-31	BAB	1	Short
2022-07-31	KLB	2	Long
2022-08-31	VCB	2	Short
2022-08-31	BID	1	Neutral
2022-08-31	CTG	1	Neutral
2022-08-31	MBB	1	Neutral
2022-08-31	TCB	1	Neutral
2022-08-31	ACB	1	Neutral
2022-08-31	VPB	0	Long
2022-08-31	HDB	1	Neutral
2022-08-31	VIB	0	Long
2022-08-31	TPB	1	Neutral
2022-08-31	STB	1	Neutral
2022-08-31	SHB	0	Long
2022-08-31	LPB	1	Neutral
2022-08-31	BAB	2	Short
2022-08-31	KLB	0	Long
2022-09-30	VCB	0	Short
2022-09-30	BID	2	Neutral
2022-09-30	CTG	2	Neutral
2022-09-30	MBB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-09-30	TCB	2	Neutral
2022-09-30	ACB	2	Neutral
2022-09-30	VPB	1	Long
2022-09-30	HDB	2	Neutral
2022-09-30	VIB	1	Long
2022-09-30	TPB	2	Neutral
2022-09-30	STB	1	Long
2022-09-30	SHB	1	Long
2022-09-30	LPB	2	Neutral
2022-09-30	BAB	0	Short
2022-09-30	KLB	1	Long
2022-09-30	VBB	0	Short
2022-10-31	VCB	0	Short
2022-10-31	BID	2	Neutral
2022-10-31	CTG	2	Neutral
2022-10-31	MBB	2	Neutral
2022-10-31	TCB	2	Neutral
2022-10-31	ACB	2	Neutral
2022-10-31	VPB	1	Long
2022-10-31	HDB	2	Neutral
2022-10-31	VIB	1	Long
2022-10-31	TPB	2	Neutral
2022-10-31	STB	1	Long
2022-10-31	SHB	1	Long
2022-10-31	LPB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-10-31	BAB	0	Short
2022-10-31	KLB	1	Long
2022-10-31	VBB	0	Short
2022-11-30	VCB	2	Short
2022-11-30	BID	0	Neutral
2022-11-30	CTG	0	Neutral
2022-11-30	MBB	0	Neutral
2022-11-30	TCB	0	Neutral
2022-11-30	ACB	0	Neutral
2022-11-30	VPB	1	Long
2022-11-30	HDB	0	Neutral
2022-11-30	VIB	1	Long
2022-11-30	TPB	0	Neutral
2022-11-30	STB	1	Long
2022-11-30	SHB	1	Long
2022-11-30	LPB	1	Long
2022-11-30	BAB	2	Short
2022-11-30	KLB	1	Long
2022-11-30	VBB	2	Short
2022-12-31	VCB	2	Short
2022-12-31	BID	0	Neutral
2022-12-31	CTG	0	Neutral
2022-12-31	MBB	0	Neutral
2022-12-31	TCB	0	Neutral
2022-12-31	ACB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2022-12-31	VPB	1	Long
2022-12-31	HDB	0	Neutral
2022-12-31	VIB	1	Long
2022-12-31	TPB	0	Neutral
2022-12-31	STB	1	Long
2022-12-31	SHB	1	Long
2022-12-31	LPB	1	Long
2022-12-31	BAB	2	Short
2022-12-31	KLB	1	Long
2022-12-31	VBB	2	Short
2023-01-31	VCB	0	Short
2023-01-31	BID	1	Neutral
2023-01-31	CTG	1	Neutral
2023-01-31	MBB	1	Neutral
2023-01-31	TCB	1	Neutral
2023-01-31	ACB	1	Neutral
2023-01-31	VPB	2	Long
2023-01-31	HDB	1	Neutral
2023-01-31	VIB	2	Long
2023-01-31	TPB	1	Neutral
2023-01-31	STB	2	Long
2023-01-31	SHB	2	Long
2023-01-31	LPB	2	Long
2023-01-31	BAB	0	Short
2023-01-31	KLB	2	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-01-31	VBB	0	Short
2023-02-28	VCB	1	Short
2023-02-28	BID	0	Neutral
2023-02-28	CTG	0	Neutral
2023-02-28	MBB	0	Neutral
2023-02-28	TCB	0	Neutral
2023-02-28	ACB	0	Neutral
2023-02-28	VPB	2	Long
2023-02-28	HDB	0	Neutral
2023-02-28	VIB	2	Long
2023-02-28	TPB	0	Neutral
2023-02-28	STB	2	Long
2023-02-28	SHB	2	Long
2023-02-28	LPB	2	Long
2023-02-28	BAB	1	Short
2023-02-28	KLB	0	Neutral
2023-02-28	VBB	1	Short
2023-03-31	VCB	0	Neutral
2023-03-31	BID	2	Short
2023-03-31	CTG	2	Short
2023-03-31	MBB	2	Short
2023-03-31	TCB	2	Short
2023-03-31	ACB	0	Neutral
2023-03-31	VPB	1	Long
2023-03-31	HDB	2	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-03-31	VIB	1	Long
2023-03-31	TPB	2	Short
2023-03-31	STB	1	Long
2023-03-31	SHB	1	Long
2023-03-31	LPB	1	Long
2023-03-31	BAB	0	Neutral
2023-03-31	KLB	2	Short
2023-03-31	VBB	2	Short
2023-04-30	VCB	2	Neutral
2023-04-30	BID	0	Short
2023-04-30	CTG	0	Short
2023-04-30	MBB	0	Short
2023-04-30	TCB	0	Short
2023-04-30	ACB	2	Neutral
2023-04-30	VPB	0	Short
2023-04-30	HDB	0	Short
2023-04-30	VIB	1	Long
2023-04-30	TPB	0	Short
2023-04-30	STB	1	Long
2023-04-30	SHB	0	Short
2023-04-30	LPB	1	Long
2023-04-30	BAB	2	Neutral
2023-04-30	KLB	0	Short
2023-04-30	VBB	0	Short
2023-05-31	VCB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-05-31	BID	0	Neutral
2023-05-31	CTG	0	Neutral
2023-05-31	MBB	0	Neutral
2023-05-31	TCB	0	Neutral
2023-05-31	ACB	0	Neutral
2023-05-31	VPB	1	Long
2023-05-31	HDB	0	Neutral
2023-05-31	VIB	1	Long
2023-05-31	TPB	0	Neutral
2023-05-31	STB	1	Long
2023-05-31	SHB	2	Short
2023-05-31	LPB	1	Long
2023-05-31	BAB	0	Neutral
2023-05-31	KLB	2	Short
2023-05-31	VBB	2	Short
2023-06-30	VCB	0	Neutral
2023-06-30	BID	0	Neutral
2023-06-30	CTG	0	Neutral
2023-06-30	MBB	0	Neutral
2023-06-30	TCB	2	Long
2023-06-30	ACB	0	Neutral
2023-06-30	VPB	2	Long
2023-06-30	HDB	0	Neutral
2023-06-30	VIB	2	Long
2023-06-30	TPB	2	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-06-30	STB	2	Long
2023-06-30	SHB	1	Short
2023-06-30	LPB	2	Long
2023-06-30	BAB	0	Neutral
2023-06-30	KLB	1	Short
2023-06-30	VBB	1	Short
2023-07-31	VCB	0	Neutral
2023-07-31	BID	0	Neutral
2023-07-31	CTG	0	Neutral
2023-07-31	MBB	0	Neutral
2023-07-31	TCB	0	Neutral
2023-07-31	ACB	0	Neutral
2023-07-31	VPB	2	Long
2023-07-31	HDB	0	Neutral
2023-07-31	VIB	2	Long
2023-07-31	TPB	2	Long
2023-07-31	STB	2	Long
2023-07-31	SHB	1	Short
2023-07-31	LPB	2	Long
2023-07-31	BAB	0	Neutral
2023-07-31	KLB	1	Short
2023-07-31	VBB	1	Short
2023-08-31	VCB	0	Neutral
2023-08-31	BID	0	Neutral
2023-08-31	CTG	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-08-31	MBB	0	Neutral
2023-08-31	TCB	0	Neutral
2023-08-31	ACB	0	Neutral
2023-08-31	VPB	1	Long
2023-08-31	HDB	0	Neutral
2023-08-31	VIB	1	Long
2023-08-31	TPB	1	Long
2023-08-31	STB	1	Long
2023-08-31	SHB	2	Short
2023-08-31	LPB	1	Long
2023-08-31	BAB	0	Neutral
2023-08-31	KLB	2	Short
2023-08-31	VBB	2	Short
2023-09-30	VCB	0	Neutral
2023-09-30	BID	0	Neutral
2023-09-30	CTG	0	Neutral
2023-09-30	MBB	0	Neutral
2023-09-30	TCB	0	Neutral
2023-09-30	ACB	0	Neutral
2023-09-30	VPB	1	Long
2023-09-30	HDB	0	Neutral
2023-09-30	VIB	1	Long
2023-09-30	TPB	1	Long
2023-09-30	STB	1	Long
2023-09-30	SHB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-09-30	LPB	1	Long
2023-09-30	BAB	0	Neutral
2023-09-30	KLB	2	Short
2023-09-30	VBB	2	Short
2023-09-30	BVB	2	Short
2023-10-31	VCB	0	Neutral
2023-10-31	BID	0	Neutral
2023-10-31	CTG	0	Neutral
2023-10-31	MBB	0	Neutral
2023-10-31	TCB	0	Neutral
2023-10-31	ACB	0	Neutral
2023-10-31	VPB	2	Long
2023-10-31	HDB	0	Neutral
2023-10-31	VIB	2	Long
2023-10-31	TPB	2	Long
2023-10-31	STB	2	Long
2023-10-31	SHB	1	Short
2023-10-31	LPB	2	Long
2023-10-31	BAB	0	Neutral
2023-10-31	KLB	1	Short
2023-10-31	VBB	1	Short
2023-10-31	BVB	1	Short
2023-11-30	VCB	0	Neutral
2023-11-30	BID	0	Neutral
2023-11-30	CTG	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-11-30	MBB	1	Long
2023-11-30	TCB	0	Neutral
2023-11-30	ACB	0	Neutral
2023-11-30	VPB	1	Long
2023-11-30	HDB	0	Neutral
2023-11-30	VIB	1	Long
2023-11-30	TPB	1	Long
2023-11-30	STB	1	Long
2023-11-30	SHB	2	Short
2023-11-30	LPB	1	Long
2023-11-30	BAB	0	Neutral
2023-11-30	KLB	2	Short
2023-11-30	VBB	2	Short
2023-11-30	BVB	2	Short
2023-12-31	VCB	0	Neutral
2023-12-31	BID	0	Neutral
2023-12-31	CTG	0	Neutral
2023-12-31	MBB	0	Neutral
2023-12-31	TCB	1	Long
2023-12-31	ACB	0	Neutral
2023-12-31	VPB	1	Long
2023-12-31	HDB	0	Neutral
2023-12-31	VIB	1	Long
2023-12-31	TPB	1	Long
2023-12-31	STB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2023-12-31	SHB	2	Short
2023-12-31	LPB	1	Long
2023-12-31	NAB	1	Long
2023-12-31	BAB	0	Neutral
2023-12-31	KLB	2	Short
2023-12-31	VBB	2	Short
2023-12-31	SGB	2	Short
2023-12-31	BVB	2	Short
2024-01-31	VCB	0	Neutral
2024-01-31	BID	0	Neutral
2024-01-31	CTG	0	Neutral
2024-01-31	MBB	0	Neutral
2024-01-31	TCB	1	Short
2024-01-31	ACB	0	Neutral
2024-01-31	VPB	2	Long
2024-01-31	HDB	0	Neutral
2024-01-31	VIB	2	Long
2024-01-31	TPB	2	Long
2024-01-31	STB	2	Long
2024-01-31	SHB	1	Short
2024-01-31	LPB	2	Long
2024-01-31	NAB	2	Long
2024-01-31	BAB	0	Neutral
2024-01-31	KLB	1	Short
2024-01-31	VBB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-01-31	SGB	1	Short
2024-01-31	BVB	1	Short
2024-02-29	VCB	2	Neutral
2024-02-29	BID	2	Neutral
2024-02-29	CTG	2	Neutral
2024-02-29	MBB	2	Neutral
2024-02-29	TCB	0	Short
2024-02-29	ACB	2	Neutral
2024-02-29	VPB	2	Neutral
2024-02-29	HDB	2	Neutral
2024-02-29	VIB	1	Long
2024-02-29	TPB	2	Neutral
2024-02-29	STB	1	Long
2024-02-29	SHB	0	Short
2024-02-29	LPB	1	Long
2024-02-29	MSB	2	Neutral
2024-02-29	NAB	1	Long
2024-02-29	BAB	0	Short
2024-02-29	KLB	0	Short
2024-02-29	PGB	1	Long
2024-02-29	VBB	0	Short
2024-02-29	SGB	0	Short
2024-02-29	ABB	0	Short
2024-02-29	BVB	0	Short
2024-03-31	VCB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-03-31	BID	2	Neutral
2024-03-31	CTG	2	Neutral
2024-03-31	MBB	2	Neutral
2024-03-31	TCB	1	Short
2024-03-31	ACB	2	Neutral
2024-03-31	VPB	2	Neutral
2024-03-31	HDB	2	Neutral
2024-03-31	VIB	0	Long
2024-03-31	TPB	2	Neutral
2024-03-31	STB	0	Long
2024-03-31	SHB	0	Long
2024-03-31	LPB	0	Long
2024-03-31	MSB	2	Neutral
2024-03-31	OCB	2	Neutral
2024-03-31	NAB	0	Long
2024-03-31	BAB	1	Short
2024-03-31	KLB	1	Short
2024-03-31	PGB	0	Long
2024-03-31	VBB	1	Short
2024-03-31	SGB	1	Short
2024-03-31	ABB	1	Short
2024-03-31	BVB	1	Short
2024-04-30	VCB	2	Neutral
2024-04-30	BID	2	Neutral
2024-04-30	CTG	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-04-30	MBB	2	Neutral
2024-04-30	TCB	1	Short
2024-04-30	ACB	2	Neutral
2024-04-30	VPB	2	Neutral
2024-04-30	HDB	2	Neutral
2024-04-30	VIB	2	Neutral
2024-04-30	TPB	2	Neutral
2024-04-30	STB	0	Long
2024-04-30	SHB	0	Long
2024-04-30	LPB	0	Long
2024-04-30	MSB	2	Neutral
2024-04-30	OCB	2	Neutral
2024-04-30	NAB	0	Long
2024-04-30	BAB	1	Short
2024-04-30	KLB	1	Short
2024-04-30	PGB	0	Long
2024-04-30	VBB	1	Short
2024-04-30	SGB	1	Short
2024-04-30	ABB	1	Short
2024-04-30	BVB	1	Short
2024-05-31	VCB	0	Neutral
2024-05-31	BID	0	Neutral
2024-05-31	CTG	0	Neutral
2024-05-31	MBB	0	Neutral
2024-05-31	TCB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-05-31	ACB	0	Neutral
2024-05-31	VPB	0	Neutral
2024-05-31	HDB	0	Neutral
2024-05-31	VIB	0	Neutral
2024-05-31	TPB	0	Neutral
2024-05-31	STB	2	Long
2024-05-31	SHB	1	Short
2024-05-31	LPB	2	Long
2024-05-31	SSB	0	Neutral
2024-05-31	MSB	0	Neutral
2024-05-31	OCB	0	Neutral
2024-05-31	NAB	2	Long
2024-05-31	BAB	1	Short
2024-05-31	KLB	1	Short
2024-05-31	PGB	2	Long
2024-05-31	VBB	1	Short
2024-05-31	SGB	1	Short
2024-05-31	ABB	1	Short
2024-05-31	BVB	1	Short
2024-06-30	VCB	0	Long
2024-06-30	BID	0	Long
2024-06-30	CTG	0	Long
2024-06-30	MBB	0	Long
2024-06-30	TCB	0	Long
2024-06-30	ACB	0	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-06-30	VPB	1	Short
2024-06-30	HDB	0	Long
2024-06-30	VIB	2	Neutral
2024-06-30	TPB	0	Long
2024-06-30	STB	2	Neutral
2024-06-30	SHB	1	Short
2024-06-30	LPB	0	Long
2024-06-30	SSB	0	Long
2024-06-30	MSB	0	Long
2024-06-30	OCB	0	Long
2024-06-30	NAB	0	Long
2024-06-30	BAB	1	Short
2024-06-30	KLB	1	Short
2024-06-30	PGB	2	Neutral
2024-06-30	VBB	2	Neutral
2024-06-30	SGB	2	Neutral
2024-06-30	ABB	1	Short
2024-06-30	BVB	2	Neutral
2024-07-31	VCB	2	Long
2024-07-31	BID	2	Long
2024-07-31	CTG	2	Long
2024-07-31	MBB	2	Long
2024-07-31	TCB	2	Long
2024-07-31	ACB	2	Long
2024-07-31	VPB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-07-31	HDB	2	Long
2024-07-31	VIB	0	Neutral
2024-07-31	TPB	2	Long
2024-07-31	STB	0	Neutral
2024-07-31	SHB	1	Short
2024-07-31	LPB	2	Long
2024-07-31	SSB	2	Long
2024-07-31	MSB	2	Long
2024-07-31	OCB	2	Long
2024-07-31	NAB	2	Long
2024-07-31	BAB	1	Short
2024-07-31	KLB	0	Neutral
2024-07-31	PGB	0	Neutral
2024-07-31	VBB	0	Neutral
2024-07-31	SGB	1	Short
2024-07-31	ABB	1	Short
2024-07-31	BVB	1	Short
2024-08-31	VCB	0	Neutral
2024-08-31	BID	1	Long
2024-08-31	CTG	0	Neutral
2024-08-31	MBB	0	Neutral
2024-08-31	TCB	1	Long
2024-08-31	ACB	0	Neutral
2024-08-31	VPB	0	Neutral
2024-08-31	HDB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-08-31	VIB	2	Short
2024-08-31	TPB	1	Long
2024-08-31	STB	1	Long
2024-08-31	SHB	2	Short
2024-08-31	LPB	1	Long
2024-08-31	SSB	0	Neutral
2024-08-31	MSB	0	Neutral
2024-08-31	OCB	0	Neutral
2024-08-31	NAB	1	Long
2024-08-31	BAB	0	Neutral
2024-08-31	KLB	2	Short
2024-08-31	PGB	1	Long
2024-08-31	VBB	2	Short
2024-08-31	SGB	0	Neutral
2024-08-31	ABB	2	Short
2024-08-31	BVB	2	Short
2024-09-30	VCB	0	Long
2024-09-30	BID	0	Long
2024-09-30	CTG	0	Long
2024-09-30	MBB	0	Long
2024-09-30	TCB	2	Neutral
2024-09-30	ACB	0	Long
2024-09-30	VPB	1	Short
2024-09-30	HDB	0	Long
2024-09-30	VIB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-09-30	TPB	0	Long
2024-09-30	STB	2	Neutral
2024-09-30	SHB	1	Short
2024-09-30	LPB	0	Long
2024-09-30	SSB	1	Short
2024-09-30	MSB	1	Short
2024-09-30	OCB	1	Short
2024-09-30	NAB	0	Long
2024-09-30	BAB	1	Short
2024-09-30	KLB	2	Neutral
2024-09-30	PGB	2	Neutral
2024-09-30	VAB	1	Short
2024-09-30	VBB	2	Neutral
2024-09-30	SGB	1	Short
2024-09-30	ABB	1	Short
2024-09-30	BVB	1	Short
2024-10-31	VCB	1	Long
2024-10-31	BID	1	Long
2024-10-31	CTG	1	Long
2024-10-31	MBB	1	Long
2024-10-31	TCB	2	Neutral
2024-10-31	ACB	1	Long
2024-10-31	VPB	0	Short
2024-10-31	HDB	1	Long
2024-10-31	VIB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-10-31	TPB	2	Neutral
2024-10-31	STB	2	Neutral
2024-10-31	SHB	0	Short
2024-10-31	LPB	1	Long
2024-10-31	SSB	0	Short
2024-10-31	MSB	0	Short
2024-10-31	OCB	0	Short
2024-10-31	NAB	1	Long
2024-10-31	BAB	0	Short
2024-10-31	KLB	2	Neutral
2024-10-31	PGB	2	Neutral
2024-10-31	VAB	0	Short
2024-10-31	VBB	2	Neutral
2024-10-31	SGB	0	Short
2024-10-31	ABB	0	Short
2024-10-31	BVB	0	Short
2024-11-30	VCB	1	Long
2024-11-30	BID	1	Long
2024-11-30	CTG	1	Long
2024-11-30	MBB	1	Long
2024-11-30	TCB	0	Neutral
2024-11-30	ACB	1	Long
2024-11-30	VPB	2	Short
2024-11-30	HDB	1	Long
2024-11-30	VIB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-11-30	TPB	0	Neutral
2024-11-30	STB	0	Neutral
2024-11-30	SHB	2	Short
2024-11-30	LPB	1	Long
2024-11-30	SSB	2	Short
2024-11-30	MSB	2	Short
2024-11-30	OCB	2	Short
2024-11-30	NAB	1	Long
2024-11-30	BAB	2	Short
2024-11-30	KLB	0	Neutral
2024-11-30	PGB	0	Neutral
2024-11-30	VAB	2	Short
2024-11-30	VBB	0	Neutral
2024-11-30	SGB	2	Short
2024-11-30	ABB	2	Short
2024-11-30	BVB	2	Short
2024-12-31	VCB	0	Long
2024-12-31	BID	0	Long
2024-12-31	CTG	0	Long
2024-12-31	MBB	0	Long
2024-12-31	TCB	0	Long
2024-12-31	ACB	0	Long
2024-12-31	VPB	1	Short
2024-12-31	HDB	0	Long
2024-12-31	VIB	0	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2024-12-31	TPB	0	Long
2024-12-31	STB	0	Long
2024-12-31	SHB	1	Short
2024-12-31	LPB	0	Long
2024-12-31	SSB	1	Short
2024-12-31	MSB	1	Short
2024-12-31	OCB	1	Short
2024-12-31	NAB	0	Long
2024-12-31	BAB	1	Short
2024-12-31	KLB	2	Neutral
2024-12-31	PGB	2	Neutral
2024-12-31	VAB	1	Short
2024-12-31	VBB	2	Neutral
2024-12-31	SGB	1	Short
2024-12-31	ABB	1	Short
2024-12-31	BVB	1	Short
2025-01-31	VCB	0	Long
2025-01-31	BID	0	Long
2025-01-31	CTG	0	Long
2025-01-31	MBB	0	Long
2025-01-31	TCB	0	Long
2025-01-31	ACB	0	Long
2025-01-31	VPB	1	Short
2025-01-31	HDB	0	Long
2025-01-31	VIB	0	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-01-31	TPB	0	Long
2025-01-31	STB	0	Long
2025-01-31	SHB	1	Short
2025-01-31	LPB	0	Long
2025-01-31	SSB	1	Short
2025-01-31	MSB	1	Short
2025-01-31	OCB	1	Short
2025-01-31	NAB	0	Long
2025-01-31	BAB	1	Short
2025-01-31	KLB	2	Neutral
2025-01-31	PGB	2	Neutral
2025-01-31	VAB	1	Short
2025-01-31	VBB	2	Neutral
2025-01-31	SGB	1	Short
2025-01-31	ABB	1	Short
2025-01-31	BVB	1	Short
2025-02-28	VCB	2	Neutral
2025-02-28	BID	1	Long
2025-02-28	CTG	1	Long
2025-02-28	MBB	1	Long
2025-02-28	TCB	1	Long
2025-02-28	ACB	2	Neutral
2025-02-28	VPB	0	Short
2025-02-28	HDB	1	Long
2025-02-28	VIB	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-02-28	TPB	0	Short
2025-02-28	STB	1	Long
2025-02-28	SHB	0	Short
2025-02-28	LPB	1	Long
2025-02-28	SSB	2	Neutral
2025-02-28	MSB	0	Short
2025-02-28	OCB	0	Short
2025-02-28	NAB	1	Long
2025-02-28	BAB	2	Neutral
2025-02-28	KLB	0	Short
2025-02-28	PGB	0	Short
2025-02-28	VAB	0	Short
2025-02-28	VBB	0	Short
2025-02-28	SGB	2	Neutral
2025-02-28	ABB	0	Short
2025-02-28	BVB	0	Short
2025-03-31	VCB	2	Neutral
2025-03-31	BID	1	Long
2025-03-31	CTG	1	Long
2025-03-31	MBB	1	Long
2025-03-31	TCB	1	Long
2025-03-31	ACB	2	Neutral
2025-03-31	VPB	0	Short
2025-03-31	HDB	1	Long
2025-03-31	VIB	1	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-03-31	TPB	0	Short
2025-03-31	STB	1	Long
2025-03-31	SHB	0	Short
2025-03-31	LPB	1	Long
2025-03-31	SSB	2	Neutral
2025-03-31	MSB	0	Short
2025-03-31	OCB	0	Short
2025-03-31	NAB	1	Long
2025-03-31	BAB	2	Neutral
2025-03-31	KLB	0	Short
2025-03-31	PGB	0	Short
2025-03-31	VAB	0	Short
2025-03-31	VBB	0	Short
2025-03-31	SGB	2	Neutral
2025-03-31	ABB	0	Short
2025-03-31	BVB	0	Short
2025-04-30	VCB	1	Neutral
2025-04-30	BID	2	Long
2025-04-30	CTG	2	Long
2025-04-30	MBB	2	Long
2025-04-30	TCB	2	Long
2025-04-30	ACB	1	Neutral
2025-04-30	VPB	0	Short
2025-04-30	HDB	2	Long
2025-04-30	VIB	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-04-30	TPB	0	Short
2025-04-30	STB	2	Long
2025-04-30	SHB	0	Short
2025-04-30	LPB	2	Long
2025-04-30	SSB	1	Neutral
2025-04-30	MSB	0	Short
2025-04-30	OCB	0	Short
2025-04-30	NAB	2	Long
2025-04-30	BAB	1	Neutral
2025-04-30	KLB	0	Short
2025-04-30	PGB	0	Short
2025-04-30	VAB	0	Short
2025-04-30	VBB	0	Short
2025-04-30	SGB	1	Neutral
2025-04-30	ABB	0	Short
2025-04-30	BVB	0	Short
2025-05-31	VCB	2	Neutral
2025-05-31	BID	1	Long
2025-05-31	CTG	1	Long
2025-05-31	MBB	1	Long
2025-05-31	TCB	1	Long
2025-05-31	ACB	2	Neutral
2025-05-31	VPB	0	Short
2025-05-31	HDB	1	Long
2025-05-31	VIB	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-05-31	TPB	0	Short
2025-05-31	STB	1	Long
2025-05-31	SHB	0	Short
2025-05-31	LPB	1	Long
2025-05-31	SSB	2	Neutral
2025-05-31	MSB	0	Short
2025-05-31	OCB	0	Short
2025-05-31	NAB	1	Long
2025-05-31	BAB	2	Neutral
2025-05-31	KLB	0	Short
2025-05-31	PGB	0	Short
2025-05-31	VAB	0	Short
2025-05-31	VBB	0	Short
2025-05-31	SGB	2	Neutral
2025-05-31	ABB	0	Short
2025-05-31	BVB	0	Short
2025-06-30	VCB	1	Neutral
2025-06-30	BID	2	Long
2025-06-30	CTG	2	Long
2025-06-30	MBB	2	Long
2025-06-30	TCB	2	Long
2025-06-30	ACB	1	Neutral
2025-06-30	VPB	0	Short
2025-06-30	HDB	2	Long
2025-06-30	VIB	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-06-30	TPB	0	Short
2025-06-30	STB	2	Long
2025-06-30	SHB	2	Long
2025-06-30	LPB	2	Long
2025-06-30	SSB	1	Neutral
2025-06-30	MSB	0	Short
2025-06-30	OCB	0	Short
2025-06-30	NAB	2	Long
2025-06-30	BAB	1	Neutral
2025-06-30	KLB	0	Short
2025-06-30	PGB	0	Short
2025-06-30	VAB	0	Short
2025-06-30	VBB	0	Short
2025-06-30	SGB	1	Neutral
2025-06-30	ABB	0	Short
2025-06-30	BVB	0	Short
2025-07-31	VCB	1	Short
2025-07-31	BID	0	Long
2025-07-31	CTG	0	Long
2025-07-31	MBB	0	Long
2025-07-31	TCB	0	Long
2025-07-31	ACB	1	Short
2025-07-31	VPB	1	Short
2025-07-31	HDB	0	Long
2025-07-31	VIB	0	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-07-31	TPB	1	Short
2025-07-31	STB	0	Long
2025-07-31	SHB	0	Long
2025-07-31	LPB	0	Long
2025-07-31	SSB	1	Short
2025-07-31	MSB	1	Short
2025-07-31	OCB	1	Short
2025-07-31	NAB	0	Long
2025-07-31	BAB	1	Short
2025-07-31	KLB	2	Neutral
2025-07-31	PGB	2	Neutral
2025-07-31	VAB	0	Long
2025-07-31	VBB	2	Neutral
2025-07-31	SGB	1	Short
2025-07-31	ABB	1	Short
2025-07-31	BVB	1	Short
2025-08-31	VCB	0	Short
2025-08-31	BID	2	Long
2025-08-31	CTG	2	Long
2025-08-31	MBB	2	Long
2025-08-31	TCB	2	Long
2025-08-31	ACB	0	Short
2025-08-31	VPB	0	Short
2025-08-31	HDB	2	Long
2025-08-31	VIB	2	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-08-31	TPB	0	Short
2025-08-31	STB	2	Long
2025-08-31	SHB	2	Long
2025-08-31	LPB	2	Long
2025-08-31	SSB	0	Short
2025-08-31	MSB	0	Short
2025-08-31	OCB	0	Short
2025-08-31	NAB	2	Long
2025-08-31	BAB	0	Short
2025-08-31	KLB	1	Neutral
2025-08-31	PGB	1	Neutral
2025-08-31	VAB	2	Long
2025-08-31	VBB	1	Neutral
2025-08-31	SGB	0	Short
2025-08-31	ABB	0	Short
2025-08-31	BVB	0	Short
2025-09-30	VCB	1	Neutral
2025-09-30	BID	1	Neutral
2025-09-30	CTG	0	Long
2025-09-30	MBB	0	Long
2025-09-30	TCB	0	Long
2025-09-30	ACB	1	Neutral
2025-09-30	VPB	0	Long
2025-09-30	HDB	0	Long
2025-09-30	VIB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-09-30	TPB	1	Neutral
2025-09-30	STB	0	Long
2025-09-30	SHB	0	Long
2025-09-30	LPB	0	Long
2025-09-30	SSB	1	Neutral
2025-09-30	MSB	1	Neutral
2025-09-30	OCB	1	Neutral
2025-09-30	NAB	0	Long
2025-09-30	BAB	1	Neutral
2025-09-30	KLB	2	Short
2025-09-30	PGB	2	Short
2025-09-30	VAB	0	Long
2025-09-30	VBB	0	Long
2025-09-30	SGB	1	Neutral
2025-09-30	ABB	2	Short
2025-09-30	BVB	1	Neutral
2025-10-31	VCB	1	Neutral
2025-10-31	BID	0	Long
2025-10-31	CTG	0	Long
2025-10-31	MBB	0	Long
2025-10-31	TCB	0	Long
2025-10-31	ACB	1	Neutral
2025-10-31	VPB	0	Long
2025-10-31	HDB	0	Long
2025-10-31	VIB	0	Long

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-10-31	TPB	0	Long
2025-10-31	STB	0	Long
2025-10-31	SHB	0	Long
2025-10-31	LPB	0	Long
2025-10-31	SSB	1	Neutral
2025-10-31	MSB	0	Long
2025-10-31	OCB	0	Long
2025-10-31	NAB	0	Long
2025-10-31	BAB	1	Neutral
2025-10-31	KLB	2	Short
2025-10-31	PGB	2	Short
2025-10-31	VAB	0	Long
2025-10-31	VBB	0	Long
2025-10-31	SGB	1	Neutral
2025-10-31	ABB	2	Short
2025-10-31	BVB	0	Long
2025-11-30	VCB	1	Short
2025-11-30	BID	1	Short
2025-11-30	CTG	2	Long
2025-11-30	MBB	2	Long
2025-11-30	TCB	2	Long
2025-11-30	ACB	1	Short
2025-11-30	VPB	0	Neutral
2025-11-30	HDB	2	Long
2025-11-30	VIB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-11-30	TPB	0	Neutral
2025-11-30	STB	2	Long
2025-11-30	SHB	2	Long
2025-11-30	LPB	2	Long
2025-11-30	SSB	1	Short
2025-11-30	MSB	0	Neutral
2025-11-30	OCB	0	Neutral
2025-11-30	NAB	2	Long
2025-11-30	BAB	1	Short
2025-11-30	KLB	0	Neutral
2025-11-30	PGB	0	Neutral
2025-11-30	VAB	0	Neutral
2025-11-30	VBB	0	Neutral
2025-11-30	SGB	1	Short
2025-11-30	ABB	0	Neutral
2025-11-30	BVB	0	Neutral
2025-12-31	VCB	0	Short
2025-12-31	BID	0	Short
2025-12-31	CTG	0	Short
2025-12-31	MBB	2	Long
2025-12-31	TCB	2	Long
2025-12-31	ACB	0	Short
2025-12-31	VPB	1	Neutral
2025-12-31	HDB	2	Long
2025-12-31	VIB	1	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2025-12-31	TPB	1	Neutral
2025-12-31	STB	2	Long
2025-12-31	SHB	2	Long
2025-12-31	LPB	2	Long
2025-12-31	SSB	0	Short
2025-12-31	MSB	1	Neutral
2025-12-31	OCB	0	Short
2025-12-31	NAB	2	Long
2025-12-31	BAB	0	Short
2025-12-31	KLB	1	Neutral
2025-12-31	PGB	1	Neutral
2025-12-31	VAB	1	Neutral
2025-12-31	VBB	1	Neutral
2025-12-31	SGB	0	Short
2025-12-31	ABB	1	Neutral
2025-12-31	BVB	0	Short
2026-01-31	VCB	1	Short
2026-01-31	BID	1	Short
2026-01-31	CTG	1	Short
2026-01-31	MBB	2	Long
2026-01-31	TCB	2	Long
2026-01-31	ACB	1	Short
2026-01-31	VPB	0	Neutral
2026-01-31	HDB	2	Long
2026-01-31	VIB	0	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-01-31	TPB	0	Neutral
2026-01-31	STB	2	Long
2026-01-31	SHB	2	Long
2026-01-31	LPB	2	Long
2026-01-31	SSB	1	Short
2026-01-31	MSB	0	Neutral
2026-01-31	OCB	1	Short
2026-01-31	NAB	2	Long
2026-01-31	BAB	1	Short
2026-01-31	KLB	0	Neutral
2026-01-31	PGB	0	Neutral
2026-01-31	VAB	2	Long
2026-01-31	VBB	0	Neutral
2026-01-31	SGB	1	Short
2026-01-31	ABB	0	Neutral
2026-01-31	BVB	1	Short
2026-02-28	VCB	1	Short
2026-02-28	BID	1	Short
2026-02-28	CTG	1	Short
2026-02-28	MBB	0	Long
2026-02-28	TCB	0	Long
2026-02-28	ACB	1	Short
2026-02-28	VPB	2	Neutral
2026-02-28	HDB	0	Long
2026-02-28	VIB	2	Neutral

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-02-28	TPB	2	Neutral
2026-02-28	STB	0	Long
2026-02-28	SHB	0	Long
2026-02-28	LPB	0	Long
2026-02-28	SSB	1	Short
2026-02-28	MSB	2	Neutral
2026-02-28	OCB	1	Short
2026-02-28	NAB	0	Long
2026-02-28	BAB	1	Short
2026-02-28	KLB	2	Neutral
2026-02-28	PGB	2	Neutral
2026-02-28	VAB	2	Neutral
2026-02-28	VBB	2	Neutral
2026-02-28	SGB	1	Short
2026-02-28	ABB	2	Neutral
2026-02-28	BVB	1	Short
2026-03-31	VCB	1	Short
2026-03-31	BID	0	Neutral
2026-03-31	CTG	2	Long
2026-03-31	MBB	2	Long
2026-03-31	TCB	2	Long
2026-03-31	ACB	1	Short
2026-03-31	VPB	0	Neutral
2026-03-31	HDB	2	Long
2026-03-31	VIB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-03-31	TPB	0	Neutral
2026-03-31	STB	2	Long
2026-03-31	SHB	2	Long
2026-03-31	LPB	2	Long
2026-03-31	SSB	1	Short
2026-03-31	MSB	0	Neutral
2026-03-31	OCB	1	Short
2026-03-31	NAB	2	Long
2026-03-31	BAB	1	Short
2026-03-31	KLB	0	Neutral
2026-03-31	PGB	0	Neutral
2026-03-31	VAB	2	Long
2026-03-31	VBB	0	Neutral
2026-03-31	SGB	1	Short
2026-03-31	ABB	0	Neutral
2026-03-31	BVB	1	Short
2026-04-30	VCB	0	Short
2026-04-30	BID	2	Neutral
2026-04-30	CTG	1	Long
2026-04-30	MBB	1	Long
2026-04-30	TCB	1	Long
2026-04-30	ACB	0	Short
2026-04-30	VPB	2	Neutral
2026-04-30	HDB	1	Long
2026-04-30	VIB	0	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-04-30	TPB	2	Neutral
2026-04-30	STB	1	Long
2026-04-30	SHB	1	Long
2026-04-30	LPB	1	Long
2026-04-30	SSB	0	Short
2026-04-30	MSB	2	Neutral
2026-04-30	OCB	0	Short
2026-04-30	NAB	1	Long
2026-04-30	BAB	0	Short
2026-04-30	KLB	2	Neutral
2026-04-30	PGB	2	Neutral
2026-04-30	VAB	1	Long
2026-04-30	VBB	0	Short
2026-04-30	SGB	0	Short
2026-04-30	ABB	2	Neutral
2026-04-30	BVB	0	Short
2026-05-31	VCB	1	Short
2026-05-31	BID	0	Neutral
2026-05-31	CTG	1	Short
2026-05-31	MBB	2	Long
2026-05-31	TCB	2	Long
2026-05-31	ACB	1	Short
2026-05-31	VPB	0	Neutral
2026-05-31	HDB	2	Long
2026-05-31	VIB	1	Short

Date	Ticker	Cluster_Label	Assigned_View_Position
2026-05-31	TPB	0	Neutral
2026-05-31	STB	2	Long
2026-05-31	SHB	2	Long
2026-05-31	LPB	2	Long
2026-05-31	SSB	1	Short
2026-05-31	MSB	0	Neutral
2026-05-31	OCB	1	Short
2026-05-31	NAB	2	Long
2026-05-31	BAB	1	Short
2026-05-31	KLB	0	Neutral
2026-05-31	PGB	0	Neutral
2026-05-31	VAB	2	Long
2026-05-31	VBB	1	Short
2026-05-31	SGB	1	Short
2026-05-31	ABB	2	Long
2026-05-31	BVB	1	Short

9. Lịch sử Trọng Danh mục

Trọng của 6 danh mục cho từng cổ phiếu tại mỗi kỳ tái cân bằng. w_{mkt} = vốn hóa thị trường; w_{TAN} = Tangency; w_{MV} = Min-Variance; w_{BL} = Black-Litterman; w_{EW} = Equal-Weight; w_{RP} = Risk-Parity.

Tổng số quan sát: 1,561 dòng

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2017-08-31	VCB	0.373349	0.300000	0.228374	0.300000	0.142857	0.142857
2017-08-31	BID	0.235950	0.078146	0.000000	0.300000	0.142857	0.142857
2017-08-31	CTG	0.138339	0.000000	0.000000	0.125668	0.142857	0.142857
2017-08-31	MBB	0.085414	0.300000	0.300000	0.000000	0.142857	0.142857
2017-08-31	ACB	0.067046	0.300000	0.180592	0.274332	0.142857	0.142857
2017-08-31	STB	0.070691	0.000000	0.000000	0.000000	0.142857	0.142857
2017-08-31	SHB	0.029211	0.021854	0.291034	0.000000	0.142857	0.142857
2017-09-30	VCB	0.376914	0.300000	0.237058	0.300000	0.142857	0.142857
2017-09-30	BID	0.233311	0.035950	0.000000	0.300000	0.142857	0.142857
2017-09-30	CTG	0.132644	0.000000	0.000000	0.144609	0.142857	0.142857
2017-09-30	MBB	0.087568	0.300000	0.300000	0.000000	0.142857	0.142857
2017-09-30	ACB	0.076128	0.300000	0.176242	0.255391	0.142857	0.142857
2017-09-30	STB	0.064861	0.000000	0.005906	0.000000	0.142857	0.142857
2017-09-30	SHB	0.028574	0.064050	0.280794	0.000000	0.142857	0.142857
2017-10-31	VCB	0.387099	0.300000	0.246125	0.300000	0.142857	0.142857
2017-10-31	BID	0.219039	0.079333	0.000000	0.300000	0.142857	0.142857
2017-10-31	CTG	0.132609	0.000000	0.000000	0.100000	0.142857	0.142857
2017-10-31	MBB	0.083441	0.300000	0.300000	0.000000	0.142857	0.142857
2017-10-31	ACB	0.080421	0.300000	0.168708	0.300000	0.142857	0.142857
2017-10-31	STB	0.068705	0.000000	0.000000	0.000000	0.142857	0.142857

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2017-10-31	SHB	0.028686	0.020667	0.285167	0.000000	0.142857	0.142857
2017-11-30	VCB	0.401989	0.300000	0.258089	0.300000	0.142857	0.142857
2017-11-30	BID	0.230218	0.034857	0.000000	0.300000	0.142857	0.142857
2017-11-30	CTG	0.126737	0.000000	0.000000	0.100000	0.142857	0.142857
2017-11-30	MBB	0.081745	0.300000	0.300000	0.000000	0.142857	0.142857
2017-11-30	ACB	0.075875	0.300000	0.160293	0.300000	0.142857	0.142857
2017-11-30	STB	0.057042	0.000000	0.000000	0.000000	0.142857	0.142857
2017-11-30	SHB	0.026393	0.065143	0.281618	0.000000	0.142857	0.142857
2017-12-31	VCB	0.397617	0.300000	0.248045	0.300000	0.142857	0.142857
2017-12-31	BID	0.228292	0.070882	0.000000	0.300000	0.142857	0.142857
2017-12-31	CTG	0.138682	0.000000	0.000000	0.100000	0.142857	0.142857
2017-12-31	MBB	0.078230	0.300000	0.300000	0.000000	0.142857	0.142857
2017-12-31	ACB	0.073008	0.300000	0.167856	0.300000	0.142857	0.142857
2017-12-31	STB	0.057355	0.000000	0.000000	0.000000	0.142857	0.142857
2017-12-31	SHB	0.026816	0.029118	0.284100	0.000000	0.142857	0.142857
2018-01-31	VCB	0.424734	0.300000	0.234821	0.300000	0.142857	0.142857
2018-01-31	BID	0.216572	0.070629	0.000000	0.300000	0.142857	0.142857
2018-01-31	CTG	0.131843	0.000000	0.000000	0.108944	0.142857	0.142857
2018-01-31	MBB	0.074099	0.300000	0.300000	0.000000	0.142857	0.142857
2018-01-31	ACB	0.073756	0.300000	0.208134	0.291056	0.142857	0.142857
2018-01-31	STB	0.053679	0.000000	0.000000	0.000000	0.142857	0.142857
2018-01-31	SHB	0.025316	0.029371	0.257046	0.000000	0.142857	0.142857
2018-02-28	VCB	0.418904	0.300000	0.232274	0.300000	0.142857	0.142857
2018-02-28	BID	0.233422	0.033103	0.000000	0.300000	0.142857	0.142857
2018-02-28	CTG	0.119111	0.000000	0.000000	0.100000	0.142857	0.142857

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2018-02-28	MBB	0.076429	0.300000	0.300000	0.000000	0.142857	0.142857
2018-02-28	ACB	0.066989	0.300000	0.209142	0.300000	0.142857	0.142857
2018-02-28	STB	0.054579	0.000000	0.000000	0.000000	0.142857	0.142857
2018-02-28	SHB	0.030568	0.066897	0.258584	0.000000	0.142857	0.142857
2018-03-31	VCB	0.414303	0.300000	0.266719	0.300000	0.142857	0.142857
2018-03-31	BID	0.237604	0.022044	0.000000	0.300000	0.142857	0.142857
2018-03-31	CTG	0.131338	0.000000	0.000000	0.194137	0.142857	0.142857
2018-03-31	MBB	0.073556	0.300000	0.300000	0.000000	0.142857	0.142857
2018-03-31	ACB	0.067390	0.300000	0.300000	0.205863	0.142857	0.142857
2018-03-31	STB	0.048267	0.000000	0.000000	0.000000	0.142857	0.142857
2018-03-31	SHB	0.027543	0.077956	0.133281	0.000000	0.142857	0.142857
2018-04-30	VCB	0.389328	0.300000	0.254140	0.300000	0.142857	0.142857
2018-04-30	BID	0.260056	0.019358	0.000000	0.300000	0.142857	0.142857
2018-04-30	CTG	0.132959	0.000000	0.000000	0.252871	0.142857	0.142857
2018-04-30	MBB	0.072920	0.300000	0.300000	0.000000	0.142857	0.142857
2018-04-30	ACB	0.072388	0.300000	0.300000	0.147129	0.142857	0.142857
2018-04-30	STB	0.045704	0.000000	0.000000	0.000000	0.142857	0.142857
2018-04-30	SHB	0.026645	0.080642	0.145860	0.000000	0.142857	0.142857
2018-05-31	VCB	0.388273	0.300000	0.261662	0.300000	0.142857	0.142857
2018-05-31	BID	0.253551	0.056793	0.000000	0.300000	0.142857	0.142857
2018-05-31	CTG	0.132002	0.000000	0.000000	0.216841	0.142857	0.142857
2018-05-31	MBB	0.073715	0.300000	0.300000	0.000000	0.142857	0.142857
2018-05-31	ACB	0.077021	0.300000	0.300000	0.183159	0.142857	0.142857
2018-05-31	STB	0.047862	0.000000	0.000000	0.000000	0.142857	0.142857
2018-05-31	SHB	0.027575	0.043207	0.138338	0.000000	0.142857	0.142857

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2018-06-30	VCB	0.395582	0.300000	0.242855	0.300000	0.142857	0.142857
2018-06-30	BID	0.232078	0.021900	0.000000	0.300000	0.142857	0.142857
2018-06-30	CTG	0.137903	0.000000	0.000000	0.212331	0.142857	0.142857
2018-06-30	MBB	0.081008	0.300000	0.300000	0.000000	0.142857	0.142857
2018-06-30	ACB	0.082125	0.300000	0.300000	0.187669	0.142857	0.142857
2018-06-30	STB	0.045875	0.000000	0.000000	0.000000	0.142857	0.142857
2018-06-30	SHB	0.025429	0.078100	0.157145	0.000000	0.142857	0.142857
2018-07-31	VCB	0.438421	0.300000	0.300000	0.300000	0.142857	0.142857
2018-07-31	BID	0.213290	0.000000	0.000000	0.300000	0.142857	0.142857
2018-07-31	CTG	0.127604	0.020642	0.000000	0.300000	0.142857	0.142857
2018-07-31	MBB	0.075376	0.300000	0.300000	0.000000	0.142857	0.142857
2018-07-31	ACB	0.075545	0.300000	0.300000	0.100000	0.142857	0.142857
2018-07-31	STB	0.046613	0.000000	0.000000	0.000000	0.142857	0.142857
2018-07-31	SHB	0.023152	0.079358	0.100000	0.000000	0.142857	0.142857
2018-08-31	VCB	0.438906	0.300000	0.300000	0.159068	0.142857	0.142857
2018-08-31	BID	0.214099	0.000000	0.000000	0.300000	0.142857	0.142857
2018-08-31	CTG	0.123236	0.047303	0.000000	0.300000	0.142857	0.142857
2018-08-31	MBB	0.079746	0.300000	0.300000	0.000000	0.142857	0.142857
2018-08-31	ACB	0.075422	0.300000	0.300000	0.240932	0.142857	0.142857
2018-08-31	STB	0.045868	0.000000	0.012860	0.000000	0.142857	0.142857
2018-08-31	SHB	0.022722	0.052697	0.087140	0.000000	0.142857	0.142857
2018-09-30	VCB	0.417540	0.300000	0.300000	0.100000	0.142857	0.142857
2018-09-30	BID	0.249364	0.000000	0.000000	0.300000	0.142857	0.142857
2018-09-30	CTG	0.126297	0.000000	0.000000	0.300000	0.142857	0.142857
2018-09-30	MBB	0.071702	0.300000	0.300000	0.000000	0.142857	0.142857

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2018-09-30	ACB	0.073380	0.300000	0.300000	0.300000	0.142857	0.142857
2018-09-30	STB	0.040486	0.000000	0.013247	0.000000	0.142857	0.142857
2018-09-30	SHB	0.021232	0.100000	0.086753	0.000000	0.142857	0.142857
2018-10-31	VCB	0.413195	0.300000	0.300000	0.268782	0.142857	0.142857
2018-10-31	BID	0.250562	0.000000	0.000000	0.300000	0.142857	0.142857
2018-10-31	CTG	0.125792	0.000000	0.000000	0.300000	0.142857	0.142857
2018-10-31	MBB	0.068582	0.300000	0.300000	0.000000	0.142857	0.142857
2018-10-31	ACB	0.072234	0.300000	0.300000	0.131218	0.142857	0.142857
2018-10-31	STB	0.047278	0.000000	0.007476	0.000000	0.142857	0.142857
2018-10-31	SHB	0.022356	0.100000	0.092524	0.000000	0.142857	0.142857
2018-11-30	VCB	0.420658	0.300000	0.300000	0.300000	0.142857	0.142857
2018-11-30	BID	0.239121	0.000000	0.000000	0.300000	0.142857	0.142857
2018-11-30	CTG	0.121656	0.000000	0.000000	0.300000	0.142857	0.142857
2018-11-30	MBB	0.074092	0.300000	0.300000	0.000000	0.142857	0.142857
2018-11-30	ACB	0.073125	0.300000	0.300000	0.100000	0.142857	0.142857
2018-11-30	STB	0.049847	0.000000	0.013261	0.000000	0.142857	0.142857
2018-11-30	SHB	0.021501	0.100000	0.086739	0.000000	0.142857	0.142857
2018-12-31	VCB	0.415930	0.300000	0.300000	0.300000	0.142857	0.142857
2018-12-31	BID	0.252662	0.000000	0.000000	0.300000	0.142857	0.142857
2018-12-31	CTG	0.118716	0.000000	0.000000	0.000000	0.142857	0.142857
2018-12-31	MBB	0.072385	0.300000	0.300000	0.300000	0.142857	0.142857
2018-12-31	ACB	0.071391	0.300000	0.300000	0.100000	0.142857	0.142857
2018-12-31	STB	0.048089	0.000000	0.039476	0.000000	0.142857	0.142857
2018-12-31	SHB	0.020828	0.100000	0.060524	0.000000	0.142857	0.142857
2019-01-31	VCB	0.409325	0.300000	0.300000	0.300000	0.142857	0.142857

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2019-01-31	BID	0.281927	0.073694	0.000000	0.300000	0.142857	0.142857
2019-01-31	CTG	0.101434	0.000000	0.000000	0.000000	0.142857	0.142857
2019-01-31	MBB	0.066575	0.300000	0.300000	0.269089	0.142857	0.142857
2019-01-31	ACB	0.072193	0.300000	0.300000	0.130911	0.142857	0.142857
2019-01-31	STB	0.048193	0.000000	0.077855	0.000000	0.142857	0.142857
2019-01-31	SHB	0.020352	0.026306	0.022145	0.000000	0.142857	0.142857
2019-02-28	VCB	0.428960	0.300000	0.300000	0.300000	0.142857	0.142857
2019-02-28	BID	0.254887	0.100000	0.000000	0.300000	0.142857	0.142857
2019-02-28	CTG	0.102289	0.000000	0.000000	0.000000	0.142857	0.142857
2019-02-28	MBB	0.073970	0.300000	0.300000	0.300000	0.142857	0.142857
2019-02-28	ACB	0.070834	0.300000	0.300000	0.100000	0.142857	0.142857
2019-02-28	STB	0.048968	0.000000	0.093891	0.000000	0.142857	0.142857
2019-02-28	SHB	0.020091	0.000000	0.006109	0.000000	0.142857	0.142857
2019-03-31	VCB	0.438656	0.300000	0.300000	0.300000	0.142857	0.142857
2019-03-31	BID	0.252572	0.100000	0.000000	0.300000	0.142857	0.142857
2019-03-31	CTG	0.103662	0.000000	0.000000	0.100000	0.142857	0.142857
2019-03-31	MBB	0.069664	0.300000	0.300000	0.000000	0.142857	0.142857
2019-03-31	ACB	0.068640	0.300000	0.300000	0.300000	0.142857	0.142857
2019-03-31	STB	0.047019	0.000000	0.100000	0.000000	0.142857	0.142857
2019-03-31	SHB	0.019786	0.000000	0.000000	0.000000	0.142857	0.142857
2019-04-30	VCB	0.448710	0.300000	0.300000	0.300000	0.142857	0.142857
2019-04-30	BID	0.255822	0.100000	0.000000	0.300000	0.142857	0.142857
2019-04-30	CTG	0.102834	0.000000	0.000000	0.100000	0.142857	0.142857
2019-04-30	MBB	0.066792	0.300000	0.300000	0.000000	0.142857	0.142857
2019-04-30	ACB	0.064460	0.300000	0.300000	0.300000	0.142857	0.142857

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2019-04-30	STB	0.042886	0.000000	0.100000	0.000000	0.142857	0.142857
2019-04-30	SHB	0.018498	0.000000	0.000000	0.000000	0.142857	0.142857
2019-05-31	VCB	0.457295	0.300000	0.300000	0.300000	0.142857	0.142857
2019-05-31	BID	0.251111	0.100000	0.000000	0.300000	0.142857	0.142857
2019-05-31	CTG	0.098402	0.000000	0.000000	0.133995	0.142857	0.142857
2019-05-31	MBB	0.067460	0.300000	0.300000	0.036848	0.142857	0.142857
2019-05-31	ACB	0.064621	0.300000	0.300000	0.229157	0.142857	0.142857
2019-05-31	STB	0.042428	0.000000	0.100000	0.000000	0.142857	0.142857
2019-05-31	SHB	0.018683	0.000000	0.000000	0.000000	0.142857	0.142857
2019-06-30	VCB	0.471576	0.300000	0.300000	0.300000	0.142857	0.142857
2019-06-30	BID	0.236853	0.100000	0.000000	0.300000	0.142857	0.142857
2019-06-30	CTG	0.097897	0.000000	0.000000	0.114141	0.142857	0.142857
2019-06-30	MBB	0.067179	0.300000	0.300000	0.000000	0.142857	0.142857
2019-06-30	ACB	0.064915	0.300000	0.300000	0.285859	0.142857	0.142857
2019-06-30	STB	0.043469	0.000000	0.100000	0.000000	0.142857	0.142857
2019-06-30	SHB	0.018111	0.000000	0.000000	0.000000	0.142857	0.142857
2019-07-31	VCB	0.485559	0.300000	0.300000	0.300000	0.142857	0.142857
2019-07-31	BID	0.233966	0.100000	0.000000	0.300000	0.142857	0.142857
2019-07-31	CTG	0.092262	0.000000	0.000000	0.100000	0.142857	0.142857
2019-07-31	MBB	0.066174	0.300000	0.300000	0.000000	0.142857	0.142857
2019-07-31	ACB	0.063519	0.300000	0.300000	0.300000	0.142857	0.142857
2019-07-31	STB	0.041214	0.000000	0.100000	0.000000	0.142857	0.142857
2019-07-31	SHB	0.017307	0.000000	0.000000	0.000000	0.142857	0.142857
2019-08-31	VCB	0.500510	0.300000	0.300000	0.300000	0.142857	0.142857
2019-08-31	BID	0.237375	0.100000	0.000000	0.300000	0.142857	0.142857

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2019-08-31	CTG	0.089039	0.000000	0.000000	0.100000	0.142857	0.142857
2019-08-31	MBB	0.063691	0.300000	0.300000	0.000000	0.142857	0.142857
2019-08-31	ACB	0.058992	0.300000	0.300000	0.300000	0.142857	0.142857
2019-08-31	STB	0.035207	0.000000	0.100000	0.000000	0.142857	0.142857
2019-08-31	SHB	0.015186	0.000000	0.000000	0.000000	0.142857	0.142857
2019-09-30	VCB	0.483369	0.300000	0.300000	0.300000	0.142857	0.142857
2019-09-30	BID	0.258367	0.100000	0.000000	0.300000	0.142857	0.142857
2019-09-30	CTG	0.086328	0.000000	0.000000	0.069807	0.142857	0.142857
2019-09-30	MBB	0.066037	0.300000	0.300000	0.030193	0.142857	0.142857
2019-09-30	ACB	0.057285	0.300000	0.300000	0.300000	0.142857	0.142857
2019-09-30	STB	0.034111	0.000000	0.100000	0.000000	0.142857	0.142857
2019-09-30	SHB	0.014501	0.000000	0.000000	0.000000	0.142857	0.142857
2019-10-31	VCB	0.489136	0.300000	0.300000	0.300000	0.142857	0.142857
2019-10-31	BID	0.251482	0.100000	0.000000	0.300000	0.142857	0.142857
2019-10-31	CTG	0.086596	0.000000	0.000000	0.050316	0.142857	0.142857
2019-10-31	MBB	0.067399	0.300000	0.300000	0.049684	0.142857	0.142857
2019-10-31	ACB	0.057760	0.300000	0.300000	0.300000	0.142857	0.142857
2019-10-31	STB	0.033133	0.000000	0.100000	0.000000	0.142857	0.142857
2019-10-31	SHB	0.014494	0.000000	0.000000	0.000000	0.142857	0.142857
2019-11-30	VCB	0.498635	0.300000	0.300000	0.300000	0.142857	0.142857
2019-11-30	BID	0.247408	0.100000	0.000000	0.300000	0.142857	0.142857
2019-11-30	CTG	0.085986	0.000000	0.000000	0.000000	0.142857	0.142857
2019-11-30	MBB	0.064914	0.300000	0.284686	0.261507	0.142857	0.142857
2019-11-30	ACB	0.056848	0.300000	0.300000	0.138493	0.142857	0.142857
2019-11-30	STB	0.032375	0.000000	0.115314	0.000000	0.142857	0.142857

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2019-11-30	SHB	0.013835	0.000000	0.000000	0.000000	0.142857	0.142857
2019-12-31	VCB	0.494942	0.300000	0.300000	0.300000	0.142857	0.142857
2019-12-31	BID	0.260411	0.100000	0.000000	0.300000	0.142857	0.142857
2019-12-31	CTG	0.081548	0.000000	0.000000	0.041888	0.142857	0.142857
2019-12-31	MBB	0.063759	0.300000	0.280297	0.058112	0.142857	0.142857
2019-12-31	ACB	0.055586	0.300000	0.300000	0.300000	0.142857	0.142857
2019-12-31	STB	0.030892	0.000000	0.119703	0.000000	0.142857	0.142857
2019-12-31	SHB	0.012863	0.000000	0.000000	0.000000	0.142857	0.142857
2020-01-31	VCB	0.494655	0.300000	0.290018	0.300000	0.142857	0.142857
2020-01-31	BID	0.277999	0.100000	0.000000	0.300000	0.142857	0.142857
2020-01-31	CTG	0.078027	0.000000	0.000000	0.034762	0.142857	0.142857
2020-01-31	MBB	0.056113	0.300000	0.269295	0.066171	0.142857	0.142857
2020-01-31	ACB	0.051371	0.300000	0.300000	0.299066	0.142857	0.142857
2020-01-31	STB	0.028797	0.000000	0.140687	0.000000	0.142857	0.142857
2020-01-31	SHB	0.013039	0.000000	0.000000	0.000000	0.142857	0.142857
2020-02-29	VCB	0.467669	0.300000	0.276845	0.300000	0.142857	0.142857
2020-02-29	BID	0.298281	0.255886	0.000000	0.300000	0.142857	0.142857
2020-02-29	CTG	0.087657	0.000000	0.000000	0.144993	0.142857	0.142857
2020-02-29	MBB	0.053952	0.280408	0.237207	0.074990	0.142857	0.142857
2020-02-29	ACB	0.049843	0.163706	0.300000	0.180018	0.142857	0.142857
2020-02-29	STB	0.028101	0.000000	0.185949	0.000000	0.142857	0.142857
2020-02-29	SHB	0.014497	0.000000	0.000000	0.000000	0.142857	0.142857
2020-03-31	VCB	0.448047	0.300000	0.293476	0.300000	0.125000	0.125000
2020-03-31	BID	0.271740	0.300000	0.000000	0.300000	0.125000	0.125000
2020-03-31	CTG	0.095336	0.000000	0.000000	0.000000	0.125000	0.125000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2020-03-31	MBB	0.054536	0.300000	0.139178	0.123754	0.125000	0.125000
2020-03-31	ACB	0.056280	0.100000	0.300000	0.000000	0.125000	0.125000
2020-03-31	VIB	0.018004	0.000000	0.061800	0.276246	0.125000	0.125000
2020-03-31	STB	0.032958	0.000000	0.205545	0.000000	0.125000	0.125000
2020-03-31	SHB	0.023100	0.000000	0.000000	0.000000	0.125000	0.125000
2020-04-30	VCB	0.465684	0.300000	0.300000	0.300000	0.125000	0.125000
2020-04-30	BID	0.254497	0.191172	0.000000	0.300000	0.125000	0.125000
2020-04-30	CTG	0.087985	0.000000	0.000000	0.000000	0.125000	0.125000
2020-04-30	MBB	0.050055	0.192232	0.147313	0.195794	0.125000	0.125000
2020-04-30	ACB	0.054967	0.251651	0.300000	0.000000	0.125000	0.125000
2020-04-30	VIB	0.017614	0.000000	0.062582	0.204206	0.125000	0.125000
2020-04-30	STB	0.028651	0.000000	0.190105	0.000000	0.125000	0.125000
2020-04-30	SHB	0.040545	0.064945	0.000000	0.000000	0.125000	0.125000
2020-05-31	VCB	0.448980	0.300000	0.300000	0.108038	0.125000	0.125000
2020-05-31	BID	0.258161	0.300000	0.000000	0.300000	0.125000	0.125000
2020-05-31	CTG	0.089926	0.000000	0.000000	0.000000	0.125000	0.125000
2020-05-31	MBB	0.051820	0.000000	0.129241	0.000000	0.125000	0.125000
2020-05-31	ACB	0.055653	0.100000	0.300000	0.000000	0.125000	0.125000
2020-05-31	VIB	0.017556	0.000000	0.003073	0.291962	0.125000	0.125000
2020-05-31	STB	0.031397	0.000000	0.000000	0.000000	0.125000	0.125000
2020-05-31	SHB	0.046506	0.300000	0.267685	0.300000	0.125000	0.125000
2020-06-30	VCB	0.481368	0.300000	0.300000	0.300000	0.125000	0.125000
2020-06-30	BID	0.248812	0.300000	0.000000	0.300000	0.125000	0.125000
2020-06-30	CTG	0.086612	0.000000	0.000000	0.038224	0.125000	0.125000
2020-06-30	MBB	0.047778	0.000000	0.092300	0.000000	0.125000	0.125000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2020-06-30	ACB	0.053168	0.100000	0.300000	0.017260	0.125000	0.125000
2020-06-30	VIB	0.016841	0.000000	0.056661	0.044516	0.125000	0.125000
2020-06-30	STB	0.030406	0.000000	0.000000	0.000000	0.125000	0.125000
2020-06-30	SHB	0.035016	0.300000	0.251040	0.300000	0.125000	0.125000
2020-07-31	VCB	0.475964	0.300000	0.300000	0.300000	0.125000	0.125000
2020-07-31	BID	0.246758	0.300000	0.000000	0.300000	0.125000	0.125000
2020-07-31	CTG	0.086994	0.000000	0.000000	0.100000	0.125000	0.125000
2020-07-31	MBB	0.048426	0.000000	0.110347	0.000000	0.125000	0.125000
2020-07-31	ACB	0.055532	0.100000	0.300000	0.300000	0.125000	0.125000
2020-07-31	VIB	0.018894	0.000000	0.023637	0.000000	0.125000	0.125000
2020-07-31	STB	0.033297	0.000000	0.000000	0.000000	0.125000	0.125000
2020-07-31	SHB	0.034135	0.300000	0.266016	0.000000	0.125000	0.125000
2020-08-31	VCB	0.470276	0.300000	0.239297	0.300000	0.111111	0.111111
2020-08-31	BID	0.247840	0.255999	0.000000	0.300000	0.111111	0.111111
2020-08-31	CTG	0.088866	0.000000	0.000000	0.100000	0.111111	0.111111
2020-08-31	MBB	0.047681	0.000000	0.000000	0.000000	0.111111	0.111111
2020-08-31	ACB	0.056874	0.004559	0.300000	0.300000	0.111111	0.111111
2020-08-31	VIB	0.020279	0.000000	0.000000	0.000000	0.111111	0.111111
2020-08-31	STB	0.032484	0.000000	0.000000	0.000000	0.111111	0.111111
2020-08-31	SHB	0.032185	0.300000	0.160703	0.000000	0.111111	0.111111
2020-08-31	KLB	0.003515	0.139442	0.300000	0.000000	0.111111	0.111111
2020-09-30	VCB	0.455432	0.300000	0.246477	0.300000	0.111111	0.111111
2020-09-30	BID	0.246771	0.144144	0.000000	0.300000	0.111111	0.111111
2020-09-30	CTG	0.096041	0.000000	0.000000	0.000000	0.111111	0.111111
2020-09-30	MBB	0.048444	0.000000	0.000000	0.000000	0.111111	0.111111

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2020-09-30	ACB	0.062104	0.060901	0.300000	0.000000	0.111111	0.111111
2020-09-30	VIB	0.022110	0.000000	0.000000	0.100000	0.111111	0.111111
2020-09-30	STB	0.032111	0.000000	0.000000	0.000000	0.111111	0.111111
2020-09-30	SHB	0.033760	0.300000	0.153523	0.300000	0.111111	0.111111
2020-09-30	KLB	0.003228	0.194955	0.300000	0.000000	0.111111	0.111111
2020-10-31	VCB	0.409789	0.300000	0.300000	0.300000	0.100000	0.100000
2020-10-31	BID	0.218408	0.121667	0.000000	0.300000	0.100000	0.100000
2020-10-31	CTG	0.088126	0.000000	0.000000	0.000000	0.100000	0.100000
2020-10-31	MBB	0.047310	0.000000	0.000000	0.000000	0.100000	0.100000
2020-10-31	ACB	0.058557	0.000000	0.240280	0.100000	0.100000	0.100000
2020-10-31	VPB	0.081657	0.000000	0.000000	0.000000	0.100000	0.100000
2020-10-31	VIB	0.024706	0.043184	0.000000	0.300000	0.100000	0.100000
2020-10-31	STB	0.035139	0.000000	0.000000	0.000000	0.100000	0.100000
2020-10-31	SHB	0.033224	0.300000	0.159720	0.000000	0.100000	0.100000
2020-10-31	KLB	0.003084	0.235149	0.300000	0.000000	0.100000	0.100000
2020-11-30	VCB	0.404752	0.300000	0.300000	0.100000	0.100000	0.100000
2020-11-30	BID	0.206215	0.013720	0.000000	0.300000	0.100000	0.100000
2020-11-30	CTG	0.096313	0.000000	0.000000	0.000000	0.100000	0.100000
2020-11-30	MBB	0.049074	0.000000	0.000000	0.000000	0.100000	0.100000
2020-11-30	ACB	0.062761	0.000000	0.240837	0.000000	0.100000	0.100000
2020-11-30	VPB	0.081175	0.000000	0.000000	0.000000	0.100000	0.100000
2020-11-30	VIB	0.029273	0.146524	0.000000	0.300000	0.100000	0.100000
2020-11-30	STB	0.033889	0.000000	0.000000	0.000000	0.100000	0.100000
2020-11-30	SHB	0.033246	0.269958	0.159163	0.300000	0.100000	0.100000
2020-11-30	KLB	0.003303	0.269798	0.300000	0.000000	0.100000	0.100000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2020-12-31	VCB	0.400213	0.300000	0.300000	0.100000	0.090909	0.090909
2020-12-31	BID	0.197126	0.000000	0.000000	0.300000	0.090909	0.090909
2020-12-31	CTG	0.098182	0.000000	0.000000	0.000000	0.090909	0.090909
2020-12-31	MBB	0.049490	0.000000	0.000000	0.000000	0.090909	0.090909
2020-12-31	ACB	0.062549	0.000000	0.240434	0.000000	0.090909	0.090909
2020-12-31	VPB	0.081282	0.000000	0.000000	0.000000	0.090909	0.090909
2020-12-31	VIB	0.026080	0.236992	0.000000	0.300000	0.090909	0.090909
2020-12-31	STB	0.032939	0.000000	0.000000	0.000000	0.090909	0.090909
2020-12-31	SHB	0.032392	0.252185	0.159566	0.300000	0.090909	0.090909
2020-12-31	LPB	0.016567	0.000000	0.000000	0.000000	0.090909	0.090909
2020-12-31	KLB	0.003180	0.210822	0.300000	0.000000	0.090909	0.090909
2021-01-31	VCB	0.385892	0.269339	0.300000	0.100000	0.090909	0.090909
2021-01-31	BID	0.209155	0.000000	0.000000	0.300000	0.090909	0.090909
2021-01-31	CTG	0.093336	0.000000	0.000000	0.000000	0.090909	0.090909
2021-01-31	MBB	0.050850	0.000000	0.000000	0.000000	0.090909	0.090909
2021-01-31	ACB	0.058667	0.000000	0.221427	0.000000	0.090909	0.090909
2021-01-31	VPB	0.089583	0.000000	0.000000	0.000000	0.090909	0.090909
2021-01-31	VIB	0.027827	0.224999	0.000000	0.300000	0.090909	0.090909
2021-01-31	STB	0.034524	0.000000	0.000000	0.000000	0.090909	0.090909
2021-01-31	SHB	0.029430	0.205662	0.164210	0.300000	0.090909	0.090909
2021-01-31	LPB	0.016509	0.000000	0.014363	0.000000	0.090909	0.090909
2021-01-31	KLB	0.004227	0.300000	0.300000	0.000000	0.090909	0.090909
2021-02-28	VCB	0.391361	0.189547	0.255318	0.147913	0.083333	0.083333
2021-02-28	BID	0.186043	0.000000	0.000000	0.287667	0.083333	0.083333
2021-02-28	CTG	0.087982	0.000000	0.000000	0.100869	0.083333	0.083333

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-02-28	MBB	0.055356	0.000000	0.000000	0.000000	0.083333	0.083333
2021-02-28	ACB	0.059791	0.000000	0.092282	0.000000	0.083333	0.083333
2021-02-28	VPB	0.088312	0.000000	0.000000	0.095251	0.083333	0.083333
2021-02-28	VIB	0.028990	0.300000	0.000000	0.183786	0.083333	0.083333
2021-02-28	STB	0.036864	0.000000	0.000000	0.054152	0.083333	0.083333
2021-02-28	SHB	0.027003	0.210453	0.121874	0.130363	0.083333	0.083333
2021-02-28	LPB	0.017974	0.000000	0.000000	0.000000	0.083333	0.083333
2021-02-28	BAB	0.016306	0.000000	0.300000	0.000000	0.083333	0.083333
2021-02-28	KLB	0.004019	0.300000	0.230526	0.000000	0.083333	0.083333
2021-03-31	VCB	0.355770	0.261641	0.300000	0.100000	0.076923	0.076923
2021-03-31	BID	0.175039	0.000000	0.000000	0.000000	0.076923	0.076923
2021-03-31	CTG	0.092474	0.000000	0.000000	0.000000	0.076923	0.076923
2021-03-31	MBB	0.056158	0.000000	0.000000	0.000000	0.076923	0.076923
2021-03-31	ACB	0.063439	0.000000	0.048220	0.000000	0.076923	0.076923
2021-03-31	VPB	0.101383	0.000000	0.000000	0.000000	0.076923	0.076923
2021-03-31	HDB	0.028668	0.000000	0.000000	0.000000	0.076923	0.076923
2021-03-31	VIB	0.031333	0.300000	0.000000	0.300000	0.076923	0.076923
2021-03-31	STB	0.034455	0.000000	0.000000	0.000000	0.076923	0.076923
2021-03-31	SHB	0.025309	0.135702	0.161904	0.000000	0.076923	0.076923
2021-03-31	LPB	0.017971	0.000000	0.015017	0.000000	0.076923	0.076923
2021-03-31	BAB	0.014561	0.002657	0.300000	0.300000	0.076923	0.076923
2021-03-31	KLB	0.003442	0.300000	0.174860	0.300000	0.076923	0.076923
2021-04-30	VCB	0.329000	0.035307	0.300000	0.000000	0.076923	0.076923
2021-04-30	BID	0.165127	0.000000	0.000000	0.000000	0.076923	0.076923
2021-04-30	CTG	0.095499	0.000000	0.000000	0.000000	0.076923	0.076923

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-04-30	MBB	0.054785	0.000000	0.000000	0.000000	0.076923	0.076923
2021-04-30	ACB	0.061210	0.000000	0.040075	0.000000	0.076923	0.076923
2021-04-30	VPB	0.108396	0.000000	0.000000	0.000000	0.076923	0.076923
2021-04-30	HDB	0.028801	0.000000	0.000000	0.000000	0.076923	0.076923
2021-04-30	VIB	0.036609	0.300000	0.000000	0.300000	0.076923	0.076923
2021-04-30	STB	0.038581	0.000000	0.000000	0.000000	0.076923	0.076923
2021-04-30	SHB	0.039178	0.098117	0.165276	0.000000	0.076923	0.076923
2021-04-30	LPB	0.020321	0.000000	0.006097	0.100000	0.076923	0.076923
2021-04-30	BAB	0.018600	0.266577	0.300000	0.300000	0.076923	0.076923
2021-04-30	KLB	0.003893	0.300000	0.188552	0.300000	0.076923	0.076923
2021-05-31	VCB	0.324410	0.004001	0.300000	0.000000	0.076923	0.076923
2021-05-31	BID	0.147356	0.000000	0.000000	0.000000	0.076923	0.076923
2021-05-31	CTG	0.090715	0.000000	0.000000	0.000000	0.076923	0.076923
2021-05-31	MBB	0.055244	0.000000	0.000000	0.000000	0.076923	0.076923
2021-05-31	ACB	0.059559	0.000000	0.132389	0.000000	0.076923	0.076923
2021-05-31	VPB	0.132753	0.000000	0.000000	0.000000	0.076923	0.076923
2021-05-31	HDB	0.027801	0.000000	0.000000	0.000000	0.076923	0.076923
2021-05-31	VIB	0.039326	0.300000	0.000000	0.000000	0.076923	0.076923
2021-05-31	STB	0.040187	0.000000	0.000000	0.000000	0.076923	0.076923
2021-05-31	SHB	0.039055	0.131285	0.074042	0.300000	0.076923	0.076923
2021-05-31	LPB	0.023553	0.000000	0.000000	0.100000	0.076923	0.076923
2021-05-31	BAB	0.015663	0.300000	0.300000	0.300000	0.076923	0.076923
2021-05-31	KLB	0.004379	0.264714	0.193569	0.300000	0.076923	0.076923
2021-06-30	VCB	0.269885	0.249672	0.292735	0.000000	0.071429	0.071429
2021-06-30	BID	0.147687	0.000000	0.000000	0.000000	0.071429	0.071429

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-06-30	CTG	0.099616	0.000000	0.000000	0.134122	0.071429	0.071429
2021-06-30	MBB	0.058322	0.000000	0.000000	0.000000	0.071429	0.071429
2021-06-30	ACB	0.060033	0.000000	0.000000	0.000000	0.071429	0.071429
2021-06-30	VPB	0.132490	0.000000	0.000000	0.000000	0.071429	0.071429
2021-06-30	HDB	0.029084	0.000000	0.000000	0.000000	0.071429	0.071429
2021-06-30	VIB	0.040871	0.300000	0.000000	0.300000	0.071429	0.071429
2021-06-30	TPB	0.025708	0.000000	0.230208	0.000000	0.071429	0.071429
2021-06-30	STB	0.047953	0.000000	0.000000	0.000000	0.071429	0.071429
2021-06-30	SHB	0.040204	0.236911	0.096537	0.300000	0.071429	0.071429
2021-06-30	LPB	0.027674	0.000000	0.000000	0.078388	0.071429	0.071429
2021-06-30	BAB	0.015926	0.071808	0.300000	0.187489	0.071429	0.071429
2021-06-30	KLB	0.004547	0.141609	0.080520	0.000000	0.071429	0.071429
2021-07-31	VCB	0.304913	0.288372	0.300000	0.000000	0.071429	0.071429
2021-07-31	BID	0.137137	0.000000	0.000000	0.000000	0.071429	0.071429
2021-07-31	CTG	0.094596	0.000000	0.000000	0.000000	0.071429	0.071429
2021-07-31	MBB	0.063680	0.000000	0.000000	0.000000	0.071429	0.071429
2021-07-31	ACB	0.061927	0.000000	0.011659	0.000000	0.071429	0.071429
2021-07-31	VPB	0.124049	0.000000	0.000000	0.300000	0.071429	0.071429
2021-07-31	HDB	0.029248	0.000000	0.000000	0.000000	0.071429	0.071429
2021-07-31	VIB	0.039521	0.300000	0.000000	0.003441	0.071429	0.071429
2021-07-31	TPB	0.024145	0.004652	0.167387	0.000000	0.071429	0.071429
2021-07-31	STB	0.041546	0.000000	0.000000	0.028774	0.071429	0.071429
2021-07-31	SHB	0.036308	0.280543	0.122372	0.172163	0.071429	0.071429
2021-07-31	LPB	0.026484	0.000000	0.000000	0.300000	0.071429	0.071429
2021-07-31	BAB	0.012647	0.000000	0.300000	0.000000	0.071429	0.071429

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-07-31	KLB	0.003799	0.126433	0.098583	0.195622	0.071429	0.071429
2021-08-31	VCB	0.254886	0.300000	0.300000	0.035919	0.066667	0.066667
2021-08-31	BID	0.118439	0.000000	0.000000	0.000000	0.066667	0.066667
2021-08-31	CTG	0.079140	0.000000	0.000000	0.064081	0.066667	0.066667
2021-08-31	MBB	0.056944	0.000000	0.000000	0.000000	0.066667	0.066667
2021-08-31	TCB	0.119425	0.000000	0.000000	0.000000	0.066667	0.066667
2021-08-31	ACB	0.062258	0.000000	0.054315	0.000000	0.066667	0.066667
2021-08-31	VPB	0.110969	0.000000	0.000000	0.000000	0.066667	0.066667
2021-08-31	HDB	0.027559	0.000000	0.000000	0.000000	0.066667	0.066667
2021-08-31	VIB	0.033019	0.300000	0.000000	0.300000	0.066667	0.066667
2021-08-31	TPB	0.022821	0.022450	0.137321	0.000000	0.066667	0.066667
2021-08-31	STB	0.040307	0.000000	0.000000	0.000000	0.066667	0.066667
2021-08-31	SHB	0.034945	0.300000	0.120161	0.300000	0.066667	0.066667
2021-08-31	LPB	0.025078	0.000000	0.000000	0.300000	0.066667	0.066667
2021-08-31	BAB	0.011077	0.000000	0.300000	0.000000	0.066667	0.066667
2021-08-31	KLB	0.003134	0.077550	0.088203	0.000000	0.066667	0.066667
2021-09-30	VCB	0.267803	0.089171	0.300000	0.025448	0.066667	0.066667
2021-09-30	BID	0.116290	0.000000	0.000000	0.000000	0.066667	0.066667
2021-09-30	CTG	0.075773	0.000000	0.000000	0.025700	0.066667	0.066667
2021-09-30	MBB	0.056907	0.000000	0.000000	0.000000	0.066667	0.066667
2021-09-30	TCB	0.117428	0.000000	0.000000	0.000000	0.066667	0.066667
2021-09-30	ACB	0.057110	0.000000	0.065548	0.000000	0.066667	0.066667
2021-09-30	VPB	0.116065	0.000000	0.000000	0.000000	0.066667	0.066667
2021-09-30	HDB	0.027482	0.000000	0.000000	0.000000	0.066667	0.066667
2021-09-30	VIB	0.029790	0.300000	0.000000	0.300000	0.066667	0.066667

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-09-30	TPB	0.023033	0.175842	0.167326	0.000000	0.066667	0.066667
2021-09-30	STB	0.038191	0.000000	0.000000	0.000000	0.066667	0.066667
2021-09-30	SHB	0.035807	0.300000	0.132744	0.300000	0.066667	0.066667
2021-09-30	LPB	0.023477	0.134987	0.000000	0.300000	0.066667	0.066667
2021-09-30	BAB	0.011567	0.000000	0.300000	0.048852	0.066667	0.066667
2021-09-30	KLB	0.003277	0.000000	0.034382	0.000000	0.066667	0.066667
2021-10-31	VCB	0.261601	0.107695	0.300000	0.000000	0.066667	0.066667
2021-10-31	BID	0.117629	0.000000	0.000000	0.000000	0.066667	0.066667
2021-10-31	CTG	0.072360	0.000000	0.000000	0.000000	0.066667	0.066667
2021-10-31	MBB	0.056760	0.000000	0.000000	0.000000	0.066667	0.066667
2021-10-31	TCB	0.119978	0.000000	0.000000	0.000000	0.066667	0.066667
2021-10-31	ACB	0.056255	0.000000	0.053615	0.000000	0.066667	0.066667
2021-10-31	VPB	0.124052	0.000000	0.000000	0.115663	0.066667	0.066667
2021-10-31	HDB	0.026883	0.000000	0.000000	0.000000	0.066667	0.066667
2021-10-31	VIB	0.029033	0.220406	0.000000	0.021821	0.066667	0.066667
2021-10-31	TPB	0.028073	0.118432	0.183153	0.000000	0.066667	0.066667
2021-10-31	STB	0.035852	0.000000	0.000000	0.000000	0.066667	0.066667
2021-10-31	SHB	0.034598	0.300000	0.136118	0.300000	0.066667	0.066667
2021-10-31	LPB	0.022039	0.226779	0.000000	0.300000	0.066667	0.066667
2021-10-31	BAB	0.011555	0.000000	0.300000	0.000000	0.066667	0.066667
2021-10-31	KLB	0.003333	0.026688	0.027115	0.262516	0.066667	0.066667
2021-11-30	VCB	0.255940	0.089712	0.300000	0.000000	0.066667	0.066667
2021-11-30	BID	0.120088	0.000000	0.000000	0.000000	0.066667	0.066667
2021-11-30	CTG	0.072758	0.000000	0.000000	0.000000	0.066667	0.066667
2021-11-30	MBB	0.056347	0.000000	0.000000	0.000000	0.066667	0.066667

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-11-30	TCB	0.121085	0.000000	0.000000	0.000000	0.066667	0.066667
2021-11-30	ACB	0.057094	0.000000	0.058296	0.000000	0.066667	0.066667
2021-11-30	VPB	0.125100	0.000000	0.000000	0.268352	0.066667	0.066667
2021-11-30	HDB	0.025934	0.000000	0.000000	0.000000	0.066667	0.066667
2021-11-30	VIB	0.029476	0.300000	0.000000	0.192649	0.066667	0.066667
2021-11-30	TPB	0.029365	0.300000	0.178534	0.000000	0.066667	0.066667
2021-11-30	STB	0.036006	0.000000	0.000000	0.000000	0.066667	0.066667
2021-11-30	SHB	0.035265	0.300000	0.136405	0.300000	0.066667	0.066667
2021-11-30	LPB	0.020548	0.010288	0.000000	0.239000	0.066667	0.066667
2021-11-30	BAB	0.011109	0.000000	0.300000	0.000000	0.066667	0.066667
2021-11-30	KLB	0.003884	0.000000	0.026766	0.000000	0.066667	0.066667
2021-12-31	VCB	0.250050	0.198537	0.300000	0.000000	0.066667	0.066667
2021-12-31	BID	0.124413	0.000000	0.000000	0.000000	0.066667	0.066667
2021-12-31	CTG	0.075031	0.000000	0.000000	0.000000	0.066667	0.066667
2021-12-31	MBB	0.055527	0.000000	0.000000	0.000000	0.066667	0.066667
2021-12-31	TCB	0.117718	0.000000	0.000000	0.000000	0.066667	0.066667
2021-12-31	ACB	0.055774	0.000000	0.086323	0.000000	0.066667	0.066667
2021-12-31	VPB	0.120011	0.000000	0.000000	0.279767	0.066667	0.066667
2021-12-31	HDB	0.030305	0.000000	0.000000	0.000000	0.066667	0.066667
2021-12-31	VIB	0.032555	0.198610	0.000000	0.212244	0.066667	0.066667
2021-12-31	TPB	0.028595	0.300000	0.151726	0.000000	0.066667	0.066667
2021-12-31	STB	0.037873	0.000000	0.000000	0.000000	0.066667	0.066667
2021-12-31	SHB	0.035498	0.300000	0.152327	0.300000	0.066667	0.066667
2021-12-31	LPB	0.021410	0.000000	0.000000	0.207989	0.066667	0.066667
2021-12-31	BAB	0.011494	0.000000	0.300000	0.000000	0.066667	0.066667

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2021-12-31	KLB	0.003745	0.002853	0.009623	0.000000	0.066667	0.066667
2022-01-31	VCB	0.252133	0.162588	0.300000	0.000000	0.066667	0.066667
2022-01-31	BID	0.128671	0.000000	0.000000	0.000000	0.066667	0.066667
2022-01-31	CTG	0.076101	0.000000	0.000000	0.000000	0.066667	0.066667
2022-01-31	MBB	0.054231	0.000000	0.000000	0.000000	0.066667	0.066667
2022-01-31	TCB	0.111282	0.000000	0.000000	0.000000	0.066667	0.066667
2022-01-31	ACB	0.056599	0.000000	0.083347	0.000000	0.066667	0.066667
2022-01-31	VPB	0.111626	0.000000	0.000000	0.297223	0.066667	0.066667
2022-01-31	HDB	0.030047	0.000000	0.000000	0.000000	0.066667	0.066667
2022-01-31	VIB	0.035747	0.237412	0.000000	0.214406	0.066667	0.066667
2022-01-31	TPB	0.034418	0.300000	0.153425	0.000000	0.066667	0.066667
2022-01-31	STB	0.040441	0.000000	0.000000	0.000000	0.066667	0.066667
2022-01-31	SHB	0.033004	0.300000	0.152971	0.300000	0.066667	0.066667
2022-01-31	LPB	0.020893	0.000000	0.000000	0.188371	0.066667	0.066667
2022-01-31	BAB	0.010921	0.000000	0.300000	0.000000	0.066667	0.066667
2022-01-31	KLB	0.003886	0.000000	0.010257	0.000000	0.066667	0.066667
2022-02-28	VCB	0.257855	0.131970	0.300000	0.000000	0.066667	0.066667
2022-02-28	BID	0.150601	0.000000	0.000000	0.000000	0.066667	0.066667
2022-02-28	CTG	0.075010	0.000000	0.000000	0.006025	0.066667	0.066667
2022-02-28	MBB	0.057240	0.000000	0.000000	0.043950	0.066667	0.066667
2022-02-28	TCB	0.106439	0.000000	0.000000	0.008438	0.066667	0.066667
2022-02-28	ACB	0.051318	0.000000	0.085852	0.000000	0.066667	0.066667
2022-02-28	VPB	0.103484	0.000000	0.000000	0.300000	0.066667	0.066667
2022-02-28	HDB	0.027254	0.000000	0.000000	0.000000	0.066667	0.066667
2022-02-28	VIB	0.032537	0.268030	0.000000	0.277330	0.066667	0.066667

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-02-28	TPB	0.031853	0.300000	0.145731	0.000000	0.066667	0.066667
2022-02-28	STB	0.041331	0.000000	0.000000	0.098196	0.066667	0.066667
2022-02-28	SHB	0.030138	0.300000	0.154746	0.266060	0.066667	0.066667
2022-02-28	LPB	0.021204	0.000000	0.000000	0.000000	0.066667	0.066667
2022-02-28	BAB	0.009849	0.000000	0.300000	0.000000	0.066667	0.066667
2022-02-28	KLB	0.003886	0.000000	0.013671	0.000000	0.066667	0.066667
2022-03-31	VCB	0.255927	0.157162	0.300000	0.000000	0.066667	0.066667
2022-03-31	BID	0.143447	0.000000	0.000000	0.000000	0.066667	0.066667
2022-03-31	CTG	0.070451	0.000000	0.000000	0.002088	0.066667	0.066667
2022-03-31	MBB	0.061096	0.000000	0.000000	0.000000	0.066667	0.066667
2022-03-31	TCB	0.106214	0.000000	0.000000	0.000000	0.066667	0.066667
2022-03-31	ACB	0.052653	0.000000	0.085562	0.000000	0.066667	0.066667
2022-03-31	VPB	0.112170	0.000000	0.000000	0.300000	0.066667	0.066667
2022-03-31	HDB	0.026544	0.000000	0.000000	0.000000	0.066667	0.066667
2022-03-31	VIB	0.033553	0.242838	0.000000	0.281862	0.066667	0.066667
2022-03-31	TPB	0.033537	0.300000	0.147568	0.000000	0.066667	0.066667
2022-03-31	STB	0.039925	0.000000	0.000000	0.000000	0.066667	0.066667
2022-03-31	SHB	0.030327	0.300000	0.155947	0.279138	0.066667	0.066667
2022-03-31	LPB	0.020087	0.000000	0.000000	0.136912	0.066667	0.066667
2022-03-31	BAB	0.010111	0.000000	0.300000	0.000000	0.066667	0.066667
2022-03-31	KLB	0.003959	0.000000	0.010922	0.000000	0.066667	0.066667
2022-04-30	VCB	0.253889	0.102956	0.300000	0.000000	0.066667	0.066667
2022-04-30	BID	0.145820	0.000000	0.000000	0.000000	0.066667	0.066667
2022-04-30	CTG	0.070414	0.000000	0.000000	0.000000	0.066667	0.066667
2022-04-30	MBB	0.059731	0.000000	0.000000	0.000000	0.066667	0.066667

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-04-30	TCB	0.106589	0.000000	0.000000	0.000000	0.066667	0.066667
2022-04-30	ACB	0.052409	0.000000	0.084006	0.000000	0.066667	0.066667
2022-04-30	VPB	0.112128	0.000000	0.000000	0.229818	0.066667	0.066667
2022-04-30	HDB	0.026687	0.000000	0.000000	0.000000	0.066667	0.066667
2022-04-30	VIB	0.034377	0.297044	0.000000	0.111869	0.066667	0.066667
2022-04-30	TPB	0.032538	0.300000	0.152237	0.000000	0.066667	0.066667
2022-04-30	STB	0.039338	0.000000	0.000000	0.000000	0.066667	0.066667
2022-04-30	SHB	0.030590	0.300000	0.153891	0.219711	0.066667	0.066667
2022-04-30	LPB	0.019605	0.000000	0.000000	0.250672	0.066667	0.066667
2022-04-30	BAB	0.010279	0.000000	0.300000	0.000000	0.066667	0.066667
2022-04-30	KLB	0.005605	0.000000	0.009866	0.187930	0.066667	0.066667
2022-05-31	VCB	0.272787	0.000000	0.300000	0.000000	0.066667	0.066667
2022-05-31	BID	0.136155	0.000000	0.000000	0.000000	0.066667	0.066667
2022-05-31	CTG	0.065632	0.000000	0.000000	0.043870	0.066667	0.066667
2022-05-31	MBB	0.058903	0.000000	0.000000	0.000000	0.066667	0.066667
2022-05-31	TCB	0.103230	0.000000	0.000000	0.000000	0.066667	0.066667
2022-05-31	ACB	0.056550	0.000000	0.091413	0.099294	0.066667	0.066667
2022-05-31	VPB	0.120598	0.000000	0.000000	0.300000	0.066667	0.066667
2022-05-31	HDB	0.025972	0.000000	0.000000	0.000000	0.066667	0.066667
2022-05-31	VIB	0.034251	0.284594	0.000000	0.101682	0.066667	0.066667
2022-05-31	TPB	0.029692	0.300000	0.126780	0.000000	0.066667	0.066667
2022-05-31	STB	0.037477	0.000000	0.000000	0.292063	0.066667	0.066667
2022-05-31	SHB	0.025415	0.297768	0.142362	0.067146	0.066667	0.066667
2022-05-31	LPB	0.017901	0.000000	0.000000	0.000000	0.066667	0.066667
2022-05-31	BAB	0.010031	0.000000	0.300000	0.000000	0.066667	0.066667

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-05-31	KLB	0.005406	0.117638	0.039445	0.095945	0.066667	0.066667
2022-06-30	VCB	0.286788	0.012871	0.300000	0.000000	0.066667	0.066667
2022-06-30	BID	0.139075	0.000000	0.000000	0.019391	0.066667	0.066667
2022-06-30	CTG	0.071132	0.000000	0.000000	0.000499	0.066667	0.066667
2022-06-30	MBB	0.059160	0.000000	0.000000	0.000000	0.066667	0.066667
2022-06-30	TCB	0.093965	0.000000	0.000000	0.000000	0.066667	0.066667
2022-06-30	ACB	0.058821	0.000000	0.187579	0.000000	0.066667	0.066667
2022-06-30	VPB	0.109821	0.039369	0.000000	0.300000	0.066667	0.066667
2022-06-30	HDB	0.028870	0.000000	0.000000	0.000000	0.066667	0.066667
2022-06-30	VIB	0.031072	0.300000	0.000000	0.171558	0.066667	0.066667
2022-06-30	TPB	0.030826	0.213955	0.047980	0.000000	0.066667	0.066667
2022-06-30	STB	0.032601	0.000000	0.000000	0.071891	0.066667	0.066667
2022-06-30	SHB	0.024883	0.235397	0.105450	0.196885	0.066667	0.066667
2022-06-30	LPB	0.017659	0.000000	0.000000	0.000000	0.066667	0.066667
2022-06-30	BAB	0.010489	0.000000	0.300000	0.000000	0.066667	0.066667
2022-06-30	KLB	0.004839	0.198409	0.058991	0.239775	0.066667	0.066667
2022-07-31	VCB	0.291893	0.014193	0.300000	0.000000	0.066667	0.066667
2022-07-31	BID	0.141710	0.000000	0.000000	0.000000	0.066667	0.066667
2022-07-31	CTG	0.071585	0.000000	0.000000	0.037480	0.066667	0.066667
2022-07-31	MBB	0.055346	0.000000	0.000000	0.000000	0.066667	0.066667
2022-07-31	TCB	0.096526	0.000000	0.000000	0.000000	0.066667	0.066667
2022-07-31	ACB	0.060027	0.181105	0.205502	0.040985	0.066667	0.066667
2022-07-31	VPB	0.110311	0.000000	0.000000	0.300000	0.066667	0.066667
2022-07-31	HDB	0.028513	0.000000	0.000000	0.000000	0.066667	0.066667
2022-07-31	VIB	0.026748	0.300000	0.000000	0.171737	0.066667	0.066667

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-07-31	TPB	0.027758	0.083043	0.039997	0.000000	0.066667	0.066667
2022-07-31	STB	0.033665	0.000000	0.000000	0.171775	0.066667	0.066667
2022-07-31	SHB	0.024347	0.275985	0.107880	0.132780	0.066667	0.066667
2022-07-31	LPB	0.016475	0.000000	0.000000	0.000000	0.066667	0.066667
2022-07-31	BAB	0.010376	0.000000	0.300000	0.000000	0.066667	0.066667
2022-07-31	KLB	0.004721	0.145674	0.046620	0.145242	0.066667	0.066667
2022-08-31	VCB	0.279114	0.000000	0.300000	0.000000	0.066667	0.066667
2022-08-31	BID	0.151057	0.000000	0.000000	0.000000	0.066667	0.066667
2022-08-31	CTG	0.071282	0.000000	0.000000	0.000000	0.066667	0.066667
2022-08-31	MBB	0.056202	0.000000	0.000000	0.000000	0.066667	0.066667
2022-08-31	TCB	0.098500	0.000000	0.000000	0.000000	0.066667	0.066667
2022-08-31	ACB	0.058911	0.256132	0.242274	0.021312	0.066667	0.066667
2022-08-31	VPB	0.104496	0.000000	0.000000	0.300000	0.066667	0.066667
2022-08-31	HDB	0.027471	0.000000	0.000000	0.000000	0.066667	0.066667
2022-08-31	VIB	0.031703	0.300000	0.000000	0.200912	0.066667	0.066667
2022-08-31	TPB	0.026055	0.000000	0.000000	0.000000	0.066667	0.066667
2022-08-31	STB	0.037035	0.000000	0.000000	0.162023	0.066667	0.066667
2022-08-31	SHB	0.025319	0.292083	0.104553	0.162099	0.066667	0.066667
2022-08-31	LPB	0.018389	0.000000	0.000000	0.000000	0.066667	0.066667
2022-08-31	BAB	0.010180	0.000000	0.300000	0.000000	0.066667	0.066667
2022-08-31	KLB	0.004285	0.151785	0.053173	0.153654	0.066667	0.066667
2022-09-30	VCB	0.291132	0.000000	0.300000	0.000000	0.062500	0.062500
2022-09-30	BID	0.150266	0.000000	0.000000	0.000000	0.062500	0.062500
2022-09-30	CTG	0.068693	0.000000	0.000000	0.000000	0.062500	0.062500
2022-09-30	MBB	0.057938	0.000000	0.000000	0.000000	0.062500	0.062500

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-09-30	TCB	0.093918	0.000000	0.000000	0.000000	0.062500	0.062500
2022-09-30	ACB	0.054757	0.251442	0.201363	0.000000	0.062500	0.062500
2022-09-30	VPB	0.107403	0.000000	0.000000	0.300000	0.062500	0.062500
2022-09-30	HDB	0.028174	0.000000	0.000000	0.000000	0.062500	0.062500
2022-09-30	VIB	0.027626	0.300000	0.000000	0.227253	0.062500	0.062500
2022-09-30	TPB	0.025435	0.000000	0.000000	0.000000	0.062500	0.062500
2022-09-30	STB	0.034491	0.000000	0.000000	0.264046	0.062500	0.062500
2022-09-30	SHB	0.025015	0.300000	0.087826	0.126527	0.062500	0.062500
2022-09-30	LPB	0.017718	0.000000	0.000000	0.000000	0.062500	0.062500
2022-09-30	BAB	0.009386	0.000000	0.300000	0.000000	0.062500	0.062500
2022-09-30	KLB	0.004279	0.148558	0.026055	0.082174	0.062500	0.062500
2022-09-30	VBB	0.003771	0.000000	0.084756	0.000000	0.062500	0.062500
2022-10-31	VCB	0.296212	0.000000	0.300000	0.000000	0.062500	0.062500
2022-10-31	BID	0.148705	0.000000	0.000000	0.000000	0.062500	0.062500
2022-10-31	CTG	0.065861	0.000000	0.000000	0.000000	0.062500	0.062500
2022-10-31	MBB	0.056920	0.000000	0.000000	0.000000	0.062500	0.062500
2022-10-31	TCB	0.091516	0.000000	0.000000	0.000000	0.062500	0.062500
2022-10-31	ACB	0.057558	0.300000	0.185046	0.016515	0.062500	0.062500
2022-10-31	VPB	0.106464	0.000000	0.000000	0.300000	0.062500	0.062500
2022-10-31	HDB	0.029476	0.000000	0.000000	0.000000	0.062500	0.062500
2022-10-31	VIB	0.028379	0.004408	0.000000	0.243974	0.062500	0.062500
2022-10-31	TPB	0.026509	0.000000	0.000000	0.000000	0.062500	0.062500
2022-10-31	STB	0.033448	0.000000	0.000000	0.300000	0.062500	0.062500
2022-10-31	SHB	0.023287	0.300000	0.084268	0.097490	0.062500	0.062500
2022-10-31	LPB	0.016904	0.186458	0.000000	0.007992	0.062500	0.062500

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-10-31	BAB	0.010298	0.000000	0.300000	0.000000	0.062500	0.062500
2022-10-31	KLB	0.004798	0.209134	0.007534	0.034029	0.062500	0.062500
2022-10-31	VBB	0.003665	0.000000	0.123152	0.000000	0.062500	0.062500
2022-11-30	VCB	0.314370	0.000000	0.300000	0.000000	0.062500	0.062500
2022-11-30	BID	0.160176	0.000000	0.000000	0.000000	0.062500	0.062500
2022-11-30	CTG	0.073815	0.000000	0.000000	0.000000	0.062500	0.062500
2022-11-30	MBB	0.053273	0.000000	0.000000	0.000000	0.062500	0.062500
2022-11-30	TCB	0.073188	0.000000	0.000000	0.000000	0.062500	0.062500
2022-11-30	ACB	0.061396	0.300000	0.201554	0.000000	0.062500	0.062500
2022-11-30	VPB	0.102197	0.000000	0.000000	0.274387	0.062500	0.062500
2022-11-30	HDB	0.026681	0.000000	0.000000	0.000000	0.062500	0.062500
2022-11-30	VIB	0.026740	0.091443	0.000000	0.233421	0.062500	0.062500
2022-11-30	TPB	0.023209	0.000000	0.000000	0.000000	0.062500	0.062500
2022-11-30	STB	0.027716	0.000000	0.000000	0.300000	0.062500	0.062500
2022-11-30	SHB	0.022651	0.286587	0.060825	0.168784	0.062500	0.062500
2022-11-30	LPB	0.016262	0.021969	0.000000	0.000000	0.062500	0.062500
2022-11-30	BAB	0.009837	0.000000	0.300000	0.000000	0.062500	0.062500
2022-11-30	KLB	0.004615	0.300000	0.033258	0.023408	0.062500	0.062500
2022-11-30	VBB	0.003873	0.000000	0.104362	0.000000	0.062500	0.062500
2022-12-31	VCB	0.319787	0.000000	0.300000	0.000000	0.062500	0.062500
2022-12-31	BID	0.175901	0.000000	0.000000	0.000000	0.062500	0.062500
2022-12-31	CTG	0.075501	0.000000	0.000000	0.000000	0.062500	0.062500
2022-12-31	MBB	0.048360	0.000000	0.000000	0.000000	0.062500	0.062500
2022-12-31	TCB	0.071129	0.000000	0.000000	0.000000	0.062500	0.062500
2022-12-31	ACB	0.055946	0.300000	0.203962	0.000000	0.062500	0.062500

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2022-12-31	VPB	0.096948	0.000000	0.000000	0.230926	0.062500	0.062500
2022-12-31	HDB	0.023363	0.000000	0.000000	0.000000	0.062500	0.062500
2022-12-31	VIB	0.025487	0.100000	0.000000	0.241246	0.062500	0.062500
2022-12-31	TPB	0.022758	0.000000	0.000000	0.000000	0.062500	0.062500
2022-12-31	STB	0.031687	0.000000	0.000000	0.300000	0.062500	0.062500
2022-12-31	SHB	0.021219	0.300000	0.055329	0.201709	0.062500	0.062500
2022-12-31	LPB	0.016820	0.000000	0.000000	0.000000	0.062500	0.062500
2022-12-31	BAB	0.008824	0.000000	0.300000	0.000000	0.062500	0.062500
2022-12-31	KLB	0.002975	0.300000	0.023589	0.026119	0.062500	0.062500
2022-12-31	VBB	0.003294	0.000000	0.117120	0.000000	0.062500	0.062500
2023-01-31	VCB	0.317999	0.000000	0.300000	0.000000	0.062500	0.062500
2023-01-31	BID	0.166343	0.000000	0.000000	0.000000	0.062500	0.062500
2023-01-31	CTG	0.076014	0.000000	0.000000	0.000000	0.062500	0.062500
2023-01-31	MBB	0.047811	0.000000	0.000000	0.000000	0.062500	0.062500
2023-01-31	TCB	0.071492	0.000000	0.000000	0.000000	0.062500	0.062500
2023-01-31	ACB	0.055805	0.300000	0.207299	0.000000	0.062500	0.062500
2023-01-31	VPB	0.104050	0.000000	0.000000	0.276669	0.062500	0.062500
2023-01-31	HDB	0.024116	0.000000	0.000000	0.000000	0.062500	0.062500
2023-01-31	VIB	0.024097	0.202438	0.000000	0.231268	0.062500	0.062500
2023-01-31	TPB	0.021929	0.000000	0.000000	0.000000	0.062500	0.062500
2023-01-31	STB	0.035890	0.000000	0.000000	0.300000	0.062500	0.062500
2023-01-31	SHB	0.021019	0.300000	0.059810	0.150820	0.062500	0.062500
2023-01-31	LPB	0.019386	0.157028	0.000000	0.000000	0.062500	0.062500
2023-01-31	BAB	0.008560	0.000000	0.300000	0.000000	0.062500	0.062500
2023-01-31	KLB	0.002337	0.040534	0.022268	0.041243	0.062500	0.062500

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-01-31	VBB	0.003153	0.000000	0.110623	0.000000	0.062500	0.062500
2023-02-28	VCB	0.318169	0.000000	0.300000	0.000000	0.062500	0.062500
2023-02-28	BID	0.169254	0.000000	0.000000	0.000000	0.062500	0.062500
2023-02-28	CTG	0.074083	0.000000	0.000000	0.000000	0.062500	0.062500
2023-02-28	MBB	0.047624	0.000000	0.000000	0.000000	0.062500	0.062500
2023-02-28	TCB	0.070808	0.000000	0.000000	0.000000	0.062500	0.062500
2023-02-28	ACB	0.057804	0.300000	0.198868	0.000000	0.062500	0.062500
2023-02-28	VPB	0.098462	0.072847	0.000000	0.270243	0.062500	0.062500
2023-02-28	HDB	0.024575	0.000000	0.000000	0.000000	0.062500	0.062500
2023-02-28	VIB	0.025994	0.027153	0.000000	0.152907	0.062500	0.062500
2023-02-28	TPB	0.022682	0.000000	0.000000	0.000000	0.062500	0.062500
2023-02-28	STB	0.037650	0.000000	0.000000	0.300000	0.062500	0.062500
2023-02-28	SHB	0.020255	0.300000	0.059475	0.158188	0.062500	0.062500
2023-02-28	LPB	0.019065	0.300000	0.000000	0.118663	0.062500	0.062500
2023-02-28	BAB	0.007794	0.000000	0.300000	0.000000	0.062500	0.062500
2023-02-28	KLB	0.002051	0.000000	0.021094	0.000000	0.062500	0.062500
2023-02-28	VBB	0.003730	0.000000	0.120563	0.000000	0.062500	0.062500
2023-03-31	VCB	0.339910	0.000000	0.300000	0.000000	0.062500	0.062500
2023-03-31	BID	0.173758	0.000000	0.000000	0.000000	0.062500	0.062500
2023-03-31	CTG	0.069895	0.000000	0.000000	0.000000	0.062500	0.062500
2023-03-31	MBB	0.044104	0.000000	0.000000	0.000000	0.062500	0.062500
2023-03-31	TCB	0.067025	0.000000	0.000000	0.000000	0.062500	0.062500
2023-03-31	ACB	0.056841	0.300000	0.216046	0.000000	0.062500	0.062500
2023-03-31	VPB	0.090619	0.000000	0.000000	0.115572	0.062500	0.062500
2023-03-31	HDB	0.023577	0.000000	0.000000	0.000000	0.062500	0.062500

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-03-31	VIB	0.024898	0.138354	0.000000	0.039905	0.062500	0.062500
2023-03-31	TPB	0.022202	0.000000	0.000000	0.000000	0.062500	0.062500
2023-03-31	STB	0.034647	0.000000	0.000000	0.244523	0.062500	0.062500
2023-03-31	SHB	0.018940	0.261646	0.072202	0.300000	0.062500	0.062500
2023-03-31	LPB	0.018886	0.300000	0.000000	0.300000	0.062500	0.062500
2023-03-31	BAB	0.008244	0.000000	0.300000	0.000000	0.062500	0.062500
2023-03-31	KLB	0.001908	0.000000	0.058678	0.000000	0.062500	0.062500
2023-03-31	VBB	0.004547	0.000000	0.053074	0.000000	0.062500	0.062500
2023-04-30	VCB	0.317308	0.006709	0.300000	0.100000	0.062500	0.062500
2023-04-30	BID	0.173867	0.000000	0.000000	0.000000	0.062500	0.062500
2023-04-30	CTG	0.071110	0.000000	0.000000	0.000000	0.062500	0.062500
2023-04-30	MBB	0.044597	0.000000	0.000000	0.000000	0.062500	0.062500
2023-04-30	TCB	0.068495	0.000000	0.000000	0.000000	0.062500	0.062500
2023-04-30	ACB	0.055646	0.300000	0.208118	0.000000	0.062500	0.062500
2023-04-30	VPB	0.106821	0.000000	0.000000	0.000000	0.062500	0.062500
2023-04-30	HDB	0.025420	0.000000	0.000000	0.000000	0.062500	0.062500
2023-04-30	VIB	0.024766	0.300000	0.000000	0.300000	0.062500	0.062500
2023-04-30	TPB	0.022412	0.047374	0.000000	0.000000	0.062500	0.062500
2023-04-30	STB	0.036499	0.000000	0.000000	0.300000	0.062500	0.062500
2023-04-30	SHB	0.019859	0.045917	0.080770	0.000000	0.062500	0.062500
2023-04-30	LPB	0.019911	0.300000	0.000000	0.300000	0.062500	0.062500
2023-04-30	BAB	0.007759	0.000000	0.300000	0.000000	0.062500	0.062500
2023-04-30	KLB	0.001787	0.000000	0.052841	0.000000	0.062500	0.062500
2023-04-30	VBB	0.003741	0.000000	0.058272	0.000000	0.062500	0.062500
2023-05-31	VCB	0.320670	0.149606	0.300000	0.300000	0.062500	0.062500

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-05-31	BID	0.167057	0.000000	0.000000	0.070467	0.062500	0.062500
2023-05-31	CTG	0.071719	0.000000	0.000000	0.000000	0.062500	0.062500
2023-05-31	MBB	0.045973	0.000000	0.000000	0.000000	0.062500	0.062500
2023-05-31	TCB	0.072724	0.000000	0.000000	0.039860	0.062500	0.062500
2023-05-31	ACB	0.054968	0.300000	0.300000	0.079266	0.062500	0.062500
2023-05-31	VPB	0.103090	0.130830	0.000000	0.137849	0.062500	0.062500
2023-05-31	HDB	0.025333	0.068560	0.030592	0.000000	0.062500	0.062500
2023-05-31	VIB	0.024780	0.051004	0.000000	0.000000	0.062500	0.062500
2023-05-31	TPB	0.024526	0.000000	0.000000	0.000000	0.062500	0.062500
2023-05-31	STB	0.035969	0.000000	0.000000	0.217974	0.062500	0.062500
2023-05-31	SHB	0.021095	0.000000	0.000000	0.000000	0.062500	0.062500
2023-05-31	LPB	0.018473	0.300000	0.000000	0.154585	0.062500	0.062500
2023-05-31	BAB	0.007803	0.000000	0.300000	0.000000	0.062500	0.062500
2023-05-31	KLB	0.001928	0.000000	0.064249	0.000000	0.062500	0.062500
2023-05-31	VBB	0.003893	0.000000	0.005159	0.000000	0.062500	0.062500
2023-06-30	VCB	0.325965	0.139853	0.300000	0.300000	0.062500	0.062500
2023-06-30	BID	0.165383	0.000000	0.000000	0.114913	0.062500	0.062500
2023-06-30	CTG	0.068139	0.000000	0.000000	0.000000	0.062500	0.062500
2023-06-30	MBB	0.045404	0.000000	0.000000	0.000000	0.062500	0.062500
2023-06-30	TCB	0.072279	0.000000	0.000000	0.186756	0.062500	0.062500
2023-06-30	ACB	0.056457	0.300000	0.300000	0.054297	0.062500	0.062500
2023-06-30	VPB	0.097861	0.071718	0.000000	0.092406	0.062500	0.062500
2023-06-30	HDB	0.025599	0.011254	0.039531	0.000000	0.062500	0.062500
2023-06-30	VIB	0.025311	0.156098	0.000000	0.000000	0.062500	0.062500
2023-06-30	TPB	0.025216	0.104135	0.000000	0.000000	0.062500	0.062500

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-06-30	STB	0.038546	0.000000	0.000000	0.179911	0.062500	0.062500
2023-06-30	SHB	0.021668	0.000000	0.000000	0.000000	0.062500	0.062500
2023-06-30	LPB	0.018721	0.216942	0.000000	0.071716	0.062500	0.062500
2023-06-30	BAB	0.007864	0.000000	0.300000	0.000000	0.062500	0.062500
2023-06-30	KLB	0.001920	0.000000	0.060469	0.000000	0.062500	0.062500
2023-06-30	VBB	0.003668	0.000000	0.000000	0.000000	0.062500	0.062500
2023-07-31	VCB	0.331378	0.011376	0.300000	0.300000	0.062500	0.062500
2023-07-31	BID	0.155696	0.000000	0.000000	0.070822	0.062500	0.062500
2023-07-31	CTG	0.068591	0.000000	0.000000	0.000000	0.062500	0.062500
2023-07-31	MBB	0.048290	0.000000	0.000000	0.010162	0.062500	0.062500
2023-07-31	TCB	0.074593	0.000000	0.000000	0.000000	0.062500	0.062500
2023-07-31	ACB	0.056082	0.300000	0.300000	0.028434	0.062500	0.062500
2023-07-31	VPB	0.096136	0.018919	0.000000	0.142948	0.062500	0.062500
2023-07-31	HDB	0.024708	0.000000	0.071517	0.000000	0.062500	0.062500
2023-07-31	VIB	0.026643	0.243322	0.000000	0.000000	0.062500	0.062500
2023-07-31	TPB	0.024149	0.139942	0.000000	0.000000	0.062500	0.062500
2023-07-31	STB	0.039623	0.000000	0.000000	0.300000	0.062500	0.062500
2023-07-31	SHB	0.022023	0.000000	0.000000	0.000000	0.062500	0.062500
2023-07-31	LPB	0.018688	0.286441	0.000000	0.147634	0.062500	0.062500
2023-07-31	BAB	0.007568	0.000000	0.300000	0.000000	0.062500	0.062500
2023-07-31	KLB	0.002190	0.000000	0.028483	0.000000	0.062500	0.062500
2023-07-31	VBB	0.003641	0.000000	0.000000	0.000000	0.062500	0.062500
2023-08-31	VCB	0.334119	0.179641	0.300000	0.300000	0.062500	0.062500
2023-08-31	BID	0.157990	0.000000	0.000000	0.124361	0.062500	0.062500
2023-08-31	CTG	0.064987	0.000000	0.000000	0.000000	0.062500	0.062500

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-08-31	MBB	0.048297	0.000000	0.000000	0.016599	0.062500	0.062500
2023-08-31	TCB	0.073702	0.000000	0.000000	0.000000	0.062500	0.062500
2023-08-31	ACB	0.054397	0.300000	0.300000	0.035642	0.062500	0.062500
2023-08-31	VPB	0.099980	0.184233	0.000000	0.148268	0.062500	0.062500
2023-08-31	HDB	0.024712	0.000000	0.079869	0.000000	0.062500	0.062500
2023-08-31	VIB	0.026301	0.180189	0.000000	0.000000	0.062500	0.062500
2023-08-31	TPB	0.023446	0.027478	0.000000	0.000000	0.062500	0.062500
2023-08-31	STB	0.035877	0.000000	0.000000	0.273721	0.062500	0.062500
2023-08-31	SHB	0.024428	0.000000	0.000000	0.000000	0.062500	0.062500
2023-08-31	LPB	0.018793	0.128459	0.000000	0.101408	0.062500	0.062500
2023-08-31	BAB	0.007053	0.000000	0.300000	0.000000	0.062500	0.062500
2023-08-31	KLB	0.002055	0.000000	0.020131	0.000000	0.062500	0.062500
2023-08-31	VBB	0.003862	0.000000	0.000000	0.000000	0.062500	0.062500
2023-09-30	VCB	0.324600	0.300000	0.300000	0.300000	0.058824	0.058824
2023-09-30	BID	0.156965	0.000000	0.000000	0.122814	0.058824	0.058824
2023-09-30	CTG	0.070231	0.000000	0.000000	0.089049	0.058824	0.058824
2023-09-30	MBB	0.047316	0.000000	0.000000	0.034709	0.058824	0.058824
2023-09-30	TCB	0.074029	0.000000	0.000000	0.000000	0.058824	0.058824
2023-09-30	ACB	0.053487	0.300000	0.300000	0.000000	0.058824	0.058824
2023-09-30	VPB	0.094491	0.197311	0.000000	0.135710	0.058824	0.058824
2023-09-30	HDB	0.023968	0.000000	0.079231	0.000000	0.058824	0.058824
2023-09-30	VIB	0.025701	0.060356	0.000000	0.010518	0.058824	0.058824
2023-09-30	TPB	0.024475	0.000000	0.000000	0.000000	0.058824	0.058824
2023-09-30	STB	0.040412	0.000000	0.000000	0.195785	0.058824	0.058824
2023-09-30	SHB	0.024104	0.000000	0.000000	0.026847	0.058824	0.058824

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-09-30	LPB	0.023732	0.142333	0.000000	0.084569	0.058824	0.058824
2023-09-30	BAB	0.006994	0.000000	0.300000	0.000000	0.058824	0.058824
2023-09-30	KLB	0.002039	0.000000	0.020769	0.000000	0.058824	0.058824
2023-09-30	VBB	0.003609	0.000000	0.000000	0.000000	0.058824	0.058824
2023-09-30	BVB	0.003848	0.000000	0.000000	0.000000	0.058824	0.058824
2023-10-31	VCB	0.329213	0.265677	0.300000	0.300000	0.058824	0.058824
2023-10-31	BID	0.153150	0.000000	0.000000	0.113442	0.058824	0.058824
2023-10-31	CTG	0.067140	0.000000	0.000000	0.000000	0.058824	0.058824
2023-10-31	MBB	0.048864	0.000000	0.000000	0.064308	0.058824	0.058824
2023-10-31	TCB	0.074582	0.000000	0.000000	0.000000	0.058824	0.058824
2023-10-31	ACB	0.053391	0.300000	0.300000	0.000000	0.058824	0.058824
2023-10-31	VPB	0.101025	0.078970	0.000000	0.124412	0.058824	0.058824
2023-10-31	HDB	0.025489	0.000000	0.095045	0.000000	0.058824	0.058824
2023-10-31	VIB	0.025229	0.000000	0.000000	0.000000	0.058824	0.058824
2023-10-31	TPB	0.023018	0.055353	0.000000	0.000000	0.058824	0.058824
2023-10-31	STB	0.039178	0.000000	0.000000	0.286180	0.058824	0.058824
2023-10-31	SHB	0.022191	0.000000	0.000000	0.000000	0.058824	0.058824
2023-10-31	LPB	0.021146	0.300000	0.000000	0.111657	0.058824	0.058824
2023-10-31	BAB	0.007119	0.000000	0.300000	0.000000	0.058824	0.058824
2023-10-31	KLB	0.001934	0.000000	0.004955	0.000000	0.058824	0.058824
2023-10-31	VBB	0.003432	0.000000	0.000000	0.000000	0.058824	0.058824
2023-10-31	BVB	0.003898	0.000000	0.000000	0.000000	0.058824	0.058824
2023-11-30	VCB	0.347333	0.235729	0.300000	0.300000	0.058824	0.058824
2023-11-30	BID	0.147656	0.000000	0.000000	0.123162	0.058824	0.058824
2023-11-30	CTG	0.065836	0.000000	0.000000	0.000000	0.058824	0.058824

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-11-30	MBB	0.048057	0.000000	0.000000	0.174852	0.058824	0.058824
2023-11-30	TCB	0.065318	0.000000	0.000000	0.000000	0.058824	0.058824
2023-11-30	ACB	0.055641	0.300000	0.300000	0.000000	0.058824	0.058824
2023-11-30	VPB	0.099044	0.300000	0.000000	0.104564	0.058824	0.058824
2023-11-30	HDB	0.026884	0.000000	0.099727	0.000000	0.058824	0.058824
2023-11-30	VIB	0.024815	0.000000	0.000000	0.000000	0.058824	0.058824
2023-11-30	TPB	0.021663	0.000000	0.000000	0.000000	0.058824	0.058824
2023-11-30	STB	0.036708	0.000000	0.000000	0.222257	0.058824	0.058824
2023-11-30	SHB	0.021463	0.000000	0.000000	0.000000	0.058824	0.058824
2023-11-30	LPB	0.023784	0.164271	0.000000	0.075166	0.058824	0.058824
2023-11-30	BAB	0.006964	0.000000	0.300000	0.000000	0.058824	0.058824
2023-11-30	KLB	0.001812	0.000000	0.000273	0.000000	0.058824	0.058824
2023-11-30	VBB	0.002997	0.000000	0.000000	0.000000	0.058824	0.058824
2023-11-30	BVB	0.004025	0.000000	0.000000	0.000000	0.058824	0.058824
2023-12-31	VCB	0.327663	0.300000	0.300000	0.300000	0.052632	0.052632
2023-12-31	BID	0.155413	0.000000	0.000000	0.075720	0.052632	0.052632
2023-12-31	CTG	0.067536	0.000000	0.000000	0.000000	0.052632	0.052632
2023-12-31	MBB	0.048370	0.000000	0.000000	0.148242	0.052632	0.052632
2023-12-31	TCB	0.067229	0.000000	0.000000	0.000000	0.052632	0.052632
2023-12-31	ACB	0.055147	0.170580	0.300000	0.000000	0.052632	0.052632
2023-12-31	VPB	0.095964	0.288639	0.000000	0.112430	0.052632	0.052632
2023-12-31	HDB	0.027187	0.000000	0.053398	0.000000	0.052632	0.052632
2023-12-31	VIB	0.025068	0.000000	0.000000	0.000000	0.052632	0.052632
2023-12-31	TPB	0.022544	0.000000	0.000000	0.000000	0.052632	0.052632
2023-12-31	STB	0.036142	0.000000	0.000000	0.300000	0.052632	0.052632

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2023-12-31	SHB	0.022193	0.000000	0.000000	0.000000	0.052632	0.052632
2023-12-31	LPB	0.024012	0.240781	0.000000	0.063608	0.052632	0.052632
2023-12-31	NAB	0.007090	0.000000	0.000000	0.000000	0.052632	0.052632
2023-12-31	BAB	0.006786	0.000000	0.300000	0.000000	0.052632	0.052632
2023-12-31	KLB	0.001831	0.000000	0.000000	0.000000	0.052632	0.052632
2023-12-31	VBB	0.003195	0.000000	0.000000	0.000000	0.052632	0.052632
2023-12-31	SGB	0.002582	0.000000	0.046602	0.000000	0.052632	0.052632
2023-12-31	BVB	0.004047	0.000000	0.000000	0.000000	0.052632	0.052632
2024-01-31	VCB	0.303513	0.226881	0.300000	0.300000	0.052632	0.052632
2024-01-31	BID	0.169631	0.000000	0.000000	0.000000	0.052632	0.052632
2024-01-31	CTG	0.067995	0.000000	0.000000	0.000000	0.052632	0.052632
2024-01-31	MBB	0.049503	0.000000	0.000000	0.000000	0.052632	0.052632
2024-01-31	TCB	0.070785	0.000000	0.000000	0.000000	0.052632	0.052632
2024-01-31	ACB	0.058673	0.173119	0.300000	0.000000	0.052632	0.052632
2024-01-31	VPB	0.094245	0.300000	0.000000	0.055702	0.052632	0.052632
2024-01-31	HDB	0.029957	0.000000	0.087687	0.000000	0.052632	0.052632
2024-01-31	VIB	0.025669	0.000000	0.000000	0.044298	0.052632	0.052632
2024-01-31	TPB	0.022548	0.000000	0.000000	0.000000	0.052632	0.052632
2024-01-31	STB	0.035889	0.000000	0.000000	0.300000	0.052632	0.052632
2024-01-31	SHB	0.021683	0.000000	0.000000	0.000000	0.052632	0.052632
2024-01-31	LPB	0.024152	0.300000	0.000000	0.300000	0.052632	0.052632
2024-01-31	NAB	0.007629	0.000000	0.000000	0.000000	0.052632	0.052632
2024-01-31	BAB	0.006630	0.000000	0.300000	0.000000	0.052632	0.052632
2024-01-31	KLB	0.001771	0.000000	0.000000	0.000000	0.052632	0.052632
2024-01-31	VBB	0.003253	0.000000	0.000000	0.000000	0.052632	0.052632

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-01-31	SGB	0.002483	0.000000	0.012313	0.000000	0.052632	0.052632
2024-01-31	BVB	0.003992	0.000000	0.000000	0.000000	0.052632	0.052632
2024-02-29	VCB	0.299357	0.049873	0.300000	0.208966	0.045455	0.045455
2024-02-29	BID	0.166889	0.000000	0.000000	0.000000	0.045455	0.045455
2024-02-29	CTG	0.070939	0.000000	0.000000	0.000000	0.045455	0.045455
2024-02-29	MBB	0.051671	0.000000	0.000000	0.000000	0.045455	0.045455
2024-02-29	TCB	0.068823	0.000000	0.000000	0.000000	0.045455	0.045455
2024-02-29	ACB	0.056584	0.300000	0.300000	0.000000	0.045455	0.045455
2024-02-29	VPB	0.085221	0.058465	0.000000	0.000000	0.045455	0.045455
2024-02-29	HDB	0.028781	0.259594	0.065650	0.000000	0.045455	0.045455
2024-02-29	VIB	0.025026	0.000000	0.000000	0.180556	0.045455	0.045455
2024-02-29	TPB	0.020696	0.000000	0.000000	0.000000	0.045455	0.045455
2024-02-29	STB	0.034362	0.000000	0.000000	0.230236	0.045455	0.045455
2024-02-29	SHB	0.020844	0.000000	0.000000	0.000000	0.045455	0.045455
2024-02-29	LPB	0.024220	0.300000	0.000000	0.300000	0.045455	0.045455
2024-02-29	MSB	0.013821	0.000000	0.000000	0.000000	0.045455	0.045455
2024-02-29	NAB	0.007003	0.032068	0.000000	0.000000	0.045455	0.045455
2024-02-29	BAB	0.006278	0.000000	0.300000	0.000000	0.045455	0.045455
2024-02-29	KLB	0.001657	0.000000	0.034350	0.000000	0.045455	0.045455
2024-02-29	PGB	0.004209	0.000000	0.000000	0.080242	0.045455	0.045455
2024-02-29	VBB	0.003016	0.000000	0.000000	0.000000	0.045455	0.045455
2024-02-29	SGB	0.002328	0.000000	0.000000	0.000000	0.045455	0.045455
2024-02-29	ABB	0.004669	0.000000	0.000000	0.000000	0.045455	0.045455
2024-02-29	BVB	0.003607	0.000000	0.000000	0.000000	0.045455	0.045455
2024-03-31	VCB	0.295866	0.229590	0.300000	0.251285	0.043478	0.043478

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-03-31	BID	0.166670	0.000000	0.000000	0.022031	0.043478	0.043478
2024-03-31	CTG	0.071739	0.000000	0.000000	0.000000	0.043478	0.043478
2024-03-31	MBB	0.051561	0.000000	0.000000	0.000000	0.043478	0.043478
2024-03-31	TCB	0.075567	0.000000	0.000000	0.000000	0.043478	0.043478
2024-03-31	ACB	0.055308	0.300000	0.300000	0.000000	0.043478	0.043478
2024-03-31	VPB	0.078559	0.000000	0.000000	0.000000	0.043478	0.043478
2024-03-31	HDB	0.027659	0.081629	0.039021	0.000000	0.043478	0.043478
2024-03-31	VIB	0.023736	0.000000	0.000000	0.191889	0.043478	0.043478
2024-03-31	TPB	0.020687	0.000000	0.000000	0.000000	0.043478	0.043478
2024-03-31	STB	0.032592	0.000000	0.000000	0.096232	0.043478	0.043478
2024-03-31	SHB	0.019070	0.000000	0.000000	0.044425	0.043478	0.043478
2024-03-31	LPB	0.021100	0.255029	0.000000	0.300000	0.043478	0.043478
2024-03-31	MSB	0.014518	0.000000	0.000000	0.000000	0.043478	0.043478
2024-03-31	OCB	0.014984	0.000000	0.022387	0.094139	0.043478	0.043478
2024-03-31	NAB	0.006618	0.042630	0.000000	0.000000	0.043478	0.043478
2024-03-31	BAB	0.005691	0.000000	0.300000	0.000000	0.043478	0.043478
2024-03-31	KLB	0.001501	0.000000	0.033466	0.000000	0.043478	0.043478
2024-03-31	PGB	0.004091	0.091121	0.000000	0.000000	0.043478	0.043478
2024-03-31	VBB	0.002738	0.000000	0.000000	0.000000	0.043478	0.043478
2024-03-31	SGB	0.002107	0.000000	0.005125	0.000000	0.043478	0.043478
2024-03-31	ABB	0.004305	0.000000	0.000000	0.000000	0.043478	0.043478
2024-03-31	BVB	0.003334	0.000000	0.000000	0.000000	0.043478	0.043478
2024-04-30	VCB	0.287435	0.300000	0.300000	0.232106	0.043478	0.043478
2024-04-30	BID	0.163195	0.000000	0.000000	0.004626	0.043478	0.043478
2024-04-30	CTG	0.071464	0.000000	0.000000	0.000000	0.043478	0.043478

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-04-30	MBB	0.054028	0.000000	0.000000	0.000000	0.043478	0.043478
2024-04-30	TCB	0.084841	0.000000	0.000000	0.000000	0.043478	0.043478
2024-04-30	ACB	0.056078	0.300000	0.300000	0.000000	0.043478	0.043478
2024-04-30	VPB	0.077694	0.000000	0.089093	0.000000	0.043478	0.043478
2024-04-30	HDB	0.028491	0.083667	0.006275	0.000000	0.043478	0.043478
2024-04-30	VIB	0.026572	0.000000	0.000000	0.000000	0.043478	0.043478
2024-04-30	TPB	0.019829	0.000000	0.000000	0.000000	0.043478	0.043478
2024-04-30	STB	0.032518	0.000000	0.000000	0.213889	0.043478	0.043478
2024-04-30	SHB	0.018354	0.000000	0.000000	0.116784	0.043478	0.043478
2024-04-30	LPB	0.021573	0.120295	0.000000	0.300000	0.043478	0.043478
2024-04-30	MSB	0.013241	0.000000	0.000000	0.000000	0.043478	0.043478
2024-04-30	OCB	0.014725	0.000000	0.000000	0.096652	0.043478	0.043478
2024-04-30	NAB	0.006653	0.098804	0.000000	0.000000	0.043478	0.043478
2024-04-30	BAB	0.005669	0.000000	0.300000	0.000000	0.043478	0.043478
2024-04-30	KLB	0.001483	0.000000	0.004632	0.000000	0.043478	0.043478
2024-04-30	PGB	0.003821	0.097234	0.000000	0.035943	0.043478	0.043478
2024-04-30	VBB	0.002623	0.000000	0.000000	0.000000	0.043478	0.043478
2024-04-30	SGB	0.002241	0.000000	0.000000	0.000000	0.043478	0.043478
2024-04-30	ABB	0.004181	0.000000	0.000000	0.000000	0.043478	0.043478
2024-04-30	BVB	0.003291	0.000000	0.000000	0.000000	0.043478	0.043478
2024-05-31	VCB	0.281444	0.300000	0.300000	0.134114	0.041667	0.041667
2024-05-31	BID	0.157029	0.015931	0.000000	0.000000	0.041667	0.041667
2024-05-31	CTG	0.067202	0.000000	0.000000	0.000000	0.041667	0.041667
2024-05-31	MBB	0.048229	0.000000	0.000000	0.000000	0.041667	0.041667
2024-05-31	TCB	0.084718	0.000000	0.000000	0.000000	0.041667	0.041667

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-05-31	ACB	0.053825	0.300000	0.300000	0.000000	0.041667	0.041667
2024-05-31	VPB	0.074355	0.000000	0.016635	0.000000	0.041667	0.041667
2024-05-31	HDB	0.028445	0.230827	0.000000	0.000000	0.041667	0.041667
2024-05-31	VIB	0.023910	0.000000	0.000000	0.000000	0.041667	0.041667
2024-05-31	TPB	0.018941	0.000000	0.000000	0.000000	0.041667	0.041667
2024-05-31	STB	0.029569	0.000000	0.000000	0.300000	0.041667	0.041667
2024-05-31	SHB	0.018769	0.000000	0.000000	0.000000	0.041667	0.041667
2024-05-31	LPB	0.026169	0.000000	0.000000	0.300000	0.041667	0.041667
2024-05-31	SSB	0.030650	0.002775	0.083365	0.000000	0.041667	0.041667
2024-05-31	MSB	0.012610	0.000000	0.000000	0.000000	0.041667	0.041667
2024-05-31	OCB	0.013701	0.000000	0.000000	0.162133	0.041667	0.041667
2024-05-31	NAB	0.006840	0.149295	0.000000	0.000000	0.041667	0.041667
2024-05-31	BAB	0.005637	0.000000	0.300000	0.000000	0.041667	0.041667
2024-05-31	KLB	0.001422	0.000000	0.000000	0.000000	0.041667	0.041667
2024-05-31	PGB	0.003651	0.001171	0.000000	0.103753	0.041667	0.041667
2024-05-31	VBB	0.002966	0.000000	0.000000	0.000000	0.041667	0.041667
2024-05-31	SGB	0.002478	0.000000	0.000000	0.000000	0.041667	0.041667
2024-05-31	ABB	0.004053	0.000000	0.000000	0.000000	0.041667	0.041667
2024-05-31	BVB	0.003384	0.000000	0.000000	0.000000	0.041667	0.041667
2024-06-30	VCB	0.270118	0.198848	0.300000	0.300000	0.041667	0.041667
2024-06-30	BID	0.150904	0.163613	0.000000	0.000385	0.041667	0.041667
2024-06-30	CTG	0.065590	0.000000	0.000000	0.057238	0.041667	0.041667
2024-06-30	MBB	0.048372	0.000000	0.000000	0.000000	0.041667	0.041667
2024-06-30	TCB	0.088421	0.000000	0.000000	0.000000	0.041667	0.041667
2024-06-30	ACB	0.059070	0.106564	0.300000	0.300000	0.041667	0.041667

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-06-30	VPB	0.076268	0.000000	0.005910	0.000000	0.041667	0.041667
2024-06-30	HDB	0.028119	0.294089	0.000000	0.000000	0.041667	0.041667
2024-06-30	VIB	0.024730	0.000000	0.000000	0.000000	0.041667	0.041667
2024-06-30	TPB	0.018745	0.000000	0.000000	0.000000	0.041667	0.041667
2024-06-30	STB	0.029309	0.000000	0.000000	0.000000	0.041667	0.041667
2024-06-30	SHB	0.018770	0.000000	0.000000	0.000000	0.041667	0.041667
2024-06-30	LPB	0.032435	0.064773	0.000000	0.199014	0.041667	0.041667
2024-06-30	SSB	0.030064	0.114681	0.094090	0.000000	0.041667	0.041667
2024-06-30	MSB	0.013309	0.000000	0.000000	0.051533	0.041667	0.041667
2024-06-30	OCB	0.014852	0.000000	0.000000	0.091829	0.041667	0.041667
2024-06-30	NAB	0.006497	0.041063	0.000000	0.000000	0.041667	0.041667
2024-06-30	BAB	0.005610	0.000000	0.300000	0.000000	0.041667	0.041667
2024-06-30	KLB	0.001517	0.000000	0.000000	0.000000	0.041667	0.041667
2024-06-30	PGB	0.003564	0.016369	0.000000	0.000000	0.041667	0.041667
2024-06-30	VBB	0.002977	0.000000	0.000000	0.000000	0.041667	0.041667
2024-06-30	SGB	0.002326	0.000000	0.000000	0.000000	0.041667	0.041667
2024-06-30	ABB	0.004541	0.000000	0.000000	0.000000	0.041667	0.041667
2024-06-30	BVB	0.003888	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	VCB	0.268964	0.300000	0.096232	0.300000	0.041667	0.041667
2024-07-31	BID	0.141208	0.000000	0.000000	0.092936	0.041667	0.041667
2024-07-31	CTG	0.064951	0.000000	0.000000	0.285457	0.041667	0.041667
2024-07-31	MBB	0.050305	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	TCB	0.089551	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	ACB	0.058096	0.300000	0.241881	0.300000	0.041667	0.041667
2024-07-31	VPB	0.080525	0.000000	0.000000	0.000000	0.041667	0.041667

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-07-31	HDB	0.028357	0.100000	0.000000	0.000000	0.041667	0.041667
2024-07-31	VIB	0.024338	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	TPB	0.019134	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	STB	0.030889	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	SHB	0.019129	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	LPB	0.036115	0.300000	0.000000	0.000000	0.041667	0.041667
2024-07-31	SSB	0.029636	0.000000	0.264903	0.000000	0.041667	0.041667
2024-07-31	MSB	0.013564	0.000000	0.000000	0.021607	0.041667	0.041667
2024-07-31	OCB	0.014477	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	NAB	0.006645	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	BAB	0.005717	0.000000	0.300000	0.000000	0.041667	0.041667
2024-07-31	KLB	0.001559	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	PGB	0.003590	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	VBB	0.002790	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	SGB	0.002388	0.000000	0.096984	0.000000	0.041667	0.041667
2024-07-31	ABB	0.004358	0.000000	0.000000	0.000000	0.041667	0.041667
2024-07-31	BVB	0.003712	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	VCB	0.268548	0.000000	0.065913	0.000000	0.041667	0.041667
2024-08-31	BID	0.148183	0.155839	0.000000	0.300000	0.041667	0.041667
2024-08-31	CTG	0.063949	0.000000	0.000000	0.018409	0.041667	0.041667
2024-08-31	MBB	0.052739	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	TCB	0.085014	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	ACB	0.057157	0.297073	0.238980	0.000000	0.041667	0.041667
2024-08-31	VPB	0.078211	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	HDB	0.032184	0.016297	0.000000	0.000000	0.041667	0.041667

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-08-31	VIB	0.023436	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	TPB	0.019250	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	STB	0.029661	0.000000	0.000000	0.300000	0.041667	0.041667
2024-08-31	SHB	0.018617	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	LPB	0.035607	0.300000	0.000000	0.228046	0.041667	0.041667
2024-08-31	SSB	0.029620	0.000000	0.265745	0.000000	0.041667	0.041667
2024-08-31	MSB	0.013245	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	OCB	0.014194	0.000000	0.000000	0.153545	0.041667	0.041667
2024-08-31	NAB	0.007521	0.230790	0.000000	0.000000	0.041667	0.041667
2024-08-31	BAB	0.005364	0.000000	0.300000	0.000000	0.041667	0.041667
2024-08-31	KLB	0.001474	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	PGB	0.003250	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	VBB	0.002661	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	SGB	0.002261	0.000000	0.129363	0.000000	0.041667	0.041667
2024-08-31	ABB	0.004105	0.000000	0.000000	0.000000	0.041667	0.041667
2024-08-31	BVB	0.003749	0.000000	0.000000	0.000000	0.041667	0.041667
2024-09-30	VCB	0.268808	0.247946	0.164723	0.300000	0.040000	0.040000
2024-09-30	BID	0.148251	0.088752	0.000000	0.300000	0.040000	0.040000
2024-09-30	CTG	0.068195	0.000000	0.000000	0.300000	0.040000	0.040000
2024-09-30	MBB	0.052286	0.000000	0.000000	0.034558	0.040000	0.040000
2024-09-30	TCB	0.083150	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	ACB	0.056223	0.000000	0.133702	0.065442	0.040000	0.040000
2024-09-30	VPB	0.075943	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	HDB	0.033006	0.063970	0.000000	0.000000	0.040000	0.040000
2024-09-30	VIB	0.023102	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-09-30	TPB	0.018478	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	STB	0.030424	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	SHB	0.017161	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	LPB	0.037352	0.299332	0.000000	0.000000	0.040000	0.040000
2024-09-30	SSB	0.028855	0.000000	0.242411	0.000000	0.040000	0.040000
2024-09-30	MSB	0.013281	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	OCB	0.013612	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	NAB	0.007976	0.300000	0.000000	0.000000	0.040000	0.040000
2024-09-30	BAB	0.005182	0.000000	0.300000	0.000000	0.040000	0.040000
2024-09-30	KLB	0.001567	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	PGB	0.003106	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	VAB	0.001697	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	VBB	0.002516	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	SGB	0.002185	0.000000	0.159164	0.000000	0.040000	0.040000
2024-09-30	ABB	0.003948	0.000000	0.000000	0.000000	0.040000	0.040000
2024-09-30	BVB	0.003696	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	VCB	0.262621	0.082972	0.147731	0.300000	0.040000	0.040000
2024-10-31	BID	0.146713	0.181170	0.000000	0.300000	0.040000	0.040000
2024-10-31	CTG	0.069998	0.000000	0.000000	0.300000	0.040000	0.040000
2024-10-31	MBB	0.052642	0.000000	0.000000	0.093419	0.040000	0.040000
2024-10-31	TCB	0.083914	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	ACB	0.056836	0.000000	0.159565	0.006581	0.040000	0.040000
2024-10-31	VPB	0.078442	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	HDB	0.032718	0.179389	0.000000	0.000000	0.040000	0.040000
2024-10-31	VIB	0.023662	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-10-31	TPB	0.020753	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	STB	0.032342	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	SHB	0.017422	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	LPB	0.037003	0.300000	0.000000	0.000000	0.040000	0.040000
2024-10-31	SSB	0.024952	0.000000	0.238152	0.000000	0.040000	0.040000
2024-10-31	MSB	0.014538	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	OCB	0.013483	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	NAB	0.007858	0.256468	0.000000	0.000000	0.040000	0.040000
2024-10-31	BAB	0.005130	0.000000	0.300000	0.000000	0.040000	0.040000
2024-10-31	KLB	0.001586	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	PGB	0.003284	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	VAB	0.001783	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	VBB	0.002545	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	SGB	0.002144	0.000000	0.154552	0.000000	0.040000	0.040000
2024-10-31	ABB	0.003940	0.000000	0.000000	0.000000	0.040000	0.040000
2024-10-31	BVB	0.003692	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	VCB	0.269394	0.176749	0.145661	0.300000	0.040000	0.040000
2024-11-30	BID	0.142140	0.015066	0.000000	0.300000	0.040000	0.040000
2024-11-30	CTG	0.068212	0.000000	0.000000	0.300000	0.040000	0.040000
2024-11-30	MBB	0.051749	0.000000	0.000000	0.100000	0.040000	0.040000
2024-11-30	TCB	0.083547	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	ACB	0.056536	0.172242	0.169369	0.000000	0.040000	0.040000
2024-11-30	VPB	0.080492	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	HDB	0.031460	0.193681	0.000000	0.000000	0.040000	0.040000
2024-11-30	VIB	0.023425	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-11-30	TPB	0.021048	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	STB	0.034419	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	SHB	0.016934	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	LPB	0.038472	0.300000	0.000000	0.000000	0.040000	0.040000
2024-11-30	SSB	0.024422	0.000000	0.227986	0.000000	0.040000	0.040000
2024-11-30	MSB	0.013877	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	OCB	0.012866	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	NAB	0.007596	0.142262	0.000000	0.000000	0.040000	0.040000
2024-11-30	BAB	0.005173	0.000000	0.300000	0.000000	0.040000	0.040000
2024-11-30	KLB	0.001432	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	PGB	0.003032	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	VAB	0.001722	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	VBB	0.002518	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	SGB	0.002096	0.000000	0.156984	0.000000	0.040000	0.040000
2024-11-30	ABB	0.003779	0.000000	0.000000	0.000000	0.040000	0.040000
2024-11-30	BVB	0.003658	0.000000	0.000000	0.000000	0.040000	0.040000
2024-12-31	VCB	0.274011	0.300000	0.134575	0.300000	0.040000	0.040000
2024-12-31	BID	0.141995	0.000000	0.000000	0.300000	0.040000	0.040000
2024-12-31	CTG	0.069690	0.000000	0.000000	0.197388	0.040000	0.040000
2024-12-31	MBB	0.050901	0.000000	0.000000	0.126267	0.040000	0.040000
2024-12-31	TCB	0.084207	0.000000	0.000000	0.000000	0.040000	0.040000
2024-12-31	ACB	0.057228	0.103517	0.173917	0.000000	0.040000	0.040000
2024-12-31	VPB	0.076929	0.000000	0.000000	0.000000	0.040000	0.040000
2024-12-31	HDB	0.030264	0.278888	0.000000	0.000000	0.040000	0.040000
2024-12-31	VIB	0.023967	0.000000	0.000000	0.019493	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2024-12-31	TPB	0.020108	0.000000	0.000000	0.006919	0.040000	0.040000
2024-12-31	STB	0.033226	0.000000	0.000000	0.049933	0.040000	0.040000
2024-12-31	SHB	0.016785	0.000000	0.000000	0.000000	0.040000	0.040000
2024-12-31	LPB	0.038436	0.300000	0.000000	0.000000	0.040000	0.040000
2024-12-31	SSB	0.025598	0.000000	0.241182	0.000000	0.040000	0.040000
2024-12-31	MSB	0.013183	0.000000	0.000000	0.000000	0.040000	0.040000
2024-12-31	OCB	0.012320	0.000000	0.000000	0.000000	0.040000	0.040000
2024-12-31	NAB	0.007628	0.017595	0.000000	0.000000	0.040000	0.040000
2024-12-31	BAB	0.005101	0.000000	0.300000	0.000000	0.040000	0.040000
2024-12-31	KLB	0.001450	0.000000	0.000000	0.000000	0.040000	0.040000
2024-12-31	PGB	0.003035	0.000000	0.000000	0.000000	0.040000	0.040000
2024-12-31	VAB	0.001720	0.000000	0.000000	0.000000	0.040000	0.040000
2024-12-31	VBB	0.002925	0.000000	0.000000	0.000000	0.040000	0.040000
2024-12-31	SGB	0.002037	0.000000	0.150325	0.000000	0.040000	0.040000
2024-12-31	ABB	0.003654	0.000000	0.000000	0.000000	0.040000	0.040000
2024-12-31	BVB	0.003605	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	VCB	0.263005	0.300000	0.124959	0.300000	0.040000	0.040000
2025-01-31	BID	0.135537	0.000000	0.000000	0.300000	0.040000	0.040000
2025-01-31	CTG	0.072372	0.000000	0.000000	0.250901	0.040000	0.040000
2025-01-31	MBB	0.051952	0.000000	0.000000	0.029047	0.040000	0.040000
2025-01-31	TCB	0.086365	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	ACB	0.057561	0.300000	0.150359	0.000000	0.040000	0.040000
2025-01-31	VPB	0.075711	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	HDB	0.035868	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	VIB	0.024403	0.000000	0.000000	0.094147	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-01-31	TPB	0.020295	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	STB	0.036156	0.000000	0.000000	0.025905	0.040000	0.040000
2025-01-31	SHB	0.016398	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	LPB	0.042713	0.300000	0.000000	0.000000	0.040000	0.040000
2025-01-31	SSB	0.024768	0.000000	0.260570	0.000000	0.040000	0.040000
2025-01-31	MSB	0.013122	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	OCB	0.012829	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	NAB	0.007583	0.100000	0.000000	0.000000	0.040000	0.040000
2025-01-31	BAB	0.005139	0.000000	0.300000	0.000000	0.040000	0.040000
2025-01-31	KLB	0.001352	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	PGB	0.002869	0.000000	0.004336	0.000000	0.040000	0.040000
2025-01-31	VAB	0.001746	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	VBB	0.002872	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	SGB	0.002083	0.000000	0.159777	0.000000	0.040000	0.040000
2025-01-31	ABB	0.003637	0.000000	0.000000	0.000000	0.040000	0.040000
2025-01-31	BVB	0.003666	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	VCB	0.262105	0.300000	0.122524	0.099708	0.040000	0.040000
2025-02-28	BID	0.141699	0.000000	0.000000	0.300000	0.040000	0.040000
2025-02-28	CTG	0.071390	0.000000	0.000000	0.300000	0.040000	0.040000
2025-02-28	MBB	0.052330	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	TCB	0.085092	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	ACB	0.055500	0.156307	0.140167	0.000000	0.040000	0.040000
2025-02-28	VPB	0.073177	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	HDB	0.031283	0.186809	0.000000	0.024247	0.040000	0.040000
2025-02-28	VIB	0.025107	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-02-28	TPB	0.019677	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	STB	0.035584	0.000000	0.000000	0.260447	0.040000	0.040000
2025-02-28	SHB	0.016178	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	LPB	0.048095	0.300000	0.000000	0.015598	0.040000	0.040000
2025-02-28	SSB	0.027213	0.000000	0.295831	0.000000	0.040000	0.040000
2025-02-28	MSB	0.012322	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	OCB	0.012378	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	NAB	0.007891	0.056883	0.000000	0.000000	0.040000	0.040000
2025-02-28	BAB	0.005044	0.000000	0.300000	0.000000	0.040000	0.040000
2025-02-28	KLB	0.001327	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	PGB	0.002778	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	VAB	0.001658	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	VBB	0.002728	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	SGB	0.002141	0.000000	0.141478	0.000000	0.040000	0.040000
2025-02-28	ABB	0.003427	0.000000	0.000000	0.000000	0.040000	0.040000
2025-02-28	BVB	0.003876	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	VCB	0.256873	0.100000	0.115961	0.028643	0.040000	0.040000
2025-03-31	BID	0.141092	0.000000	0.000000	0.300000	0.040000	0.040000
2025-03-31	CTG	0.075829	0.000000	0.000000	0.300000	0.040000	0.040000
2025-03-31	MBB	0.052259	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	TCB	0.087283	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	ACB	0.055356	0.300000	0.179171	0.000000	0.040000	0.040000
2025-03-31	VPB	0.072668	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	HDB	0.030948	0.000000	0.000000	0.039519	0.040000	0.040000
2025-03-31	VIB	0.024537	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-03-31	TPB	0.019440	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	STB	0.035778	0.000000	0.000000	0.300000	0.040000	0.040000
2025-03-31	SHB	0.016945	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	LPB	0.047163	0.300000	0.000000	0.019481	0.040000	0.040000
2025-03-31	SSB	0.027385	0.000000	0.268444	0.012357	0.040000	0.040000
2025-03-31	MSB	0.012204	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	OCB	0.012088	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	NAB	0.008036	0.300000	0.000000	0.000000	0.040000	0.040000
2025-03-31	BAB	0.005205	0.000000	0.300000	0.000000	0.040000	0.040000
2025-03-31	KLB	0.001405	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	PGB	0.002791	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	VAB	0.001843	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	VBB	0.002802	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	SGB	0.002068	0.000000	0.136424	0.000000	0.040000	0.040000
2025-03-31	ABB	0.003569	0.000000	0.000000	0.000000	0.040000	0.040000
2025-03-31	BVB	0.004436	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	VCB	0.263224	0.220671	0.134200	0.009407	0.040000	0.040000
2025-04-30	BID	0.133396	0.000000	0.000000	0.300000	0.040000	0.040000
2025-04-30	CTG	0.075692	0.000000	0.000000	0.238153	0.040000	0.040000
2025-04-30	MBB	0.054730	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	TCB	0.091910	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	ACB	0.055314	0.300000	0.159560	0.000000	0.040000	0.040000
2025-04-30	VPB	0.071473	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	HDB	0.029655	0.068048	0.000000	0.122149	0.040000	0.040000
2025-04-30	VIB	0.023455	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-04-30	TPB	0.016635	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	STB	0.035798	0.000000	0.000000	0.300000	0.040000	0.040000
2025-04-30	SHB	0.021085	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	LPB	0.043351	0.300000	0.000000	0.030292	0.040000	0.040000
2025-04-30	SSB	0.027576	0.000000	0.278663	0.000000	0.040000	0.040000
2025-04-30	MSB	0.013110	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	OCB	0.011968	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	NAB	0.007553	0.111280	0.000000	0.000000	0.040000	0.040000
2025-04-30	BAB	0.005201	0.000000	0.300000	0.000000	0.040000	0.040000
2025-04-30	KLB	0.001324	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	PGB	0.002843	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	VAB	0.001858	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	VBB	0.002949	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	SGB	0.002081	0.000000	0.127577	0.000000	0.040000	0.040000
2025-04-30	ABB	0.003659	0.000000	0.000000	0.000000	0.040000	0.040000
2025-04-30	BVB	0.004161	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	VCB	0.255298	0.300000	0.126505	0.000000	0.040000	0.040000
2025-05-31	BID	0.128853	0.000000	0.000000	0.300000	0.040000	0.040000
2025-05-31	CTG	0.073572	0.000000	0.000000	0.202992	0.040000	0.040000
2025-05-31	MBB	0.057943	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	TCB	0.095199	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	ACB	0.055423	0.300000	0.158165	0.000000	0.040000	0.040000
2025-05-31	VPB	0.067453	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	HDB	0.031181	0.091731	0.000000	0.150951	0.040000	0.040000
2025-05-31	VIB	0.023088	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-05-31	TPB	0.017126	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	STB	0.039588	0.000000	0.000000	0.290486	0.040000	0.040000
2025-05-31	SHB	0.023124	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	LPB	0.045965	0.300000	0.000000	0.055570	0.040000	0.040000
2025-05-31	SSB	0.028114	0.000000	0.288013	0.000000	0.040000	0.040000
2025-05-31	MSB	0.013028	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	OCB	0.012091	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	NAB	0.008131	0.008269	0.000000	0.000000	0.040000	0.040000
2025-05-31	BAB	0.005300	0.000000	0.300000	0.000000	0.040000	0.040000
2025-05-31	KLB	0.001495	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	PGB	0.002855	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	VAB	0.002111	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	VBB	0.003098	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	SGB	0.002204	0.000000	0.127317	0.000000	0.040000	0.040000
2025-05-31	ABB	0.003758	0.000000	0.000000	0.000000	0.040000	0.040000
2025-05-31	BVB	0.004001	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	VCB	0.241100	0.057003	0.089438	0.080552	0.040000	0.040000
2025-06-30	BID	0.128818	0.000000	0.000000	0.300000	0.040000	0.040000
2025-06-30	CTG	0.072701	0.138049	0.000000	0.210065	0.040000	0.040000
2025-06-30	MBB	0.057487	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	TCB	0.105768	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	ACB	0.055856	0.000000	0.133803	0.000000	0.040000	0.040000
2025-06-30	VPB	0.072126	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	HDB	0.030264	0.211354	0.000000	0.062522	0.040000	0.040000
2025-06-30	VIB	0.022861	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-06-30	TPB	0.017183	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	STB	0.039542	0.000000	0.000000	0.251575	0.040000	0.040000
2025-06-30	SHB	0.023865	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	LPB	0.046262	0.300000	0.000000	0.027612	0.040000	0.040000
2025-06-30	SSB	0.026978	0.000000	0.281121	0.067674	0.040000	0.040000
2025-06-30	MSB	0.012890	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	OCB	0.011831	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	NAB	0.007901	0.293593	0.000000	0.000000	0.040000	0.040000
2025-06-30	BAB	0.005268	0.000000	0.300000	0.000000	0.040000	0.040000
2025-06-30	KLB	0.002142	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	PGB	0.002858	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	VAB	0.002613	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	VBB	0.003528	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	SGB	0.002099	0.000000	0.195638	0.000000	0.040000	0.040000
2025-06-30	ABB	0.003899	0.000000	0.000000	0.000000	0.040000	0.040000
2025-06-30	BVB	0.004159	0.000000	0.000000	0.000000	0.040000	0.040000
2025-07-31	VCB	0.234626	0.078296	0.081971	0.000000	0.040000	0.040000
2025-07-31	BID	0.125075	0.000000	0.000000	0.097761	0.040000	0.040000
2025-07-31	CTG	0.076580	0.000000	0.000000	0.119586	0.040000	0.040000
2025-07-31	MBB	0.058651	0.000000	0.000000	0.123618	0.040000	0.040000
2025-07-31	TCB	0.114387	0.010777	0.000000	0.129430	0.040000	0.040000
2025-07-31	ACB	0.054289	0.300000	0.113315	0.000000	0.040000	0.040000
2025-07-31	VPB	0.071571	0.000000	0.000000	0.000000	0.040000	0.040000
2025-07-31	HDB	0.029278	0.007236	0.000000	0.019546	0.040000	0.040000
2025-07-31	VIB	0.022499	0.000000	0.000000	0.058023	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-07-31	TPB	0.016793	0.000000	0.000000	0.000000	0.040000	0.040000
2025-07-31	STB	0.043685	0.000000	0.000000	0.186244	0.040000	0.040000
2025-07-31	SHB	0.022696	0.000000	0.000000	0.057919	0.040000	0.040000
2025-07-31	LPB	0.045180	0.300000	0.000000	0.103388	0.040000	0.040000
2025-07-31	SSB	0.025622	0.000000	0.288162	0.000000	0.040000	0.040000
2025-07-31	MSB	0.012901	0.000000	0.000000	0.000000	0.040000	0.040000
2025-07-31	OCB	0.012578	0.000000	0.000883	0.000000	0.040000	0.040000
2025-07-31	NAB	0.007607	0.300000	0.000000	0.093129	0.040000	0.040000
2025-07-31	BAB	0.005334	0.000000	0.300000	0.000000	0.040000	0.040000
2025-07-31	KLB	0.001959	0.000000	0.000000	0.000000	0.040000	0.040000
2025-07-31	PGB	0.002637	0.000000	0.000000	0.000000	0.040000	0.040000
2025-07-31	VAB	0.002693	0.003692	0.000000	0.000000	0.040000	0.040000
2025-07-31	VBB	0.003497	0.000000	0.000000	0.011356	0.040000	0.040000
2025-07-31	SGB	0.002006	0.000000	0.215669	0.000000	0.040000	0.040000
2025-07-31	ABB	0.003944	0.000000	0.000000	0.000000	0.040000	0.040000
2025-07-31	BVB	0.003912	0.000000	0.000000	0.000000	0.040000	0.040000
2025-08-31	VCB	0.218100	0.135595	0.087518	0.000000	0.040000	0.040000
2025-08-31	BID	0.114771	0.000000	0.000000	0.127075	0.040000	0.040000
2025-08-31	CTG	0.073824	0.070569	0.000000	0.114382	0.040000	0.040000
2025-08-31	MBB	0.072464	0.000000	0.000000	0.128013	0.040000	0.040000
2025-08-31	TCB	0.100086	0.000000	0.000000	0.127249	0.040000	0.040000
2025-08-31	ACB	0.051594	0.057543	0.118684	0.000000	0.040000	0.040000
2025-08-31	VPB	0.087833	0.000000	0.000000	0.000000	0.040000	0.040000
2025-08-31	HDB	0.031271	0.000000	0.000000	0.034696	0.040000	0.040000
2025-08-31	VIB	0.027338	0.000000	0.000000	0.044357	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-08-31	TPB	0.017249	0.000000	0.000000	0.000000	0.040000	0.040000
2025-08-31	STB	0.041123	0.000000	0.000000	0.163449	0.040000	0.040000
2025-08-31	SHB	0.030095	0.000000	0.000000	0.074948	0.040000	0.040000
2025-08-31	LPB	0.042164	0.300000	0.000000	0.096160	0.040000	0.040000
2025-08-31	SSB	0.024227	0.000000	0.272315	0.000000	0.040000	0.040000
2025-08-31	MSB	0.013058	0.000000	0.000000	0.000000	0.040000	0.040000
2025-08-31	OCB	0.014165	0.000000	0.000964	0.000000	0.040000	0.040000
2025-08-31	NAB	0.009426	0.300000	0.000000	0.084031	0.040000	0.040000
2025-08-31	BAB	0.005700	0.000000	0.300000	0.000000	0.040000	0.040000
2025-08-31	KLB	0.003656	0.000000	0.000000	0.000000	0.040000	0.040000
2025-08-31	PGB	0.002673	0.000000	0.000000	0.000000	0.040000	0.040000
2025-08-31	VAB	0.003961	0.136293	0.000000	0.000000	0.040000	0.040000
2025-08-31	VBB	0.003991	0.000000	0.000000	0.005638	0.040000	0.040000
2025-08-31	SGB	0.001848	0.000000	0.220520	0.000000	0.040000	0.040000
2025-08-31	ABB	0.005411	0.000000	0.000000	0.000000	0.040000	0.040000
2025-08-31	BVB	0.003974	0.000000	0.000000	0.000000	0.040000	0.040000
2025-09-30	VCB	0.208596	0.141576	0.108528	0.212445	0.040000	0.040000
2025-09-30	BID	0.109067	0.000000	0.000000	0.300000	0.040000	0.040000
2025-09-30	CTG	0.069264	0.063901	0.000000	0.000000	0.040000	0.040000
2025-09-30	MBB	0.081930	0.000000	0.000000	0.056813	0.040000	0.040000
2025-09-30	TCB	0.097831	0.000000	0.000000	0.010921	0.040000	0.040000
2025-09-30	ACB	0.052341	0.008688	0.105533	0.000000	0.040000	0.040000
2025-09-30	VPB	0.100008	0.000000	0.000000	0.000000	0.040000	0.040000
2025-09-30	HDB	0.033181	0.053700	0.000000	0.026250	0.040000	0.040000
2025-09-30	VIB	0.026738	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-09-30	TPB	0.019280	0.000000	0.000000	0.000000	0.040000	0.040000
2025-09-30	STB	0.038419	0.000000	0.000000	0.281833	0.040000	0.040000
2025-09-30	SHB	0.031203	0.000000	0.000000	0.000000	0.040000	0.040000
2025-09-30	LPB	0.046557	0.300000	0.000000	0.088776	0.040000	0.040000
2025-09-30	SSB	0.022316	0.000000	0.253360	0.000000	0.040000	0.040000
2025-09-30	MSB	0.014057	0.000000	0.000000	0.000000	0.040000	0.040000
2025-09-30	OCB	0.013324	0.000000	0.000000	0.022963	0.040000	0.040000
2025-09-30	NAB	0.008408	0.300000	0.000000	0.000000	0.040000	0.040000
2025-09-30	BAB	0.004990	0.000000	0.300000	0.000000	0.040000	0.040000
2025-09-30	KLB	0.002975	0.000000	0.000000	0.000000	0.040000	0.040000
2025-09-30	PGB	0.002382	0.000000	0.000000	0.000000	0.040000	0.040000
2025-09-30	VAB	0.003845	0.132136	0.000000	0.000000	0.040000	0.040000
2025-09-30	VBB	0.003494	0.000000	0.000000	0.000000	0.040000	0.040000
2025-09-30	SGB	0.001668	0.000000	0.232579	0.000000	0.040000	0.040000
2025-09-30	ABB	0.004370	0.000000	0.000000	0.000000	0.040000	0.040000
2025-09-30	BVB	0.003758	0.000000	0.000000	0.000000	0.040000	0.040000
2025-10-31	VCB	0.200932	0.000000	0.174164	0.141446	0.040000	0.040000
2025-10-31	BID	0.108522	0.000000	0.000000	0.259123	0.040000	0.040000
2025-10-31	CTG	0.073095	0.074087	0.000000	0.000000	0.040000	0.040000
2025-10-31	MBB	0.082453	0.000000	0.000000	0.253206	0.040000	0.040000
2025-10-31	TCB	0.102321	0.000000	0.000000	0.000000	0.040000	0.040000
2025-10-31	ACB	0.051175	0.208478	0.000000	0.000000	0.040000	0.040000
2025-10-31	VPB	0.094728	0.000000	0.000000	0.062297	0.040000	0.040000
2025-10-31	HDB	0.032776	0.000000	0.000000	0.000000	0.040000	0.040000
2025-10-31	VIB	0.025096	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-10-31	TPB	0.018528	0.000000	0.000000	0.000000	0.040000	0.040000
2025-10-31	STB	0.041909	0.000000	0.000000	0.154836	0.040000	0.040000
2025-10-31	SHB	0.029904	0.000000	0.000000	0.000000	0.040000	0.040000
2025-10-31	LPB	0.054598	0.300000	0.000000	0.121324	0.040000	0.040000
2025-10-31	SSB	0.021453	0.000000	0.258903	0.000000	0.040000	0.040000
2025-10-31	MSB	0.013358	0.000000	0.000000	0.007768	0.040000	0.040000
2025-10-31	OCB	0.013162	0.000000	0.000000	0.000000	0.040000	0.040000
2025-10-31	NAB	0.008211	0.300000	0.000000	0.000000	0.040000	0.040000
2025-10-31	BAB	0.004950	0.000000	0.300000	0.000000	0.040000	0.040000
2025-10-31	KLB	0.003571	0.000000	0.000000	0.000000	0.040000	0.040000
2025-10-31	PGB	0.002346	0.000000	0.000000	0.000000	0.040000	0.040000
2025-10-31	VAB	0.003700	0.117435	0.000000	0.000000	0.040000	0.040000
2025-10-31	VBB	0.003509	0.000000	0.000000	0.000000	0.040000	0.040000
2025-10-31	SGB	0.001628	0.000000	0.266933	0.000000	0.040000	0.040000
2025-10-31	ABB	0.004545	0.000000	0.000000	0.000000	0.040000	0.040000
2025-10-31	BVB	0.003530	0.000000	0.000000	0.000000	0.040000	0.040000
2025-11-30	VCB	0.192849	0.000000	0.210785	0.000000	0.040000	0.040000
2025-11-30	BID	0.101691	0.000000	0.000000	0.000000	0.040000	0.040000
2025-11-30	CTG	0.101902	0.298112	0.000000	0.002807	0.040000	0.040000
2025-11-30	MBB	0.073615	0.000000	0.000000	0.185508	0.040000	0.040000
2025-11-30	TCB	0.094343	0.000000	0.000000	0.111332	0.040000	0.040000
2025-11-30	ACB	0.050724	0.000000	0.000000	0.000000	0.040000	0.040000
2025-11-30	VPB	0.086642	0.000000	0.000000	0.003349	0.040000	0.040000
2025-11-30	HDB	0.047837	0.000000	0.000000	0.156585	0.040000	0.040000
2025-11-30	VIB	0.023187	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-11-30	TPB	0.017886	0.000000	0.000000	0.058453	0.040000	0.040000
2025-11-30	STB	0.040518	0.000000	0.000000	0.279499	0.040000	0.040000
2025-11-30	SHB	0.028839	0.000395	0.000000	0.049298	0.040000	0.040000
2025-11-30	LPB	0.055423	0.300000	0.056895	0.073032	0.040000	0.040000
2025-11-30	SSB	0.019500	0.000000	0.217299	0.000000	0.040000	0.040000
2025-11-30	MSB	0.014559	0.000000	0.000000	0.048843	0.040000	0.040000
2025-11-30	OCB	0.013149	0.000000	0.000000	0.009363	0.040000	0.040000
2025-11-30	NAB	0.007886	0.288984	0.000000	0.000000	0.040000	0.040000
2025-11-30	BAB	0.004798	0.000000	0.300000	0.000000	0.040000	0.040000
2025-11-30	KLB	0.003765	0.000000	0.026435	0.021931	0.040000	0.040000
2025-11-30	PGB	0.002790	0.000000	0.016447	0.000000	0.040000	0.040000
2025-11-30	VAB	0.003588	0.021532	0.000000	0.000000	0.040000	0.040000
2025-11-30	VBB	0.004721	0.090977	0.000000	0.000000	0.040000	0.040000
2025-11-30	SGB	0.001650	0.000000	0.172138	0.000000	0.040000	0.040000
2025-11-30	ABB	0.004762	0.000000	0.000000	0.000000	0.040000	0.040000
2025-11-30	BVB	0.003375	0.000000	0.000000	0.000000	0.040000	0.040000
2025-12-31	VCB	0.189498	0.000000	0.176834	0.000000	0.040000	0.040000
2025-12-31	BID	0.102922	0.000000	0.000000	0.000000	0.040000	0.040000
2025-12-31	CTG	0.103969	0.000000	0.000000	0.000000	0.040000	0.040000
2025-12-31	MBB	0.073995	0.000000	0.000000	0.267803	0.040000	0.040000
2025-12-31	TCB	0.092534	0.170506	0.000000	0.071180	0.040000	0.040000
2025-12-31	ACB	0.049216	0.000000	0.000000	0.000000	0.040000	0.040000
2025-12-31	VPB	0.090092	0.000000	0.000000	0.000000	0.040000	0.040000
2025-12-31	HDB	0.048807	0.017732	0.000000	0.149644	0.040000	0.040000
2025-12-31	VIB	0.023590	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2025-12-31	TPB	0.018961	0.000000	0.000000	0.027651	0.040000	0.040000
2025-12-31	STB	0.036200	0.143214	0.051154	0.224683	0.040000	0.040000
2025-12-31	SHB	0.030060	0.000000	0.000000	0.081851	0.040000	0.040000
2025-12-31	LPB	0.054211	0.300000	0.073663	0.102389	0.040000	0.040000
2025-12-31	SSB	0.019446	0.000000	0.185126	0.000000	0.040000	0.040000
2025-12-31	MSB	0.016210	0.000000	0.000000	0.000000	0.040000	0.040000
2025-12-31	OCB	0.012942	0.000000	0.000000	0.000000	0.040000	0.040000
2025-12-31	NAB	0.008080	0.257511	0.000000	0.042214	0.040000	0.040000
2025-12-31	BAB	0.004598	0.000000	0.300000	0.000000	0.040000	0.040000
2025-12-31	KLB	0.003519	0.000000	0.032932	0.032585	0.040000	0.040000
2025-12-31	PGB	0.002760	0.038384	0.034501	0.000000	0.040000	0.040000
2025-12-31	VAB	0.003500	0.006197	0.000000	0.000000	0.040000	0.040000
2025-12-31	VBB	0.004468	0.066456	0.000000	0.000000	0.040000	0.040000
2025-12-31	SGB	0.001672	0.000000	0.145791	0.000000	0.040000	0.040000
2025-12-31	ABB	0.005382	0.000000	0.000000	0.000000	0.040000	0.040000
2025-12-31	BVB	0.003367	0.000000	0.000000	0.000000	0.040000	0.040000
2026-01-31	VCB	0.185156	0.000000	0.154779	0.000000	0.040000	0.040000
2026-01-31	BID	0.105259	0.000000	0.000000	0.000000	0.040000	0.040000
2026-01-31	CTG	0.107008	0.000000	0.000000	0.000000	0.040000	0.040000
2026-01-31	MBB	0.078537	0.000000	0.000000	0.154738	0.040000	0.040000
2026-01-31	TCB	0.093342	0.167819	0.000000	0.118043	0.040000	0.040000
2026-01-31	ACB	0.047509	0.000000	0.000000	0.000000	0.040000	0.040000
2026-01-31	VPB	0.086071	0.000000	0.000000	0.000000	0.040000	0.040000
2026-01-31	HDB	0.057289	0.067623	0.000000	0.134618	0.040000	0.040000
2026-01-31	VIB	0.022078	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2026-01-31	TPB	0.018281	0.000000	0.000000	0.045092	0.040000	0.040000
2026-01-31	STB	0.042138	0.000000	0.000000	0.235764	0.040000	0.040000
2026-01-31	SHB	0.028523	0.000000	0.000000	0.086680	0.040000	0.040000
2026-01-31	LPB	0.045474	0.300000	0.056810	0.082144	0.040000	0.040000
2026-01-31	SSB	0.019132	0.000000	0.204459	0.000000	0.040000	0.040000
2026-01-31	MSB	0.014910	0.000000	0.000000	0.000000	0.040000	0.040000
2026-01-31	OCB	0.012264	0.000000	0.000000	0.000000	0.040000	0.040000
2026-01-31	NAB	0.007881	0.300000	0.000000	0.000000	0.040000	0.040000
2026-01-31	BAB	0.004415	0.000000	0.300000	0.000000	0.040000	0.040000
2026-01-31	KLB	0.003769	0.000000	0.045887	0.029394	0.040000	0.040000
2026-01-31	PGB	0.002628	0.058273	0.028130	0.000000	0.040000	0.040000
2026-01-31	VAB	0.003272	0.048768	0.000000	0.113527	0.040000	0.040000
2026-01-31	VBB	0.004316	0.030685	0.000000	0.000000	0.040000	0.040000
2026-01-31	SGB	0.001619	0.000000	0.209935	0.000000	0.040000	0.040000
2026-01-31	ABB	0.005868	0.026833	0.000000	0.000000	0.040000	0.040000
2026-01-31	BVB	0.003260	0.000000	0.000000	0.000000	0.040000	0.040000
2026-02-28	VCB	0.205519	0.000000	0.148265	0.000000	0.040000	0.040000
2026-02-28	BID	0.136900	0.000000	0.000000	0.000000	0.040000	0.040000
2026-02-28	CTG	0.105003	0.000000	0.000000	0.000000	0.040000	0.040000
2026-02-28	MBB	0.076439	0.000000	0.000000	0.244560	0.040000	0.040000
2026-02-28	TCB	0.086925	0.182342	0.000000	0.060268	0.040000	0.040000
2026-02-28	ACB	0.043190	0.000000	0.000000	0.000000	0.040000	0.040000
2026-02-28	VPB	0.076148	0.000000	0.000000	0.000000	0.040000	0.040000
2026-02-28	HDB	0.049419	0.186305	0.000000	0.134815	0.040000	0.040000
2026-02-28	VIB	0.020047	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2026-02-28	TPB	0.016647	0.000000	0.000000	0.016759	0.040000	0.040000
2026-02-28	STB	0.041436	0.000000	0.007043	0.300000	0.040000	0.040000
2026-02-28	SHB	0.025277	0.000000	0.000000	0.091074	0.040000	0.040000
2026-02-28	LPB	0.041022	0.300000	0.044972	0.116802	0.040000	0.040000
2026-02-28	SSB	0.017023	0.000000	0.213821	0.000000	0.040000	0.040000
2026-02-28	MSB	0.013498	0.000000	0.000000	0.000000	0.040000	0.040000
2026-02-28	OCB	0.010963	0.000000	0.000000	0.000000	0.040000	0.040000
2026-02-28	NAB	0.007057	0.240903	0.000000	0.035722	0.040000	0.040000
2026-02-28	BAB	0.004515	0.000000	0.300000	0.000000	0.040000	0.040000
2026-02-28	KLB	0.003087	0.033220	0.046682	0.000000	0.040000	0.040000
2026-02-28	PGB	0.002949	0.000000	0.023988	0.000000	0.040000	0.040000
2026-02-28	VAB	0.003218	0.000000	0.000000	0.000000	0.040000	0.040000
2026-02-28	VBB	0.003983	0.012059	0.000000	0.000000	0.040000	0.040000
2026-02-28	SGB	0.001586	0.000000	0.215229	0.000000	0.040000	0.040000
2026-02-28	ABB	0.005019	0.045171	0.000000	0.000000	0.040000	0.040000
2026-02-28	BVB	0.003130	0.000000	0.000000	0.000000	0.040000	0.040000
2026-03-31	VCB	0.193651	0.000000	0.069068	0.000000	0.040000	0.040000
2026-03-31	BID	0.124527	0.000000	0.000000	0.000000	0.040000	0.040000
2026-03-31	CTG	0.106090	0.000000	0.000000	0.000000	0.040000	0.040000
2026-03-31	MBB	0.081979	0.000000	0.000000	0.134013	0.040000	0.040000
2026-03-31	TCB	0.089833	0.178748	0.000000	0.192372	0.040000	0.040000
2026-03-31	ACB	0.045032	0.000000	0.002997	0.000000	0.040000	0.040000
2026-03-31	VPB	0.080577	0.000000	0.000000	0.000000	0.040000	0.040000
2026-03-31	HDB	0.049958	0.138152	0.000000	0.127838	0.040000	0.040000
2026-03-31	VIB	0.019935	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2026-03-31	TPB	0.018227	0.000000	0.000000	0.041071	0.040000	0.040000
2026-03-31	STB	0.044096	0.027699	0.060705	0.294892	0.040000	0.040000
2026-03-31	SHB	0.025790	0.000000	0.000000	0.000000	0.040000	0.040000
2026-03-31	LPB	0.043396	0.300000	0.065898	0.063256	0.040000	0.040000
2026-03-31	SSB	0.017170	0.000000	0.221723	0.000000	0.040000	0.040000
2026-03-31	MSB	0.013593	0.000000	0.000000	0.049757	0.040000	0.040000
2026-03-31	OCB	0.011079	0.000000	0.000000	0.000000	0.040000	0.040000
2026-03-31	NAB	0.007150	0.262583	0.045402	0.000000	0.040000	0.040000
2026-03-31	BAB	0.004594	0.000000	0.300000	0.000000	0.040000	0.040000
2026-03-31	KLB	0.003170	0.000000	0.008083	0.006902	0.040000	0.040000
2026-03-31	PGB	0.003115	0.000000	0.025683	0.000000	0.040000	0.040000
2026-03-31	VAB	0.003148	0.000000	0.000000	0.089898	0.040000	0.040000
2026-03-31	VBB	0.004038	0.067016	0.068944	0.000000	0.040000	0.040000
2026-03-31	SGB	0.001624	0.000000	0.131497	0.000000	0.040000	0.040000
2026-03-31	ABB	0.005138	0.025803	0.000000	0.000000	0.040000	0.040000
2026-03-31	BVB	0.003089	0.000000	0.000000	0.000000	0.040000	0.040000
2026-04-30	VCB	0.191340	0.000000	0.034094	0.000000	0.040000	0.040000
2026-04-30	BID	0.113053	0.000000	0.000000	0.000000	0.040000	0.040000
2026-04-30	CTG	0.105919	0.000000	0.000000	0.000000	0.040000	0.040000
2026-04-30	MBB	0.083973	0.000000	0.000000	0.135348	0.040000	0.040000
2026-04-30	TCB	0.083984	0.192158	0.000000	0.167530	0.040000	0.040000
2026-04-30	ACB	0.047678	0.000000	0.042229	0.000000	0.040000	0.040000
2026-04-30	VPB	0.082023	0.000000	0.000000	0.000000	0.040000	0.040000
2026-04-30	HDB	0.050207	0.070429	0.000000	0.160492	0.040000	0.040000
2026-04-30	VIB	0.021560	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2026-04-30	TPB	0.017822	0.000000	0.000000	0.092014	0.040000	0.040000
2026-04-30	STB	0.046217	0.100852	0.072188	0.300000	0.040000	0.040000
2026-04-30	SHB	0.027759	0.000000	0.000000	0.000000	0.040000	0.040000
2026-04-30	LPB	0.046625	0.300000	0.055763	0.105675	0.040000	0.040000
2026-04-30	SSB	0.019006	0.000000	0.231462	0.000000	0.040000	0.040000
2026-04-30	MSB	0.014203	0.000000	0.000000	0.038940	0.040000	0.040000
2026-04-30	OCB	0.011703	0.000000	0.000000	0.000000	0.040000	0.040000
2026-04-30	NAB	0.007350	0.258381	0.051370	0.000000	0.040000	0.040000
2026-04-30	BAB	0.004691	0.000000	0.300000	0.000000	0.040000	0.040000
2026-04-30	KLB	0.002994	0.000000	0.023740	0.000000	0.040000	0.040000
2026-04-30	PGB	0.003224	0.000000	0.012073	0.000000	0.040000	0.040000
2026-04-30	VAB	0.003250	0.000000	0.000000	0.000000	0.040000	0.040000
2026-04-30	VBB	0.004499	0.000000	0.027886	0.000000	0.040000	0.040000
2026-04-30	SGB	0.001749	0.000000	0.149194	0.000000	0.040000	0.040000
2026-04-30	ABB	0.006038	0.078180	0.000000	0.000000	0.040000	0.040000
2026-04-30	BVB	0.003132	0.000000	0.000000	0.000000	0.040000	0.040000
2026-05-31	VCB	0.191279	0.000000	0.000000	0.000000	0.040000	0.040000
2026-05-31	BID	0.111754	0.000000	0.000000	0.000000	0.040000	0.040000
2026-05-31	CTG	0.103916	0.000000	0.000000	0.000000	0.040000	0.040000
2026-05-31	MBB	0.080326	0.120225	0.000000	0.000000	0.040000	0.040000
2026-05-31	TCB	0.089926	0.014423	0.000000	0.022297	0.040000	0.040000
2026-05-31	ACB	0.046210	0.000000	0.051955	0.000000	0.040000	0.040000
2026-05-31	VPB	0.079088	0.000000	0.000000	0.000000	0.040000	0.040000
2026-05-31	HDB	0.050968	0.000000	0.000000	0.278046	0.040000	0.040000
2026-05-31	VIB	0.021240	0.000000	0.000000	0.000000	0.040000	0.040000

Date	Ticker	w_mkt	w_TAN	w_MV	w_BL	w_EW	w_RP
2026-05-31	TPB	0.017256	0.000000	0.000000	0.161473	0.040000	0.040000
2026-05-31	STB	0.048713	0.131561	0.077103	0.271547	0.040000	0.040000
2026-05-31	SHB	0.025589	0.000000	0.000000	0.030358	0.040000	0.040000
2026-05-31	LPB	0.051117	0.300000	0.045324	0.028583	0.040000	0.040000
2026-05-31	SSB	0.018188	0.000000	0.265335	0.000000	0.040000	0.040000
2026-05-31	MSB	0.014930	0.000000	0.000000	0.000000	0.040000	0.040000
2026-05-31	OCB	0.011469	0.000000	0.000000	0.000000	0.040000	0.040000
2026-05-31	NAB	0.007717	0.259867	0.016185	0.000000	0.040000	0.040000
2026-05-31	BAB	0.004638	0.000000	0.300000	0.000000	0.040000	0.040000
2026-05-31	KLB	0.003265	0.000000	0.000000	0.086979	0.040000	0.040000
2026-05-31	PGB	0.003053	0.000000	0.028042	0.000000	0.040000	0.040000
2026-05-31	VAB	0.003203	0.000000	0.000000	0.120716	0.040000	0.040000
2026-05-31	VBB	0.005524	0.000000	0.009129	0.000000	0.040000	0.040000
2026-05-31	SGB	0.001685	0.000000	0.195197	0.000000	0.040000	0.040000
2026-05-31	ABB	0.005904	0.173923	0.011730	0.000000	0.040000	0.040000
2026-05-31	BVB	0.003042	0.000000	0.000000	0.000000	0.040000	0.040000

Phụ lục – Lãi suất Phi rủi ro VN10Y theo Năm

Lãi suất trái phiếu chính phủ Việt Nam kỳ hạn 10 năm (VN10Y) được dùng làm lãi suất phi rủi ro. Lãi suất tháng = VN10Y_năm / 12.

Năm	VN10Y (% năm)	RF tháng (%)
2014	6.50	0.5417
2015	6.50	0.5417
2016	6.20	0.5167
2017	5.80	0.4833
2018	4.58	0.3817
2019	4.58	0.3817
2020	3.00	0.2500
2021	2.53	0.2108
2022	4.00	0.3333
2023	3.50	0.2917
2024	3.06	0.2550
2025	3.10	0.2583
2026	4.35	0.3625